



DATA ANALYTICS

UE24CS342AA2

UNIT-1

**Lecture 1 : Introduction to DA , Data
Sources and Representations**

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 1: Introduction to Data Analytics, Data Sources and Representations

Slides excerpted from: U. Dinesh Kumar,
“Business Analytics”, Wiley, 2nd Edition 2022

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With grateful thanks for contribution of slides to:

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DATA ANALYTICS

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Component	Marks
ISA 1 (CBT)	40 marks (scaled to 20 marks)
ISA 2 (CBT)	40 marks (scaled to 20 marks)
Worksheets (Experiential Learning)	Best 3 of 4 (3 marks)
Hackathon (Experiential Learning)	7 marks
ESA	100 marks (scaled to 50 marks)

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Handholding/ Capacity Building Sessions



Component	Marks
Mini Labs (Experiential Learning)	In class/ at home
Session 1: Introduction to R	Online: Thursday 7:00-8:00pm
Session 2: Prompt Engineering for Data Analytics	Online: TBA
Session 3: Introduction to Kaggle	Online: TBA
Session 4: Introduction to Plotly Express	Online: TBA
Session 5: Power BI	Online: TBA
Session 6: Prophet and Neural Prophet	Online: TBA
Session 7: Snowflake	Online: TBA

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Prerequisite For Course: Mathematics for Computer Science



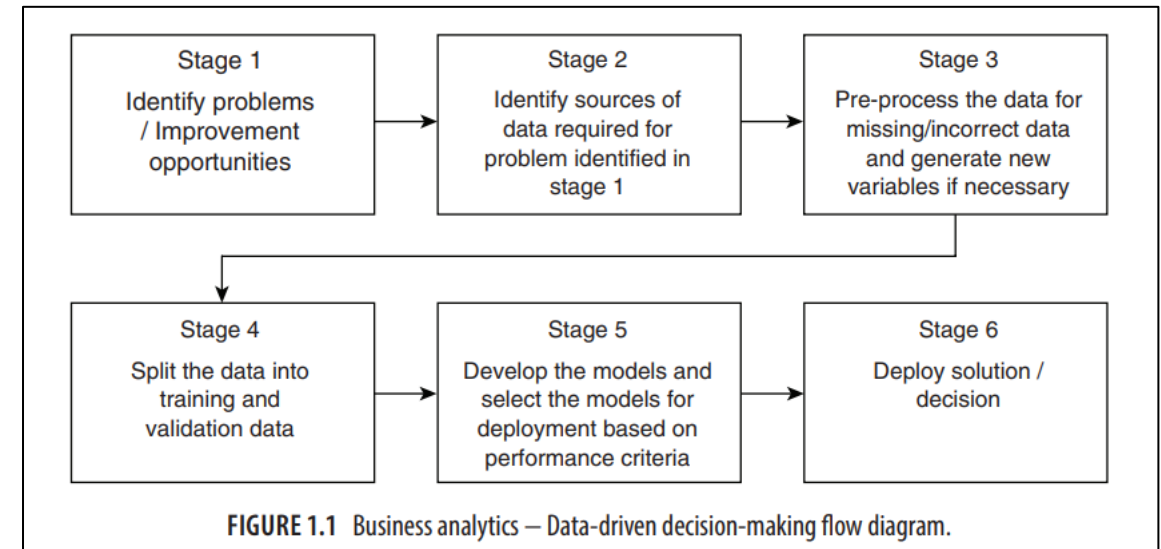
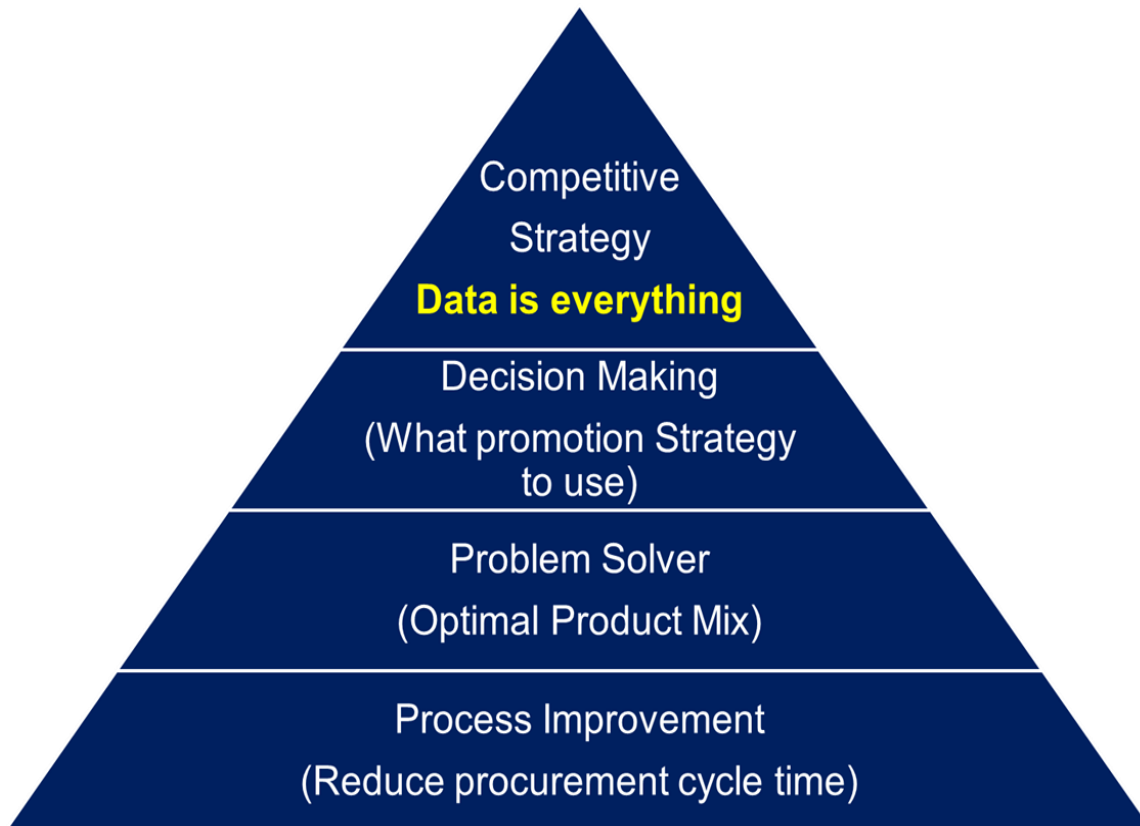
Brush your skills and take the Test!!!!

[Click Here](#)

- The science of examining raw data to **elicit patterns, develop insights** , and **draw conclusions** to help take a **business decision**.
- **The need** : Business decisions are very complex. There exist several alternate solutions, complex interdependent factors and lack of available time to take a decision.

Analysis vs Analytics

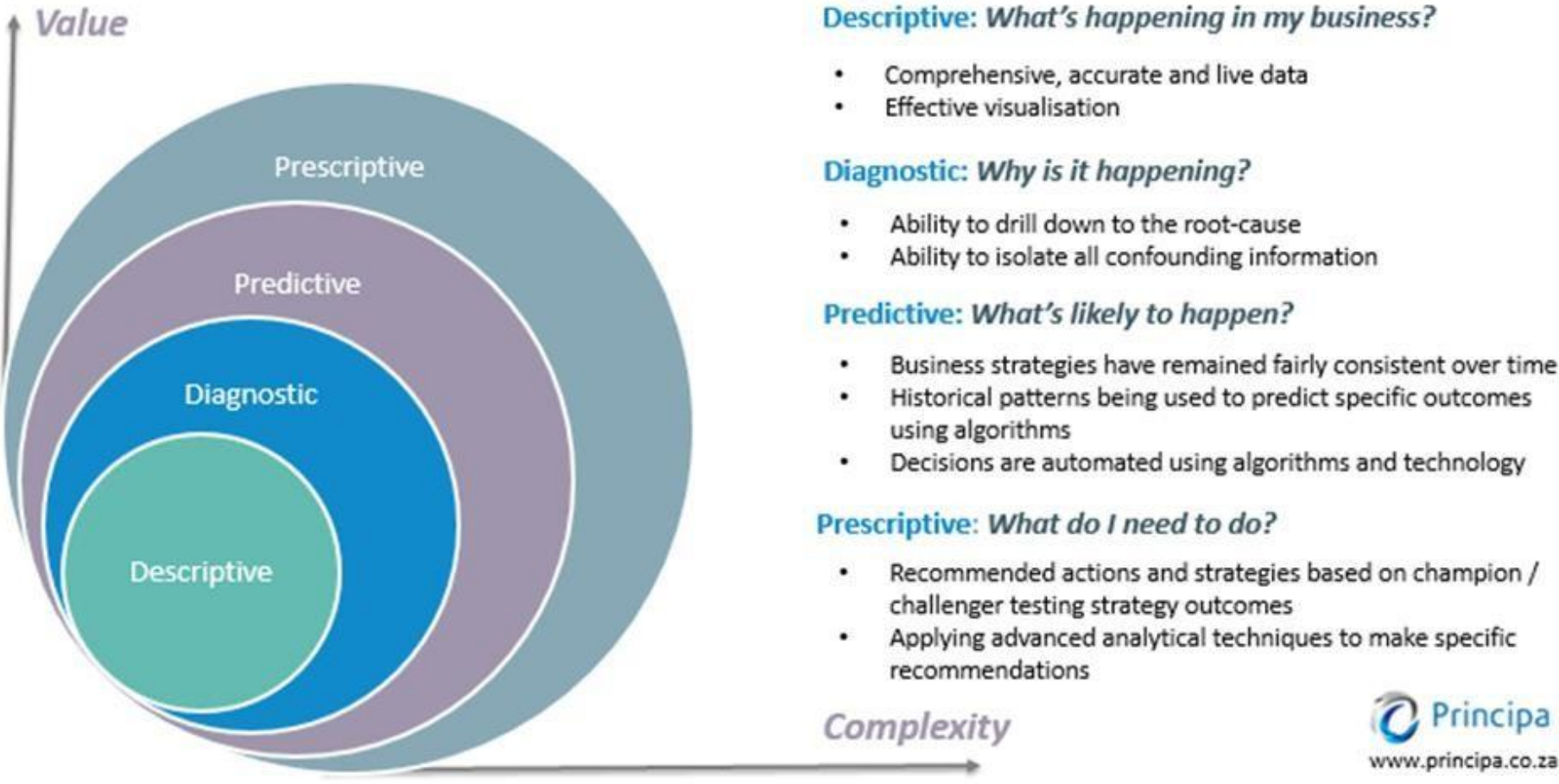
- Analysis – Examining and understanding past data.
- Analytics – Analysis + forecasting (or predictive modeling).



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What does it involve?

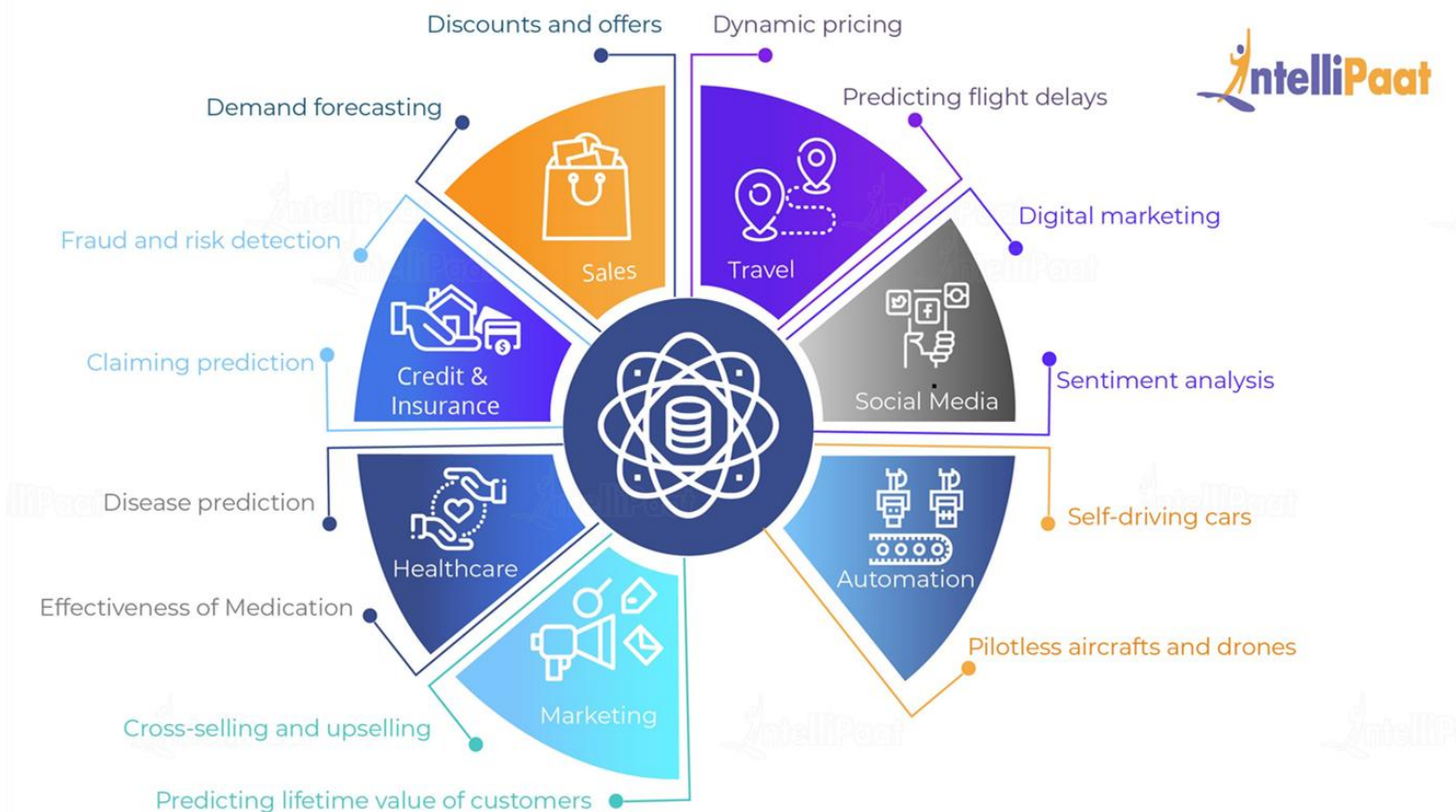
4 types of Data Analytics



DATA ANALYTICS

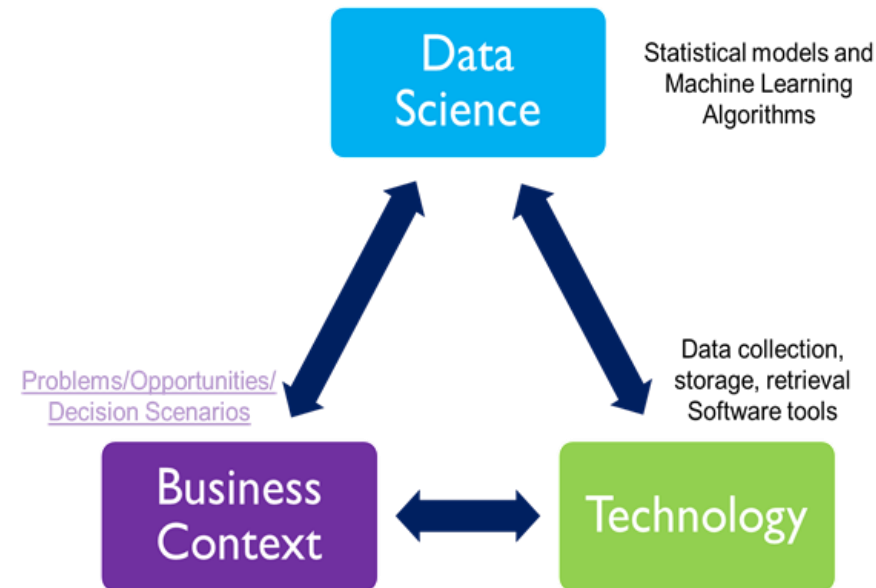
Why is it important?

Data Analytics is used in all these application areas...



... and more!

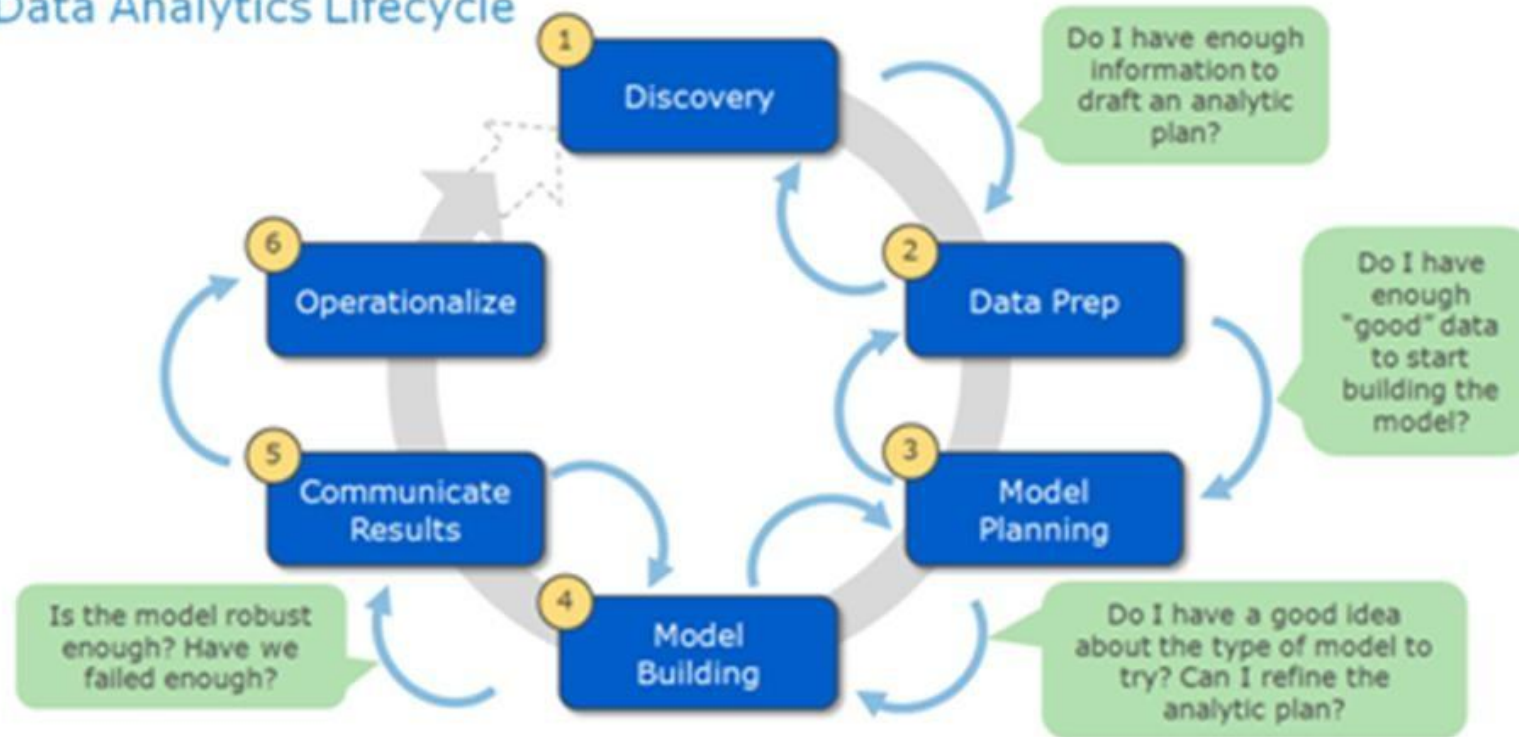
- **Banking** : To reduce cheque clearance time, in determining loan approval and interest rate.
- **E-Commerce** : To analyze buyer behavior to plan inventory and recommend products.
- **Retail stores** : Shelf space allocation to drive the profits up.
- **OTT Platforms** : Recommend content a user would like.



DATA ANALYTICS

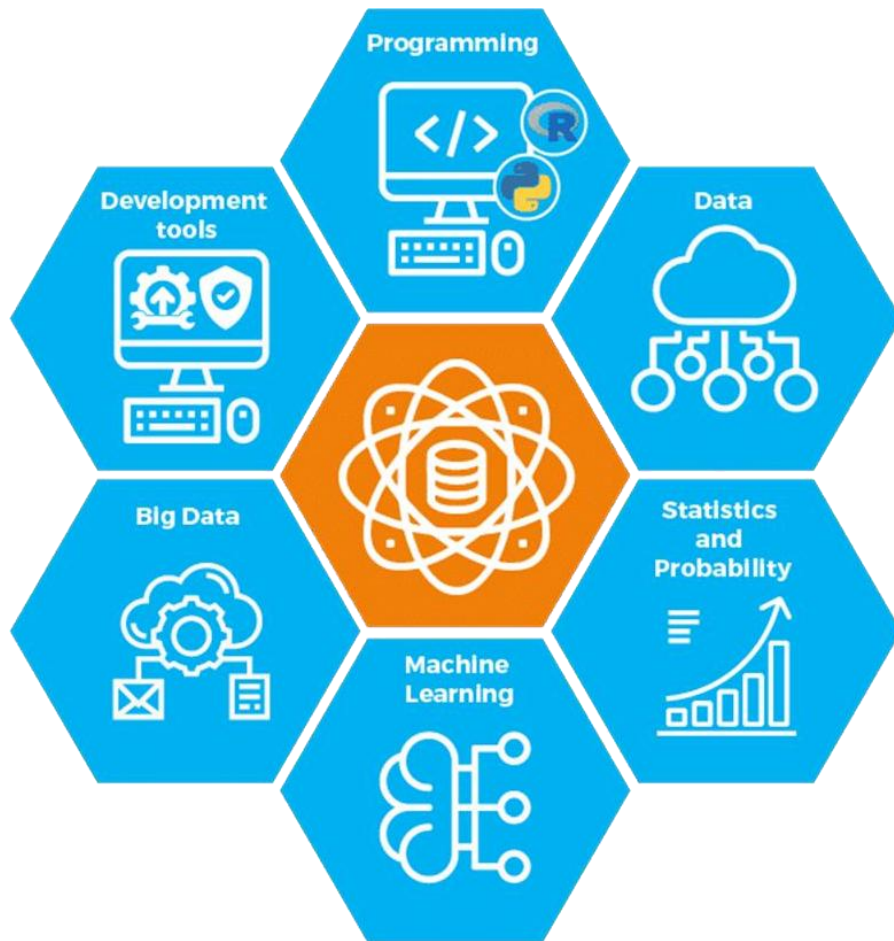
Lifecycle

Data Analytics Lifecycle



DATA ANALYTICS

Skills Required



And...
a secret ingredient



Intuition or
deductive reasoning
and domain knowledge

Indian online grocery store [bigbasket.com](https://www.bigbasket.com)

Problem context driving analytics : “Did you forget?” feature

- The ability to predict the items that a customer may have forgotten to order can have a significant impact on the profits of online grocers such as [bigbasket.com](https://www.bigbasket.com)
- The ability to ask right questions is an important success criteria for analytics projects.

Indian online grocery store bigbasket.com

Technology:

- To find out whether a customer has forgotten to place an order for an item
- Information technology is used for data capture, data storage, data preparation, data analysis, data share and to deploy solution
- An important output of analytics is automation of actionable items derived from analytical models which is usually achieved using IT

Indian online grocery store bigbasket.com

Data science is the most important component of analytics, it consists of statistical and operations research techniques, machine learning and deep learning algorithms.

The objective of the data science component of analytics is to identify the most appropriate statistical model/machine learning algorithm that is best based on a measure of accuracy.

Example: “did you forget?” prediction is a classification problem in which customers are classified into:

1. Forget
2. Not forget

DATA ANALYTICS

Data Sources

- There has been enormous data growth in both commercial and scientific databases due to advances in data generation and collection technologies.
- New mantra
 - Gather whatever data you can whenever and wherever possible
- Expectations
 - Gathered data will have value either for the purpose collected or for a purpose not envisioned



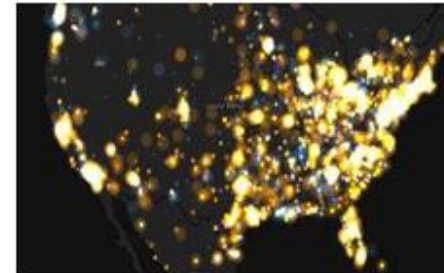
Cyber Security



E-Commerce



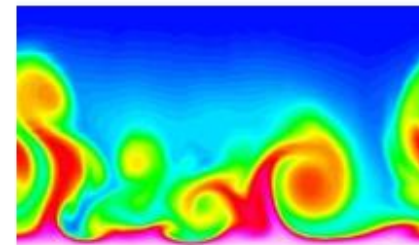
Traffic Patterns



Social Networking: Twitter



Sensor Networks



Computational Simulations

- Lots of data is collected and warehoused every day
- Yahoo has peta bytes of web data
- Facebook has billions of active users
- Purchases at department/ grocery stores, e-commerce
 - Amazon handles millions of visits/day
- Bank/Credit Card transactions



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How large is *big* (data)?

- 1 bit
- 1 byte = 8 bits
- 1 KB = 1024 bytes
- 1 MB = 1024 KB (kilobytes)
- 1 GB = 1024 MB (megabytes)
- 1 TB = 1024 GB (gigabytes) $\approx 10^{12}$ bytes
- 1 PB = 1024 TB (terabytes) $\approx 10^{15}$ bytes

20 PB = amt of data processed by Google per day!

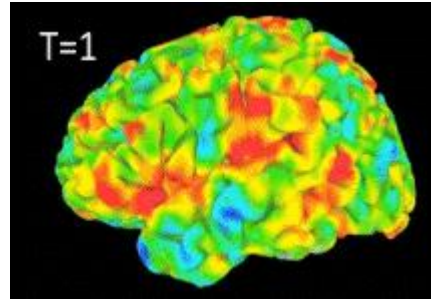
- 1 EB = 1024 PB (petabytes)
- 1 ZB = 1024 EB (exabytes)
- 1 YB = 1024 ZB (zettabytes)
- What is a Domegemegrottebyte?

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Data Sources

Data collected and stored at enormous speeds

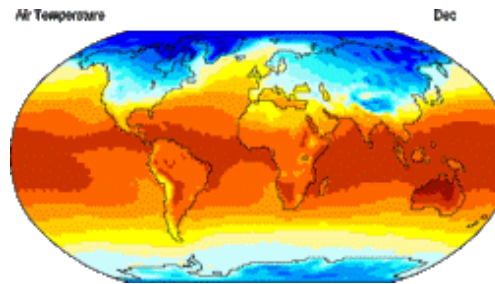
- Remote sensors on a satellite
 - NASA EOSDIS archives over petabytes of earth science data / year
- Telescopes scanning the skies
 - Sky survey data
- High-throughput biological data
- Scientific simulations -
terabytes of data generated in a few hours



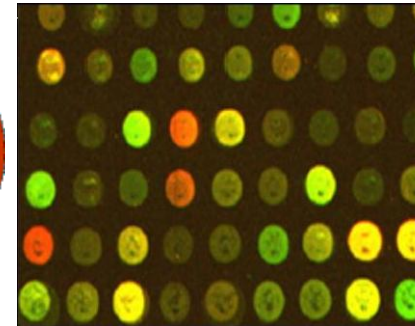
fMRI Data from Brain



Sky Survey Data



Surface Temperature of Earth



Gene Expression Data

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What is Data?

- Collection of **data objects** and their **attributes**.
- An **attribute** is a property or characteristic of an object.
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, dimension, or feature.
- A collection of attributes describe an **object**.
 - An object is also known as a record, point, case, sample, entity, or instance.

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- **Attribute values** are numbers or symbols assigned to an attribute for a particular object
- Distinction between attributes and attribute values
 - Same attribute can be mapped to different attribute values
 - Example: height can be measured in feet or meters
- Different attributes can be mapped to the same set of values
 - Example: Attribute values for ID and age are integers
 - But properties of attribute values can be different

DATA ANALYTICS

Types of Attributes

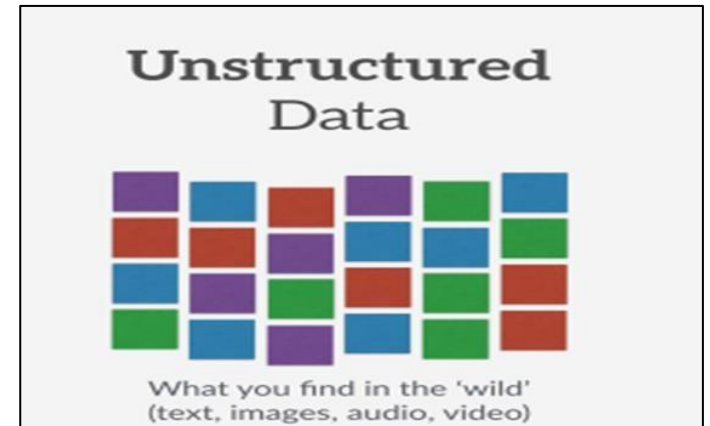
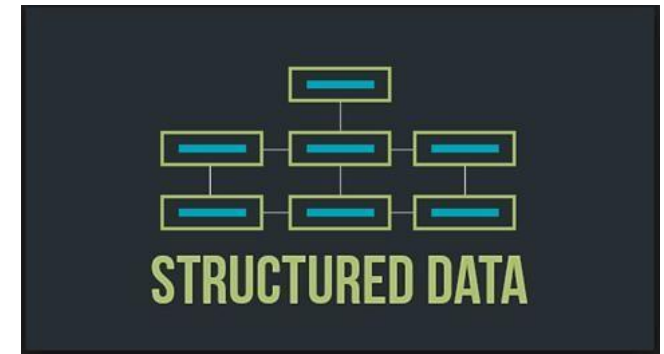


- There are different types of attributes
 - **Nominal**
 - Examples: ID numbers, eye color, zip codes
 - **Ordinal**
 - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height {tall, medium, short}
 - **Interval**
 - Examples: Calendar dates, temperatures in Celsius or Fahrenheit.
 - **Ratio**
 - Examples: Temperature in Kelvin, length, counts, elapsed time (e.g., time to run a race)

- Discrete Attribute
 - Has only a finite or countably infinite set of values
 - Examples: zip codes, counts, or the set of words in a collection of documents
 - Often represented as integer variables.
 - Note: **binary attributes** are a special case of discrete attributes
- Continuous Attribute
 - Has real numbers as attribute values
 - Examples: temperature, height, or weight.
 - Practically, real values can only be measured and represented using a finite number of digits.
 - Continuous attributes are typically represented as floating point variables.

Data Representations

- **Structured Data:** Structured data means that the data is described in a matrix form with labelled rows and columns.
- **Unstructured Data:** Any data that is not originally in the matrix form with rows and columns is an unstructured data.
- **Semi structured:** Semi-structured data (also known as partially structured data) is a type of data that doesn't follow the tabular structure associated with relational databases or other forms of data tables but does contain tags and metadata to separate semantic elements and establish hierarchies of records and fields.



DATA ANALYTICS

Data Representations

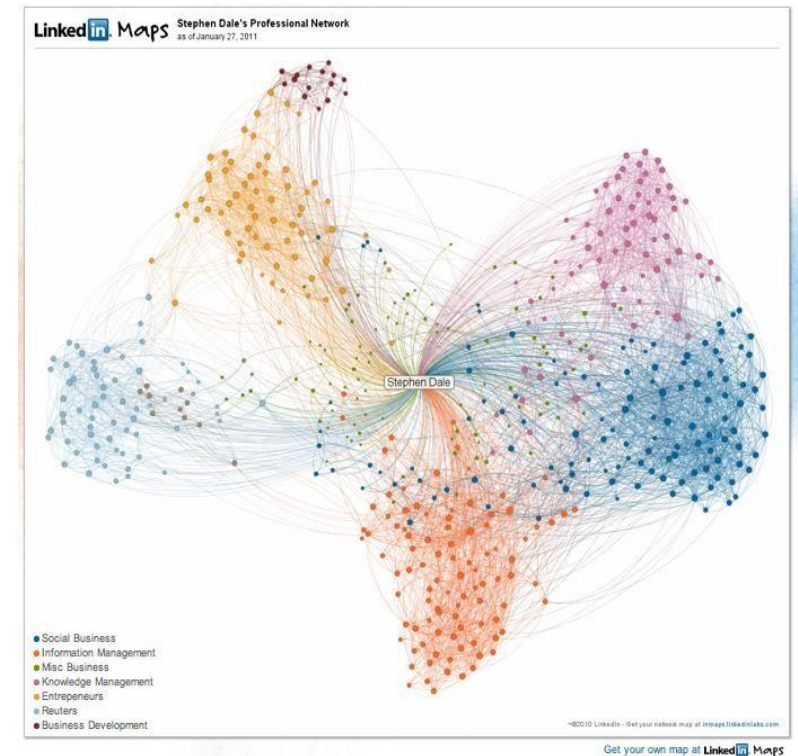


- Relational databases and spreadsheets. – **Structured Data**
- Text and multimedia content. Photos and graphic images, videos, streaming instrument data, webpages, PDF files, PowerPoint presentations, emails, blog entries, wikis and word processing documents. - **Unstructured Data**
- XML documents and NoSQL databases. – **Semi structured Data**
- For example, word processing software now can include metadata showing the author's name and the date created, with the bulk of the document just being unstructured text.

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Data Representations

- Record
 - Relational records
 - Data matrix, e.g., numerical matrix, crosstabs
 - Document data: text documents: term-frequency vector
 - Transaction data
- Graph and network
 - World Wide Web
 - Social or information networks
 - Molecular Structures



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Data Representations



- Ordered
 - Video data: sequence of images
 - Temporal data: time-series
 - Sequential Data: transaction sequences
 - Genetic sequence data

- Spatial, image and multimedia
 - Spatial data: maps
 - Image data
 - Video data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
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5	No	Divorced	95K	Yes
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Data Representations-Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

- Each document becomes a 'term' vector,
 - each term is a component (attribute) of the vector,
 - the value of each component is the number of times the corresponding term occurs in the document.

	team	coach	Play	ball	score	game	w i n	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Data Representations-Transaction data

- A special type of record data, where
 - each record (transaction) involves a set of items.
 - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

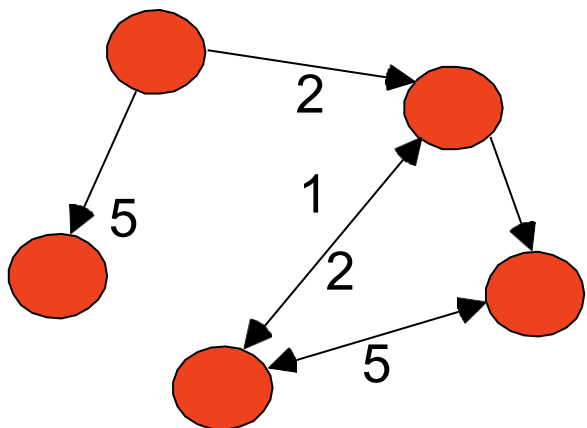
<i>TID</i>	<i>Items</i>
1	Bread,Coke,Milk
2	Beer,Bread
3	Beer,Coke,Diaper,Milk
4	Beer,Bread,Diaper,Milk
5	Coke,Diaper,Milk

DATA ANALYTICS

Data Representations

■ Graph Data

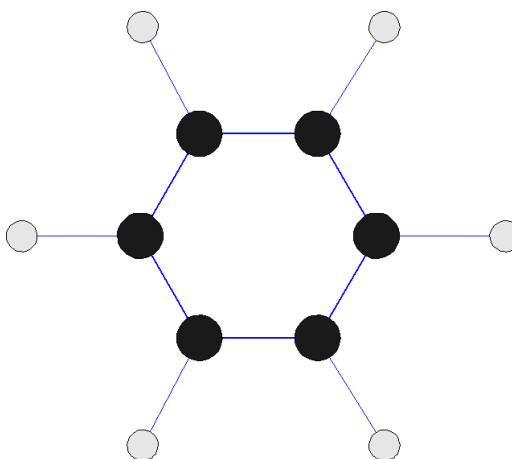
■ Examples: Generic graph and HTML Links



```
<a>Graph Partitioning </a>  
<li>  
<a href="papers/papers.html#aaaa">  
Parallel Solution of Sparse Linear System of Equations </a>  
<li>  
<a href="papers/papers.html#ffff">  
N-Body Computation and Dense Linear System Solvers
```

■ Chemical Data

■ Benzene Molecule: C_6H_6



■ Sequences of transactions

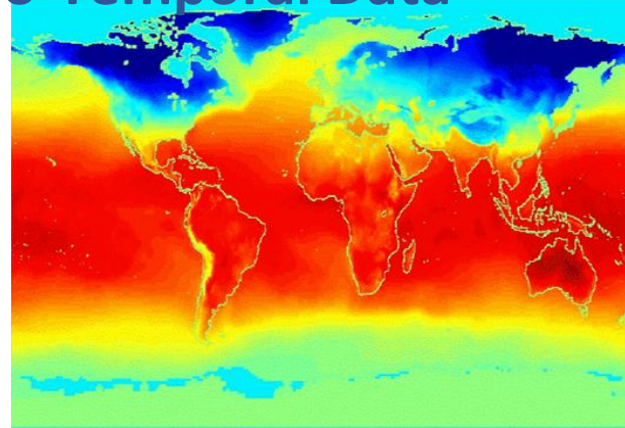
Items/Events

(A B) (D) (C E)
(B D) (C) (E)
(C D) (B) (A E)

(C D)

An element of
the sequence

■ Spatio-Temporal Data



■ Genomic sequence data

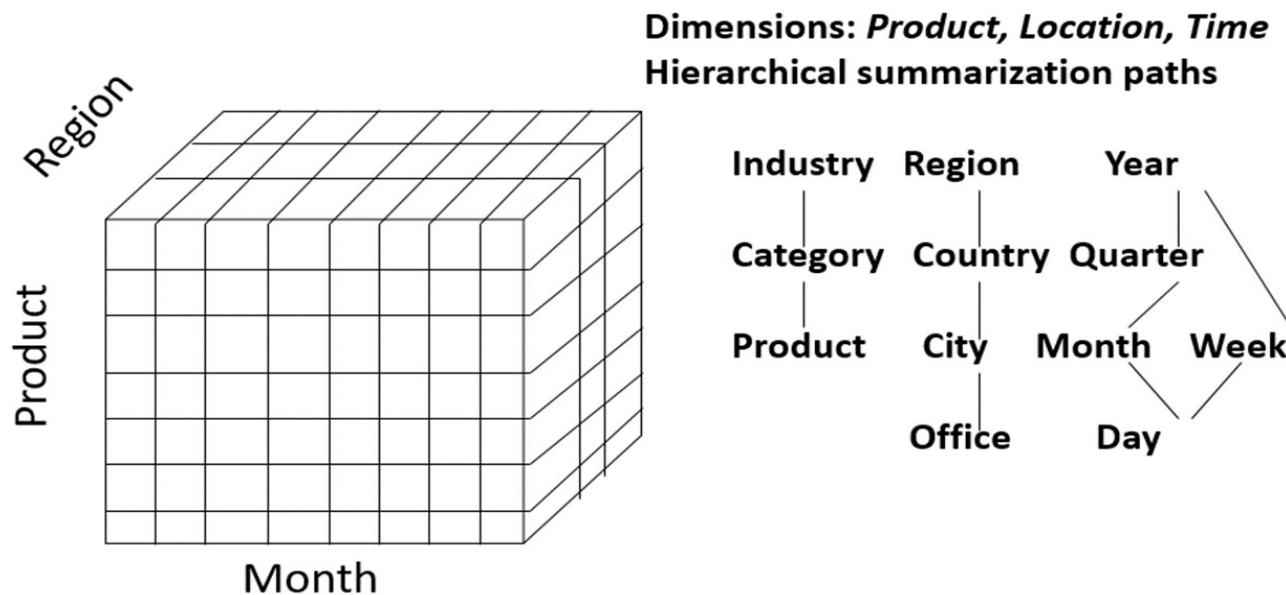
```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

DATA ANALYTICS

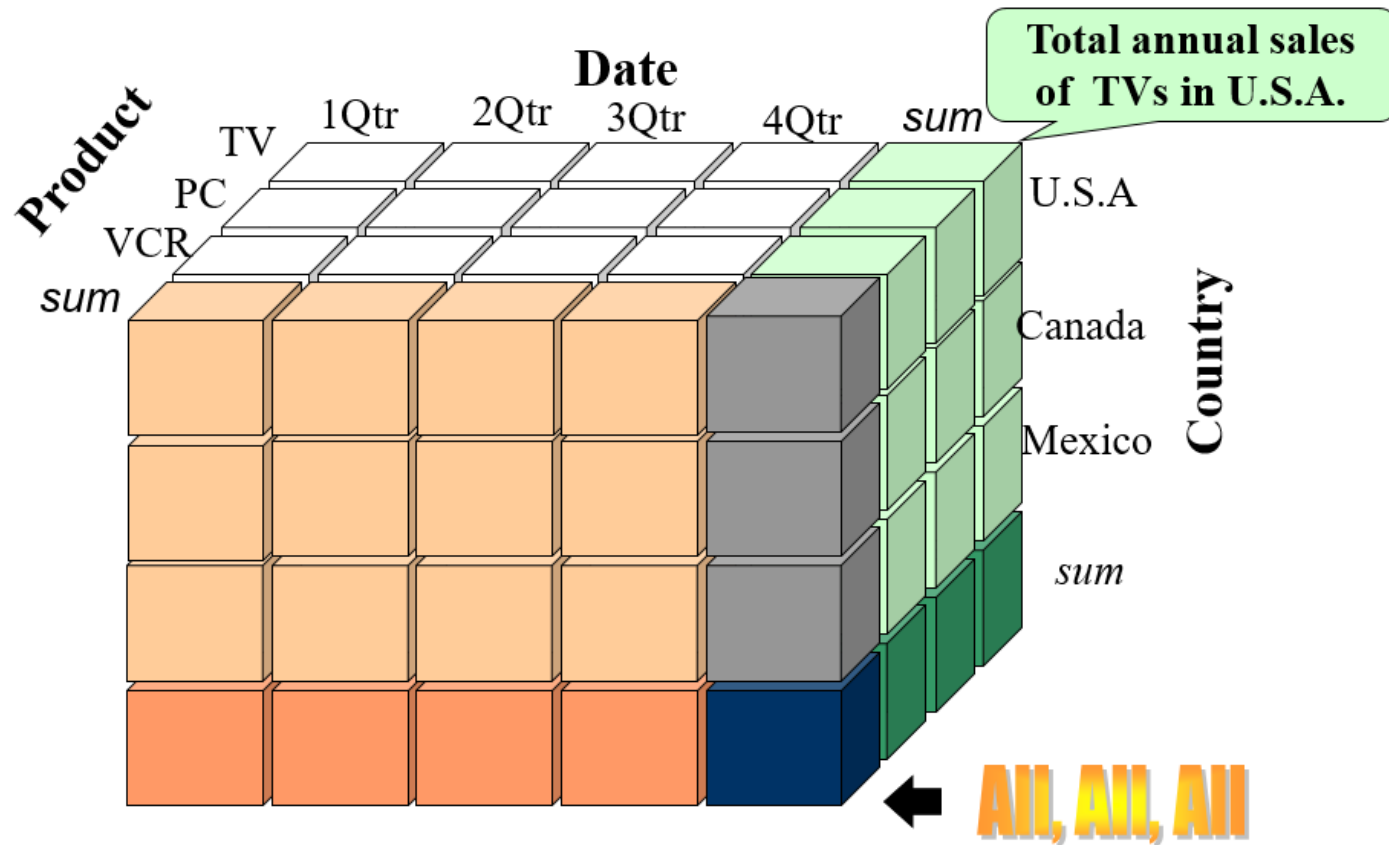
Data Representations- Data Warehouse

“A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management’s decision-making process.” —W. H. Inmon

Sales volume as a function of product, month, and region

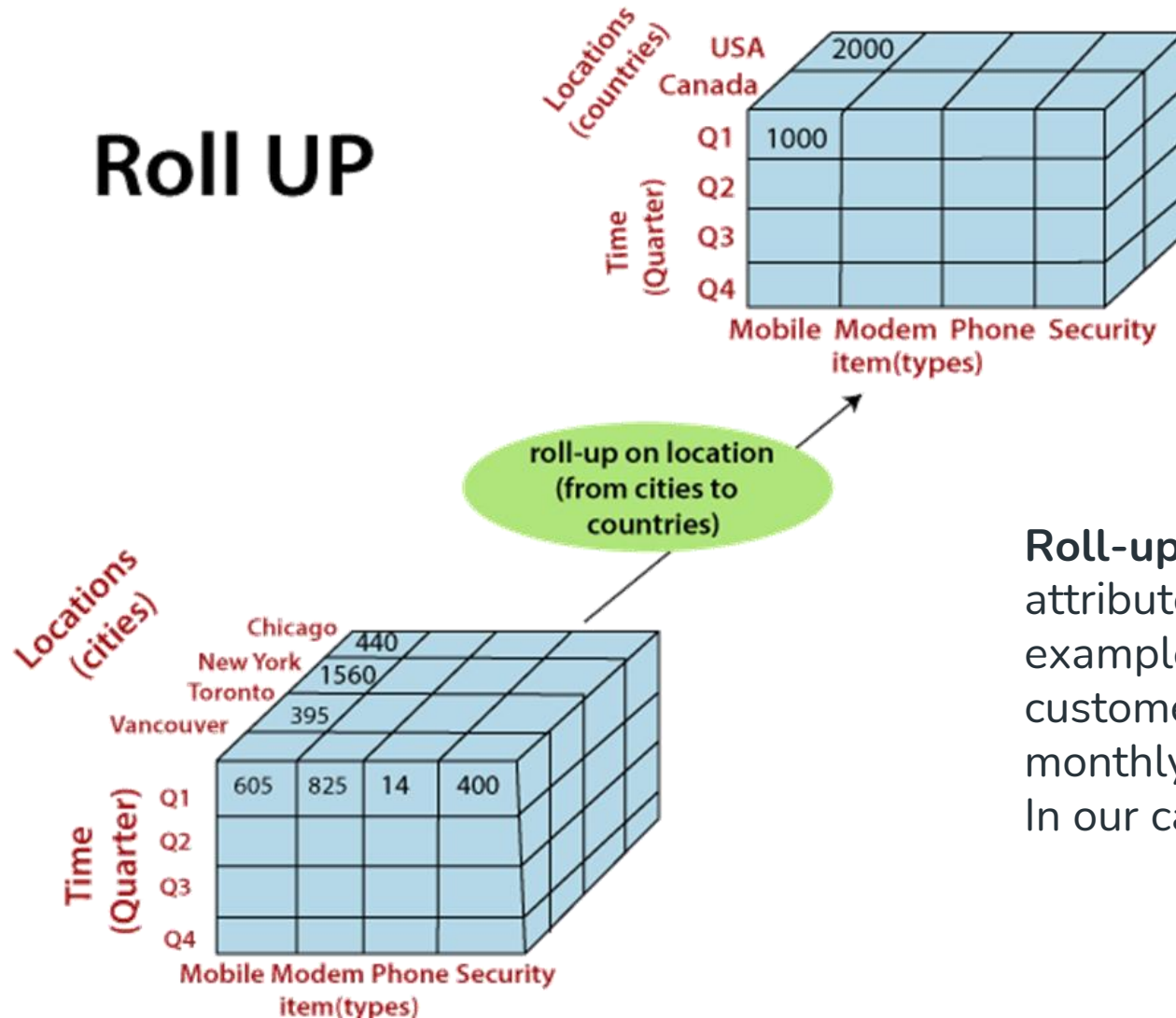


A Sample Data Cube

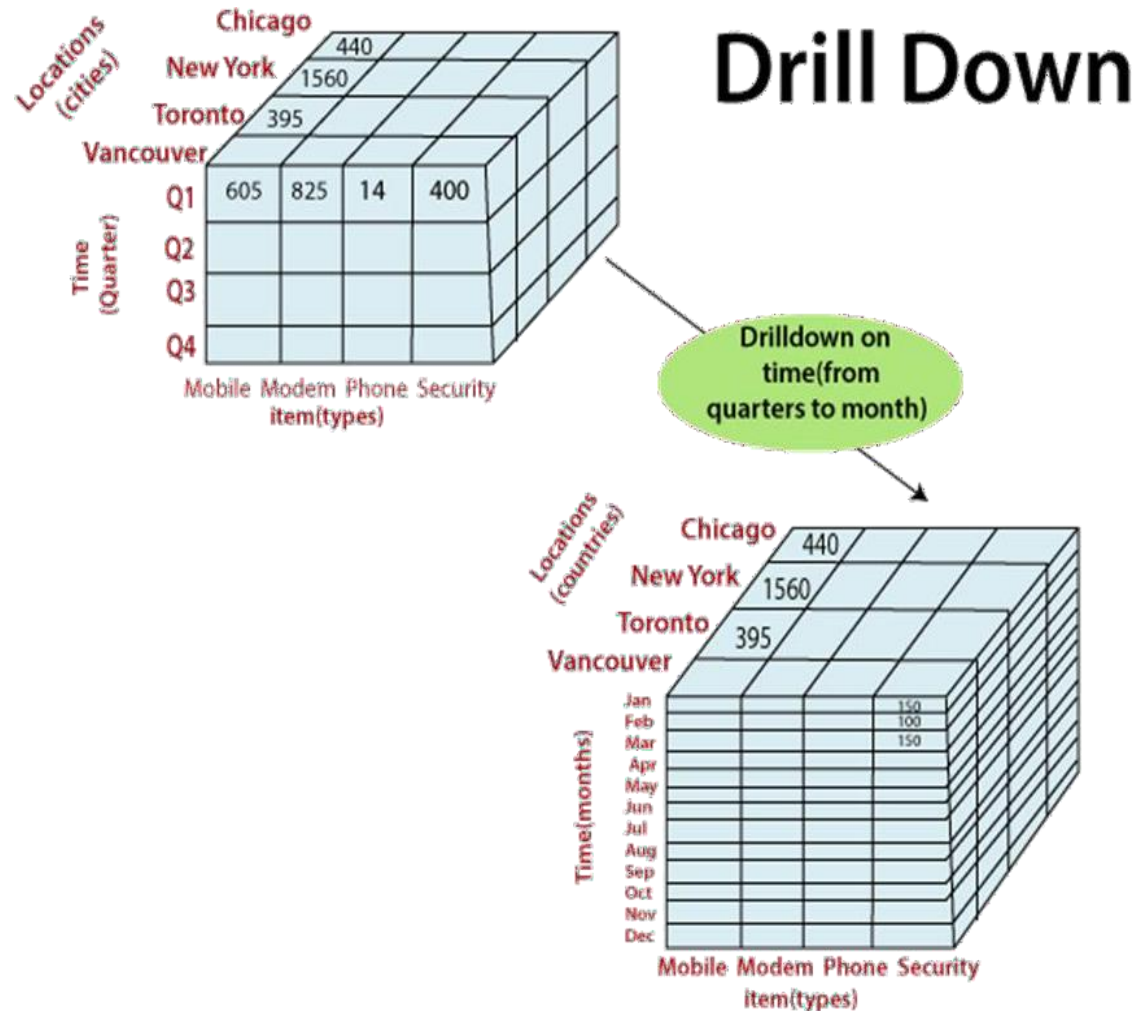


- **Roll up (drill-up):** summarize data
 - by climbing up hierarchy or by dimension reduction
- **Drill down (roll down):** reverse of roll-up
 - from higher level summary to lower level summary or detailed data, or introducing new dimensions
- **Slice and dice:** project and select
- **Pivot (rotate):**
 - reorient the cube, visualization, 3D to series of 2D planes
- Other operations
 - **drill across:** involving (across) more than one fact table
 - **drill through:** through the bottom level of the cube to its back-end relational tables (using SQL)

Roll UP

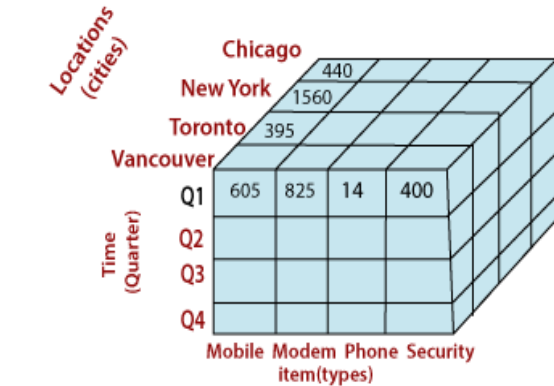


Roll-up: operation and aggregate certain similar data attributes having the same dimension together. For example, if the data cube displays the daily income of a customer, we can use a roll-up operation to find the monthly income of his salary. In our case we roll-up from cities to countries as shown

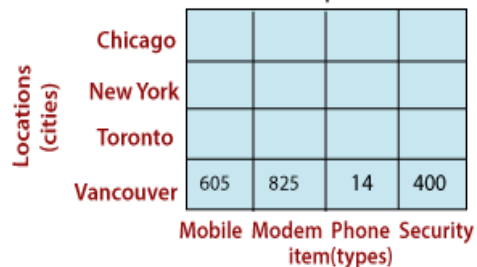


Drill-down: this operation is the reverse of the roll-up operation. It allows us to take particular information and then subdivide it further for coarser granularity analysis. It zooms into more detail. For example- if India is an attribute of a country column and we wish to see villages in India, then the drill-down operation splits India into states, districts, towns, cities, villages and then displays the required information. In our case we drill down from quarters to months.

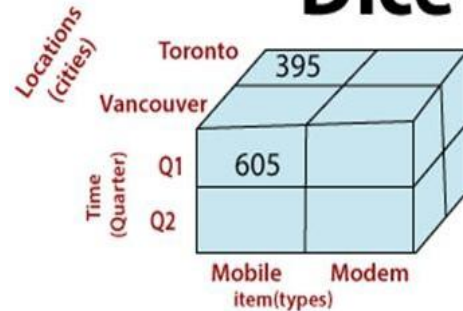
Slice



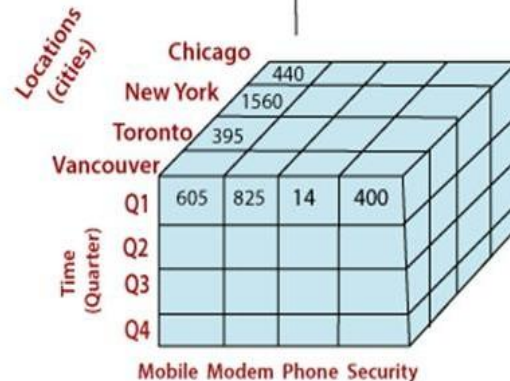
slice
for time
="Q1"



Dice

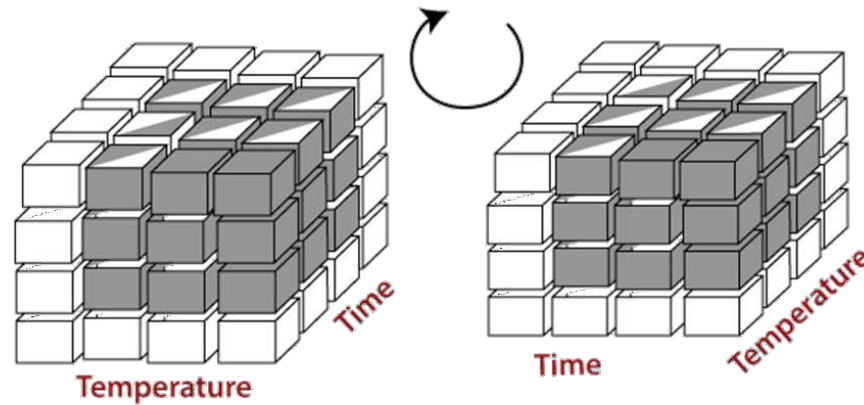
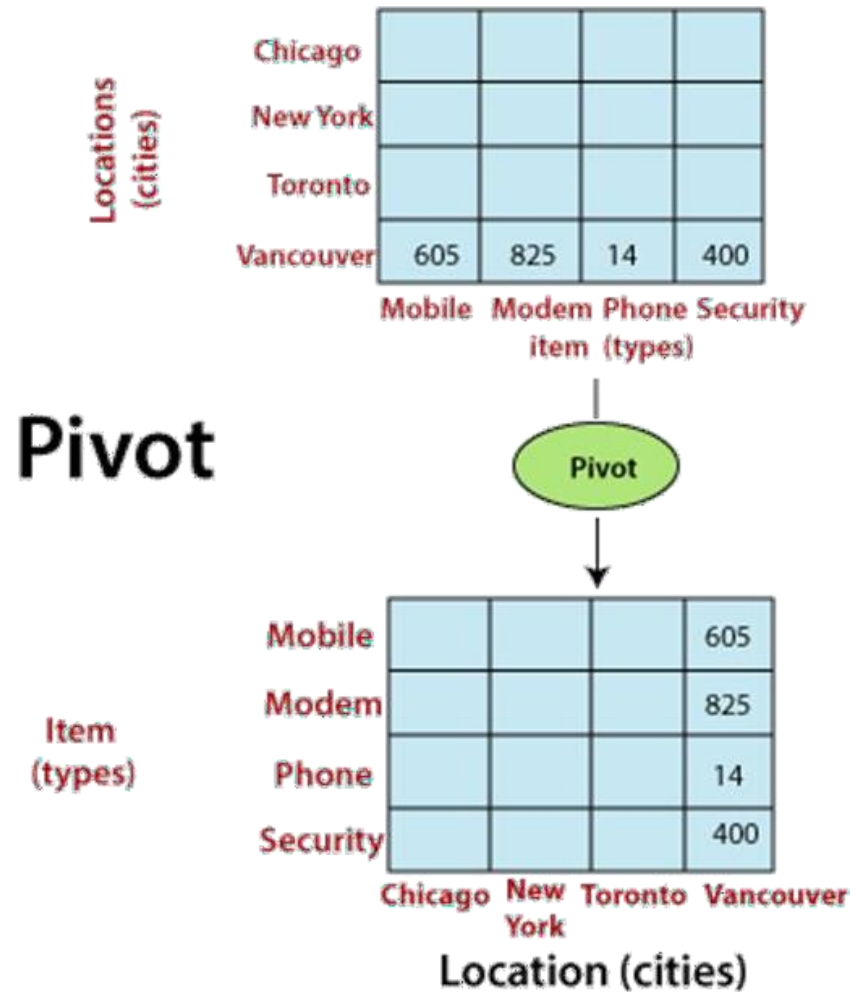


Dice for (location="Toronto"
or "Vancouver")
and (time="Q1" or "Q2") and
(item="Mobile" or "Modem")

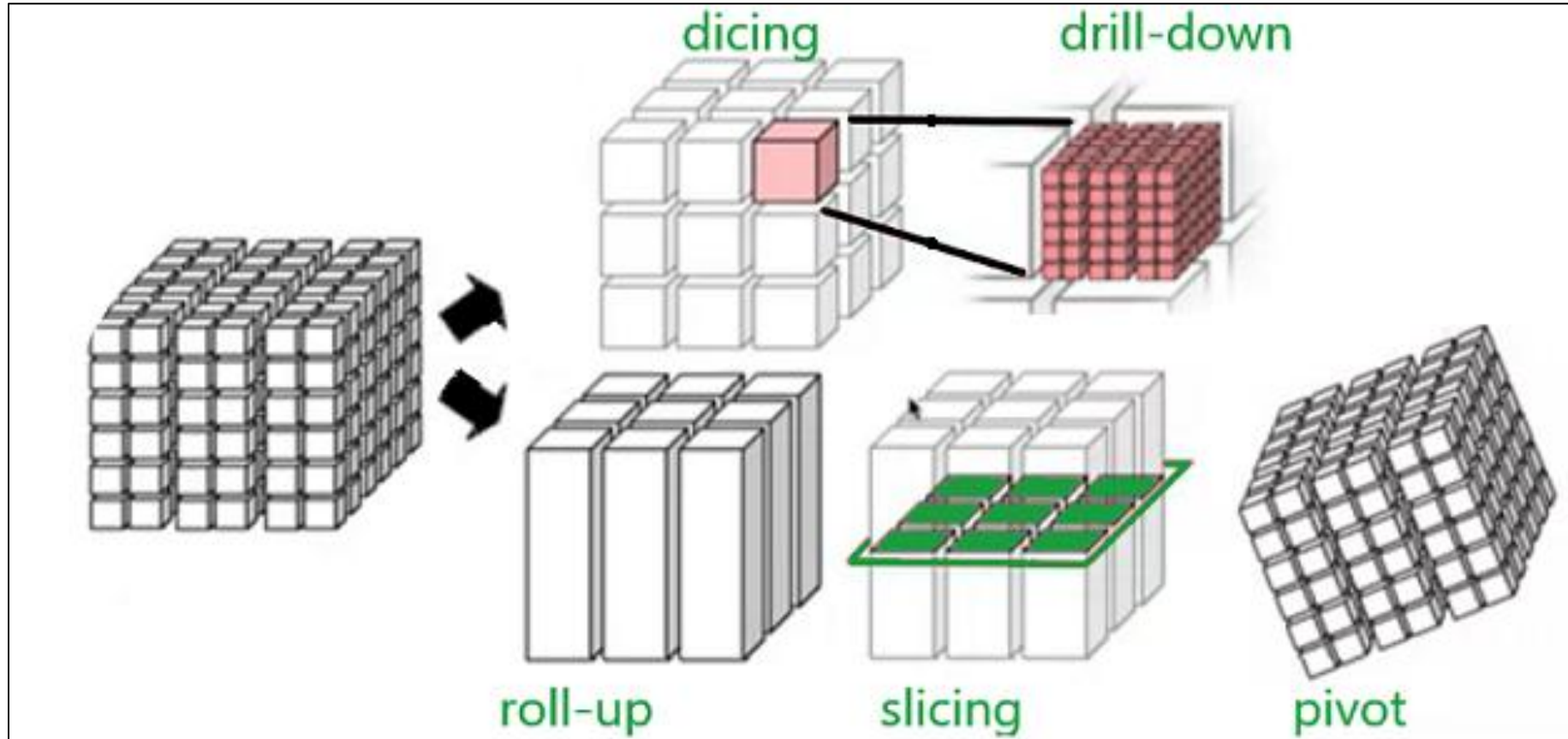


•**Slicing**: this operation filters the unnecessary portions. Suppose in a particular dimension, the user doesn't need everything for analysis, rather a particular attribute.

•**Dicing**: this operation does a multidimensional cutting, that not only cuts only one dimension but also can go to another dimension and cut a certain range of it. As a result, it looks more like a subcube out of the whole cube (as depicted in the figure).



- **Pivot:** this operation is very important from a viewing point of view. It basically transforms the data cube in terms of view. It doesn't change the data present in the data cube. For example, if the user is comparing year versus branch, using the pivot operation, the user can change the viewpoint and now compare branch versus item type.



- Which of the following is an example of a ratio attribute?

a. Temperature in Celsius **Interval**

b. Calendar date **Interval**

c. Height (cm) **Ratio**

d. Letter grade **Ordinal**

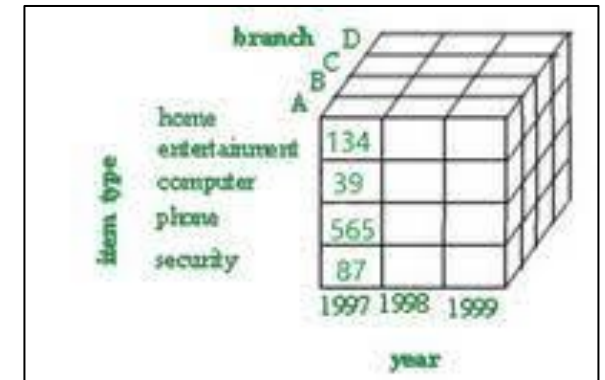
c) Height (cm)

- Which OLAP operation are you likely to perform at the **end** of the financial year?

Roll-Up

- The example here is a 3D cube having attributes like branch(A,B,C,D), item type (home, entertainment, computer, phone, security), year(1997,1998,1999). If user wants to observe only “branch A” data then which OLAP operation must be performed?

Slicing



- Business Analytics by U. Dinesh Kumar – Wiley 2nd Edition, 2022
Chapter : 1.1-1.7
- [Data Mining : Concepts and Techniques](#) by Han, Kamber and Pei ,
The Morgan Kaufmann Series in Data Management Systems ,3rd
Edition Chapter : 4.2.5
- <https://www.geeksforgeeks.org/data-cube-or-olap-approach-in-data-mining/>



THANK YOU

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DATA ANALYTICS

UE24CS342AA2

UNIT-1

Lecture 2: Data Preprocessing – Data Quality and Missing Data

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 2: Data Preprocessing – Data Quality and Missing Data

Slides excerpted from: Data Mining : Concepts and Techniques by Han, Kamber and Pei, 3rd Edition

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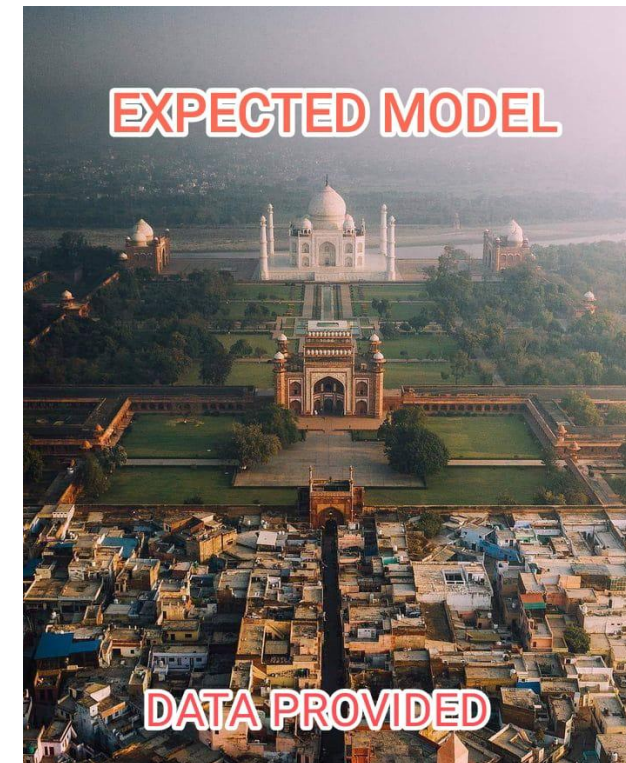
With grateful thanks for contribution of slides to:

Dr. Mamatha H R, Professor at the Department of CSE, PESU

DATA ANALYTICS

Data Preprocessing

- Analysis on data can only be as good as the data itself. Low quality data will lead to low quality analysis.
- Real world databases are highly susceptible to noisy, missing and inconsistent data owing to their huge size and multiple heterogeneous sources.
- Data processing techniques when applied before analysis can substantially improve the overall quality of analysis and/or the time required for the actual analysis.



1) **Accuracy** : Data must not contain errors or a lot of noise.

Example of inaccurate data : Date = 30/02/2002.

Reasons for inaccurate data :

- Data collection instruments may be faulty.
- Human errors occur during data entry.
- Disguised missing data : Users may purposefully submit incorrect data values for mandatory fields when they don't want to share their personal information. Example : Choosing the default value of January 1st for date of birth.

- 2) **Completeness** : Data must not lack attribute values. It must contain attributes of interest and relevance to the problem at hand.

Reasons for incompleteness :

- Attributes of interest were not considered important at the time of entry.
- Data might not be recorded due to equipment malfunction resulting in missing data.

- 3) **Consistency** : Should not contain any discrepancies in the data or the naming convention of the attributes.

Examples of inconsistency:

- Age is recorded as 50 but Date of Birth = 03/04/2005.
- In the result column of students' marks , few entries are in GPA format and rest in percentage.
- Discrepancies can exist between duplicate records.

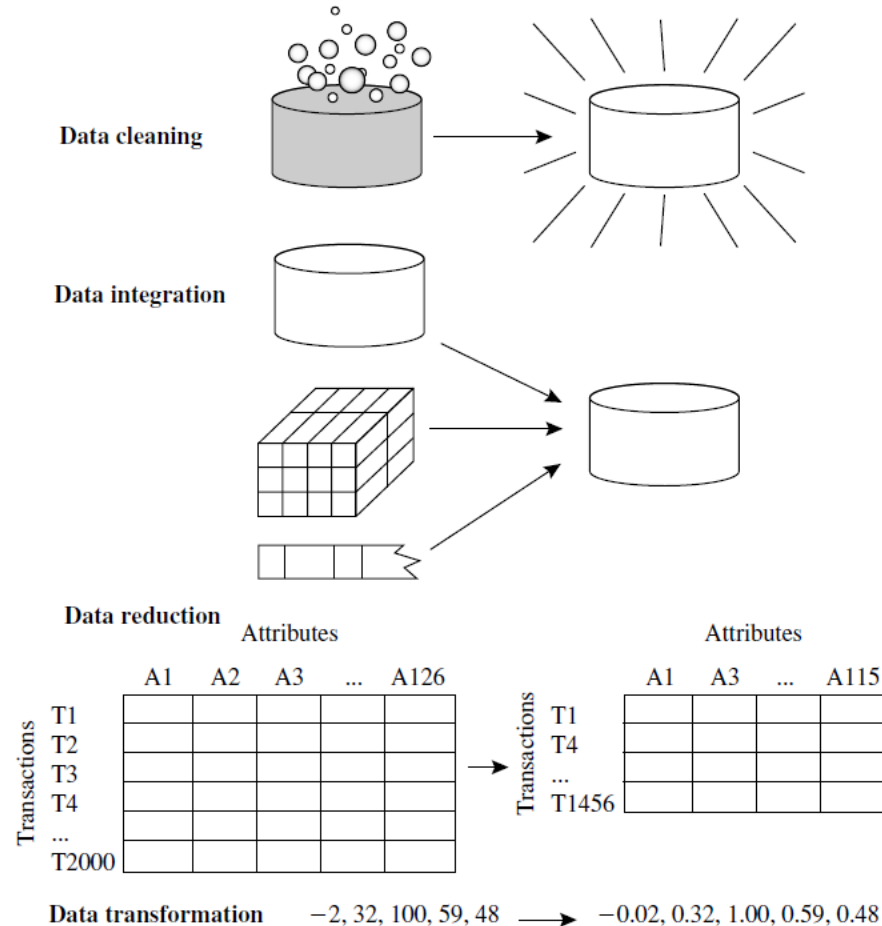
- 4) **Timeliness** : The data must be updated in a timely fashion. For example , for an analysis run on the first day of every month, previous month's data must be up to date for accurate analysis.
- 4) **Interpretability** : The data must be easily understood. If the attributes of the data aren't easily understandable , the analysis is going to be hindered.
- 4) **Believability** : The data and its source must be trusted by the users. If this data or the source caused problems in the past , current users will find it hard to trust it.

NOTE : The quality of data is subjective and depends on the intended use of data. The data needs of each problem is different.

DATA ANALYTICS

Major Tasks in Data Preprocessing

- Data Cleaning
- Data Integration
- Data Reduction
- Data Transformation



DATA ANALYTICS

Data Cleaning

Data cleaning entails :

- Filling in missing values
- Smoothing noisy data
- Identifying and removing outliers

If users believe the data is dirty, they are unlikely to trust the outcome of the analysis.



Data is not always complete. Missing data maybe due to :

- Equipment malfunction
- Inconsistent with other recorded data leading to its deletion
- Data not recorded due to a misunderstanding
- Certain data may not be considered useful at the time of entry

Missing data may need to be inferred.

1. **Ignore the tuple** : Usually done when the class label is missing (for a classification task). This is not effective when the percentage of missing values per attribute varies considerably
2. **Fill in the missing value manually** : Time consuming and infeasible for a large data set.
3. **Fill it with a global constant** : Replace it with a global constant like the word “Unknown”. The downside is that the model might learn patterns with respect to the occurrence of the word “Unknown”.
4. **Fill it with a central tendency** : For symmetric data distributions , replace it with the mean and for skewed data distributions, replace it with the median. A smarter way is to use attribute mean or median(based on the distribution)for all samples belonging to the same class.
1. **Use most probable model** : Use models like regression , decision tree or inference-based Bayesian formalism to infer the missing value.

1. Missing Completely At Random (MCAR)

- The missing data is independent of the observed and unobserved data. In other words, no systematic differences exist between records with missing data and those with complete data.
- For example : A weighing scale running out of batteries. This is not dependent on the person and the probability of this happening is equal to everyone.
- Assuming the data as MCAR is a strong and often unrealistic assumption as “true randomness” is rare in the real world.
- MCAR data doesn't add bias to the analysis.
- Ways to deal with it :
 - Delete the records : If it is a small fraction of data
 - Delete the attributes : If it is a small fraction of attributes
 - Mean imputation
 - Pairwise deletion : Compute the mean, variance and covariance with another variable available.

2. Missing At Random (MAR)

- MAR assumes that the missing value can be predicted based on the other observed data. The missingness is still random.
- Example : Employed people are less likely to answer all questions of a survey when compared to unemployed people. Data is MAR if the likelihood of completing the survey is dependent of the employment status but not on the topic of survey.
- Almost always produces a bias in the analysis.
- MCAR implies MAR but the converse isn't true.
- Ways to deal with it :
 - Regression imputation : unbiased if it considers the factor which influences the missingness.
 - Last observation carried forward (LOCF) and Baseline observation carried forward(BOCF) : Yields biased estimates. Must be used only if the underlying assumptions are scientifically justifiable.
 - Use of multiple imputation (Packages mice and amelia in R)

3. Missing Not At random (MNAR)

- The missingness of the data depends on the value of the data. The mechanism for why the data is missing is known. Yet , the values can't be effectively inferred.
- Examples :
 - Censored data
 - People belonging to certain income brackets might not wish to disclose their assets.
 - A weighing machine can only measure weights in a particular range.
- Ways to deal with this :
 - One must model the missingness explicitly, jointly modelling the response and missingness.
 - Generally , the data is assumed to be MAR whenever feasible to avoid this situation.

NOTE : There is no statistical way to determine under which category your missing data will fall under.

Missing Completely at Random, MCAR, means there is no relationship between the missingness of the data and any values, observed or missing. Those missing data points are a random subset of the data. There is nothing systematic going on that makes some data more likely to be missing than others.

Missing at Random, MAR, means there is a systematic relationship between the propensity of missing values and the observed data, but not the missing data.

Whether an observation is missing has nothing to do with the missing values, but it does have to do with the values of an individual's observed variables. So, for example, if men are more likely to tell you their weight than women, weight is MAR.

Missing Not at Random, MNAR, means there is a relationship between the propensity of a value to be missing and its values.

- Imagine you are collecting some information from your classmates. For many reasons , not everyone will answer every question of yours. And that is okay!
- Well the next step is replacing missing values right? We can use any one of the methods we have discussed till now after some analysis of the data.
- But wait! Don't you think the fact that they did not answer is some kind of information per se which can be beneficial to our analysis?
- So the next time you build a model , before dealing with the missing values , create an additional variable (preferably a binary variable) in which you store if the particular student answered or not.
- This may (or may not!) help you gain more insights about the population or improve the analytics model you are building!

Test your understanding!



- Which of the following is an example of inconsistent data?
 - a) Age = 25, Year of Birth = 1999
 - b) Age = 50, Date of Birth = 03/04/2005
 - c) Salary field is empty
 - d) Weight = 70 kg

b) Age and Date of Birth contradict each other.

- A numerical attribute has a highly skewed distribution with some missing values. Which is the most appropriate replacement?
 - a) Mean
 - b) Median
 - c) Global constant ("Unknown")
 - d) Delete the entire dataset

b) Median, since it is more robust to skewed distributions.

- The statement “Most of the missing people from work are sickest people” denotes what type of missingness?

MNAR

- Which type of missingness is called “non-ignorable”?

MNAR

Because the missing data mechanism itself has to be modelled as you deal with the missing data. You have to include some model for why the data are missing and what the likely values are.

- Mean imputation can be done for which type of missing data?

MCAR

- [Data Mining : Concepts and Techniques](#) by Han, Kamber and Pei , The Morgan Kaufmann Series in Data Management Systems ,3rd Edition
Chapter : 3.1-3.2
- <http://dept.stat.lsa.umich.edu/~jerrick/courses/stat701/notes/mi.html>
- <https://www.scribbr.com/statistics/missing-data/>
- <https://www.theanalysisfactor.com/missing-data-mechanism/>



THANK YOU

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DATA ANALYTICS

UE24CS342AA2

UNIT-1

**Lecture 3: Data Preprocessing – Cleaning
as a Process, Integration and Reduction**

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 3: Data Preprocessing – Cleaning as a Process, Integration and Reduction

Slides excerpted from: Data Mining : Concepts and Techniques by Han, Kamber and Pei, 3rd Edition

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With grateful thanks for contribution of slides to:

Dr. Mamatha H R, Professor at the Department of CSE, PESU

Noise is a random error or variance in a measured variable.

Data smoothing techniques to combat noise :

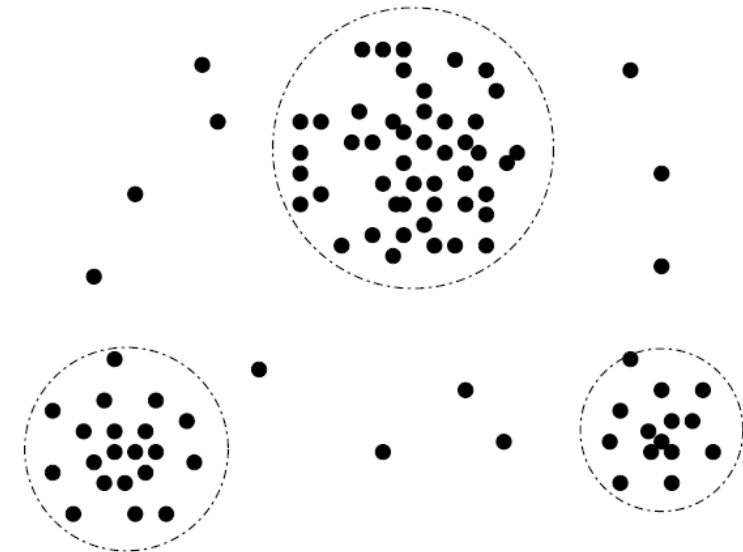
- **Binning**
 - Sort the data and partition into bins(equal-width, equal-frequency, etc.)
 - Smooth by bin means, bin medians, by bin boundaries etc.
 - More on binning in further lectures.
- **Regression** - Data can be smoothened by fitting it to a regression model.
- **Clustering** - Outliers can be detected with the help of clustering and can be removed to smoothen the data.
- **Combined computer and human inspection** – Computer detects suspicious values and is validated by a human. Is useful when dealing with possible outliers.

Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set.

Case 1 : Outliers are noise that interferes with data analysis.

Case 2 : Outliers are the main goal of our analysis. Examples :

- Credit card fraud
- Spam detection
- Intrusion detection



- **Data discrepancy detection**
 - Refer to the metadata of data to gain knowledge regarding its properties.
 - Perform summary statistics for all attributes and discover the distributions, dependencies , outliers and so on.
 - Look for inconsistent representation of data. For example , make sure all dates are following the same format , for instance , DD-MM-YYYY.
 - Check for field overloading – practice of coupling two or more data elements to a single field. It ensures efficient memory utilization.
 - Check for uniqueness rule, consecutive rule and null rule.
 - Uniqueness rule : Each value of the given attribute must be unique.
 - Consecutive rule : There can't be any missing values between lowest and highest value for that attribute. All values must be unique. Example - cheque number.
 - Null rule : Specifies how to record a null value. For example , use 0 for numeric attribute and '?' for nominal attribute.

- Data discrepancy detection
 - Use commercial tools that can aid in this step.
 - Data scrubbing tools : Use simple domain knowledge (example, knowledge of postal zip code and spell-check) to detect errors and make corrections.
 - Data auditing tools : Find discrepancies by analyzing the data to discover rules and relationships , and detect data that violates the discovered rules. For example , it employs statistical analysis to find correlations or clustering to detect outliers.

- **Data transformation**
 - Some data inconsistencies can be corrected manually but most errors require data transformations.
 - Data migration tools allow transformations to be specified.
 - ETL (Extraction/Transformation/Loading) tools allow users to specify transformations through a graphical user interface.
- Data transformations may introduce more discrepancies.
- The 2-step process of discrepancy detection and data transformation occurs iteratively until no further anomalies are found.
- New approaches to data cleaning emphasize increased interactivity. Potter's wheel is a publicly available data cleaning tool that integrates both the steps.

- Data analysis often requires data integration – the merging of data from multiple data stores into a coherent store.
- Careful integration can help reduce and avoid redundancies and inconsistencies in the resulting dataset. This can help improve the accuracy and speed of the subsequent data analysis process.
- The semantic heterogeneity and structure of data pose great challenges in data integration.
- How can we match schema and objects from different sources?
- **Schema Integration!**
 - Example : How can a data analyst be sure that the attribute `customer_id` in table A and `customer_number` in table B refer to the same attribute?
 - With the help of metadata! It provides all possible information regarding the attributes , thus ensuring error free schema integration.

- **Entity identification problem** : Identify real world entities from multiple data sources. Example : Bill Clinton = William Clinton
- **Detecting and resolving data value conflicts**
 - For the same real world entity, attribute values from different sources are different.
 - Possible reasons : different representations, different scales, example – metric vs British units
- During integration , special attention must be paid to the structure of the data. This is to ensure that any attribute functional dependencies and referential constraints in the source system match those in the target system. For example , in one system, a discount may be applied to the entire order whereas in another system , it is applied to each individual line item. If this is not caught before integration, items in the target system may be improperly discounted.

Redundant data often occur during the integration of multiple databases.

- Object identification : The same attribute or object may have different names in different databases which causes redundancy.
- Derivable data : An attribute may be redundant if it can be *derived* from another attribute or set of attributes. For example , annual revenue can be derived from monthly revenue.
- Few redundancies can be detected by correlation analysis.
- For nominal data , χ^2 (chi-square) test is employed.
- For numeric data , correlation coefficient and covariance is used.

- In addition to detecting redundancies between attributes, duplication should be detected at the tuple level (Example , where there are two or more identical tuples for a unique data entry case)
- The use of denormalized tables (often done to improve performance by avoiding joins) is another source of data redundancy.
- Inconsistencies often arise between various duplicates, due to inaccurate data entry or updating some but not all data occurrences.

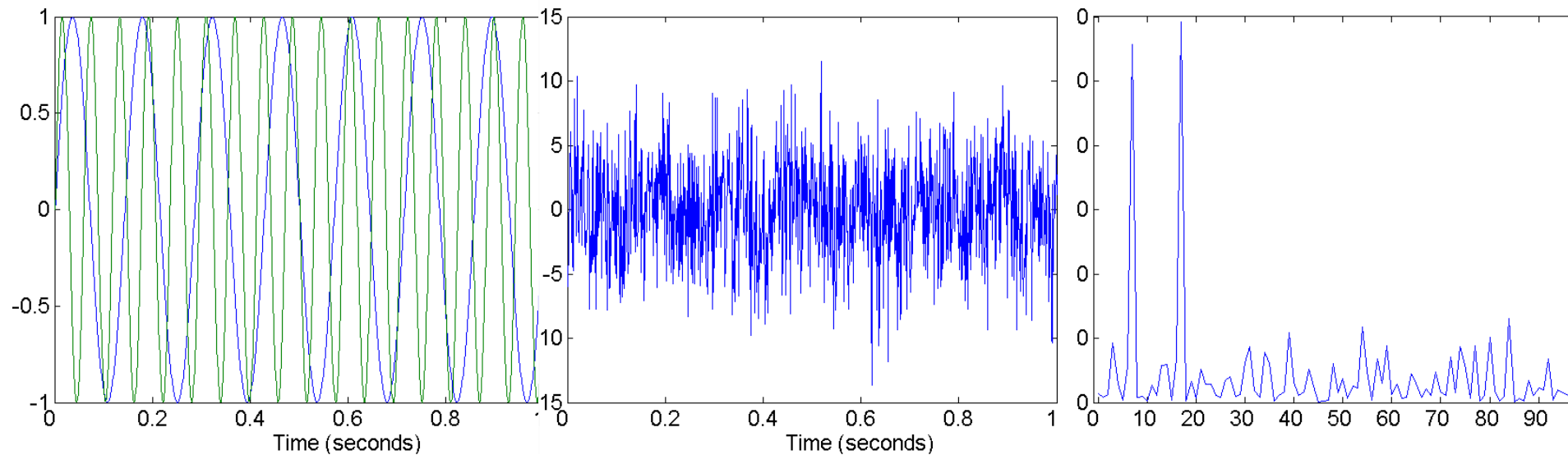
- Data integration also involves the detection and resolution of data value conflicts.
- For example, for the same real-world entity, attribute values from different sources may differ.
- This may be due to differences in representation, scaling or encoding.
- For instance , a weight attribute may be stored in metric units in once system and British imperial units in another.

- Data reduction techniques are applied to obtain a reduced representation of the dataset that is much smaller in volume , yet closely maintains the integrity of the original data.
- Analysis on the reduced dataset should be more efficient yet produce the same or almost the same analytical results.
- Why do we need data reduction? A database or a data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set.

- **Dimensionality reduction** – process of removing unimportant attributes
 - Wavelet transforms
 - Principal Component Analysis (PCA)
 - Attribute subset selection
- **Numerosity reduction** – replaces the original data volume by an alternative, smaller forms of data representation
 - Regression and log-linear models
 - Histograms, clustering and sampling
 - Data cube aggregation
- **Data compression** – transformations are applied on the data to obtain a *reduced* or a *compressed* representation of the original data.

- **Curse of dimensionality**
 - When dimensionality increases, data becomes increasingly sparse.
 - Density and distance between points , which are critical to clustering and outlier analysis become less meaningful.
 - The possible combinations of subspaces will grow exponentially.
- **Dimensionality reduction**
 - Avoids the curse of dimensionality.
 - Helps to eliminate irrelevant attributes and reduce noise.
 - Reduces time and space required for data analytics.
 - Enables easier visualization.

- **Fourier transform**
- **Wavelet transform**



Two Sine Waves

Two Sine Waves + Noise

Frequency

Test your understanding!



- Which of these is **not** a method to deal with noisy data?
 - a) Binning
 - b) Regression
 - c) Principal Component Analysis
 - d) Clustering

c) Principal Component Analysis

- Outliers need to be removed in every dataset , regardless of the problem statement.

False

- What does ETL stand for?

Extraction, Transformation, Loading

Test your understanding!



- Which of the following is an example of a dimensionality reduction technique?
 - a) Regression
 - b) Wavelet Transform
 - c) Data Cube Aggregation
 - d) Histograms

b) Wavelet Transform

- Which statistical test is used to detect redundancy in nominal data?

Chi-square (χ^2)

- What is the first step in data integration?

Understanding the metadata

- [Data Mining : Concepts and Techniques](#) by Han, Kamber and Pei , The Morgan Kaufmann Series in Data Management Systems ,3rd Edition
Chapter : 3.1-3.4
- <http://dept.stat.lsa.umich.edu/~jerrick/courses/stat701/notes/mi.html>
- <https://www.scribbr.com/statistics/missing-data/>
- <https://www.theanalysisfactor.com/missing-data-mechanism/>
- <https://www.st-andrews.ac.uk/~wjh/dataview/tutorials/dwt.html>
- <https://towardsdatascience.com/the-wavelet-transform-e9cfa85d7b34>
- https://www.cs.unm.edu/~mueen/Teaching/CS_521/Lectures/Lecture2.pdf
- <https://medium.com/analytics-vidhya/understanding-principle-component-analysis-pca-step-by-step-e7a4bb4031d9>



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DATA ANALYTICS

UE24CS342AA2

UNIT-1

Lecture 4: Data Reduction – PCA, Numerosity Reduction and Compression

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Data Analytics

Unit 1

Lecture 4: Data Reduction – PCA, Numerosity Reduction and Compression

Slides excerpted from: Data Mining : Concepts and Techniques by Han, Kamber and Pei, 3rd Edition

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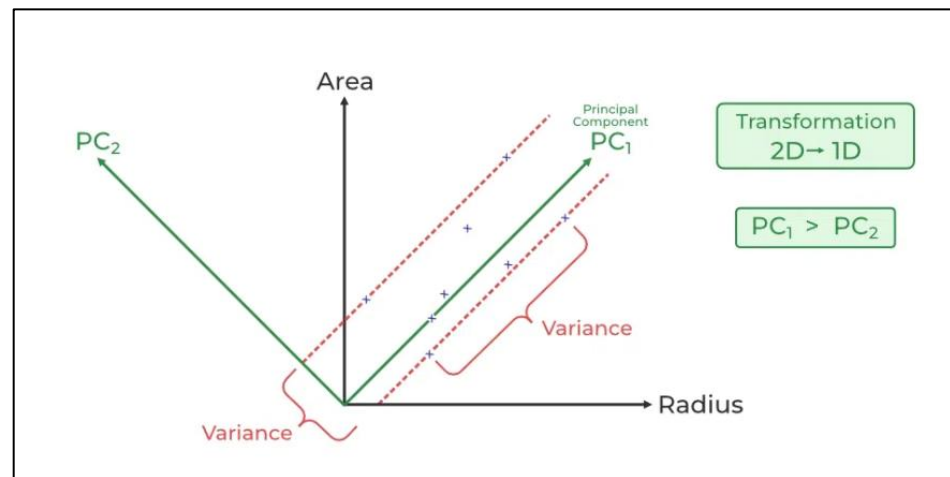
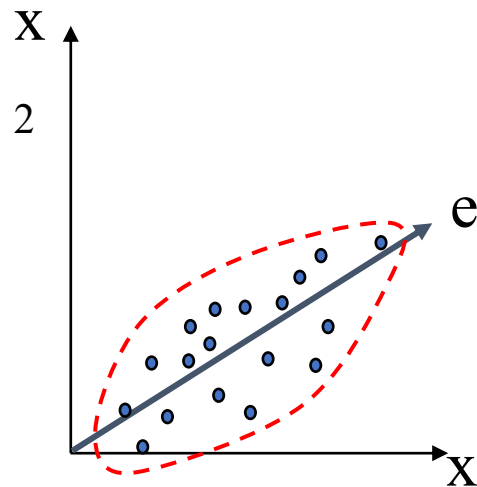
What is PCA?

Assume there are 50 questions in all in the survey. The following three are among them:

1. I feel comfortable around people
2. I easily make friends
3. I like going out

These queries could appear different now. There is a catch, though. They aren't, generally speaking. They all gauge how extroverted you are. Therefore, combining them makes it logical, right? That's where linear algebra and dimensionality reduction methods come in! We want to lessen the complexity of the problem by minimizing the number of variables since we have much too many variables that aren't all that different. That is the main idea behind dimensionality reduction. And it just so happens that PCA is one of the most straightforward and popular techniques in this field.

- Finds a projection that captures the largest amount of variation in data.
- The original data is projected onto much smaller space , resulting in dimensionality reduction.
- We find eigenvectors of the covariance matrix and these eigenvectors define the new space.



Step 1: Standardize the dataset.

Step 2: Calculate the covariance matrix for the features in the dataset.

Step 3: Calculate the eigenvalues and eigenvectors for the covariance matrix.

Step 4: Sort eigenvalues and their corresponding eigenvectors.

Step 5: Pick k eigenvalues and form a matrix of eigenvectors.

Step 6: Transform the original matrix

Lets go step by step

DATA ANALYTICS

PCA using R (factoMineR , factoextra)



name	100m	Long.jump	//	Javeline	1500m	Rank	Points	Competition
SEBRLE	11.04	7.58		63.19	291.7	1	8217	Decastar
CLAY	10.76	7.4		60.15	301.5	2	8122	Decastar
Macey	10.89	7.47		58.46	265.42	4	8414	OlympicG
Warners	10.62	7.74		55.39	278.05	5	8343	OlympicG
\\								
Zsivoczky	10.91	7.14		63.45	269.54	6	8287	OlympicG
Hernu	10.97	7.19		57.76	264.35	7	8237	OlympicG
Pogorelov	10.95	7.31		53.45	287.63	11	8084	OlympicG
Schoenbeck	10.9	7.3		60.89	278.82	12	8077	OlympicG
Barras	11.14	6.99		64.55	267.09	13	8067	OlympicG
KARPOV	11.02	7.3		50.31	300.2	3	8099	Decastar
WARNERS	11.11	7.6		51.77	278.1	6	8030	Decastar
Nool	10.8	7.53		61.33	276.33	8	8235	OlympicG
Drews	10.87	7.38		51.53	274.21	19	7926	OlympicG

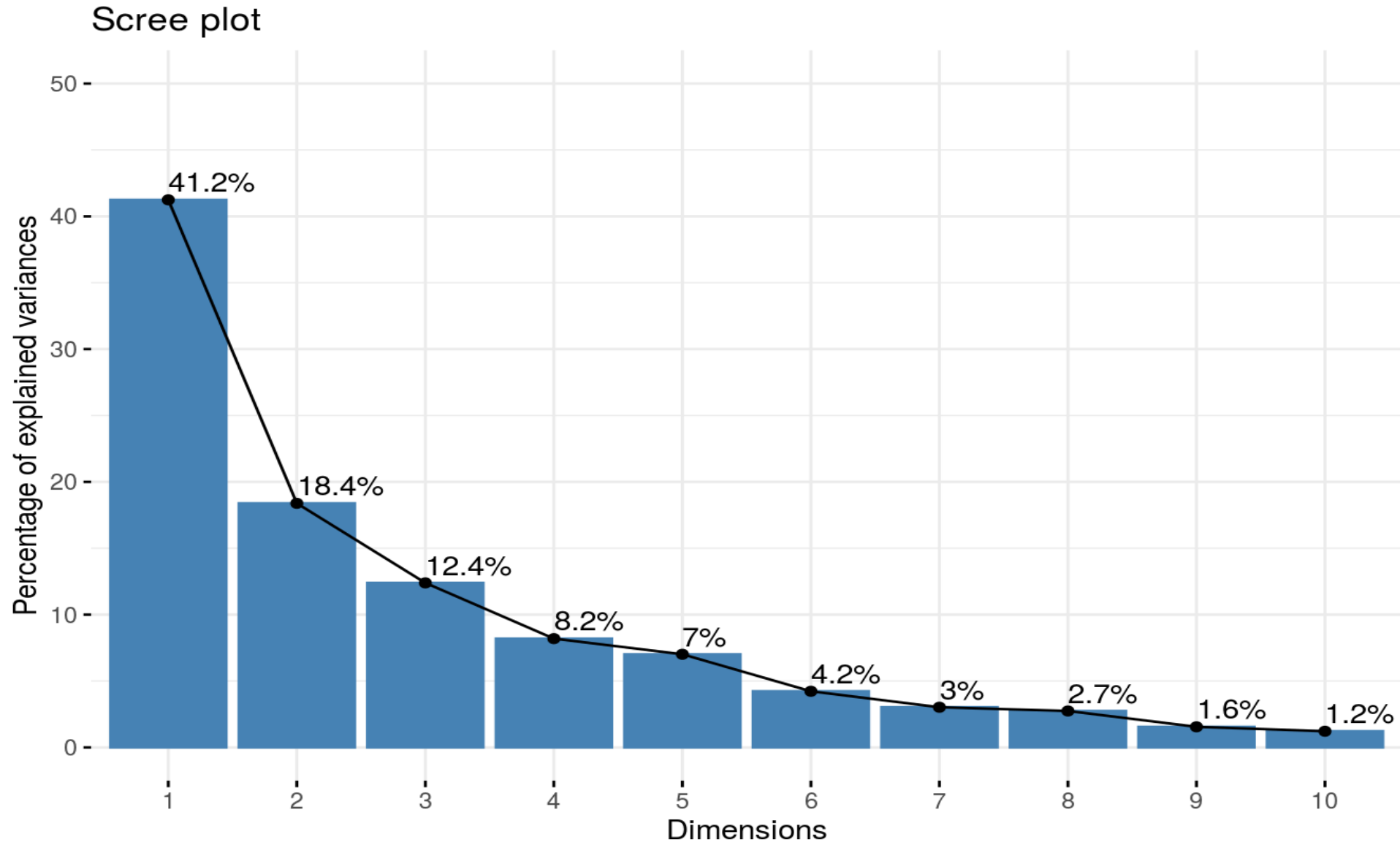
	Active individuals
	Active variables
	Supplementary quantitative variables
	Supplementary qualitative variable
	Supplementary individuals

<http://www.sthda.com/english/articles/31-principal-component-methods-in-r-practical-guide/112-pca-principal-component-analysis-essentials/>

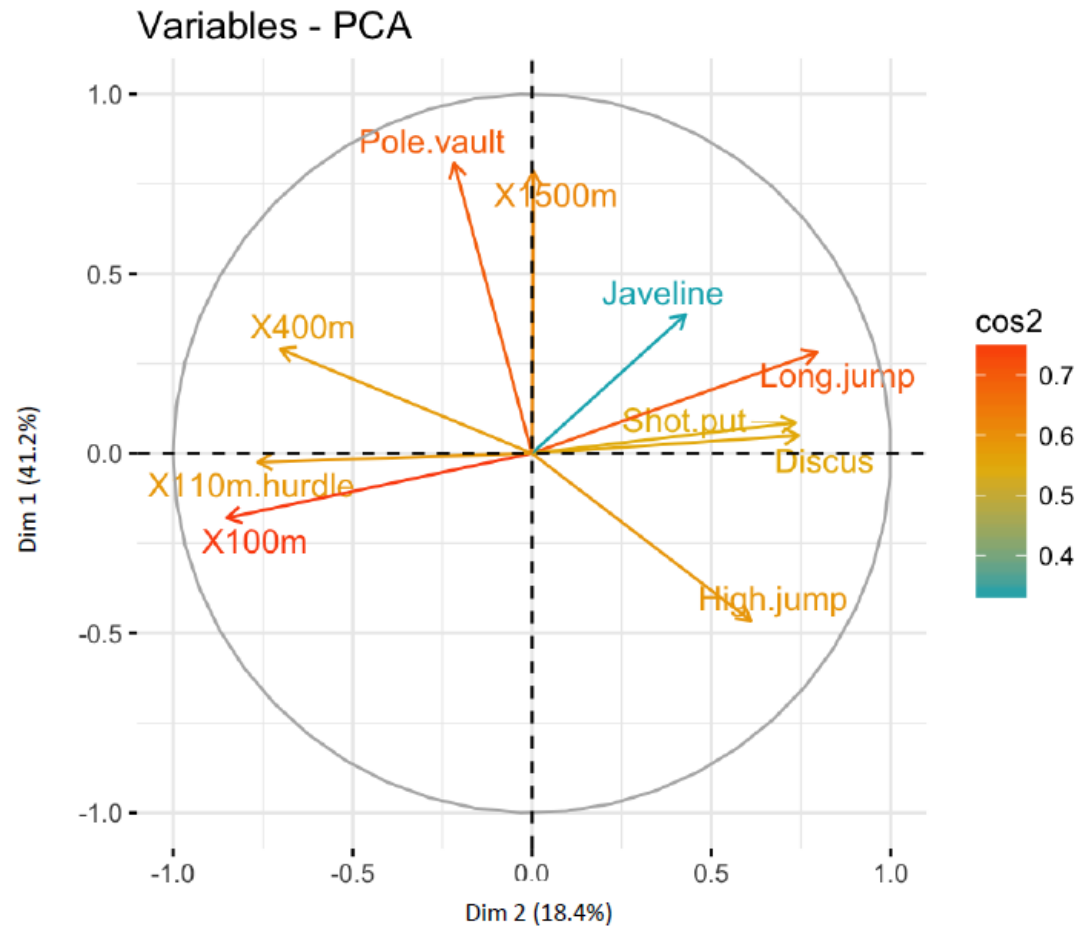
- Selecting the principal components

```
library("factoextra") eig.val <- get_eigenvalue(res.pca)
eig.val
```

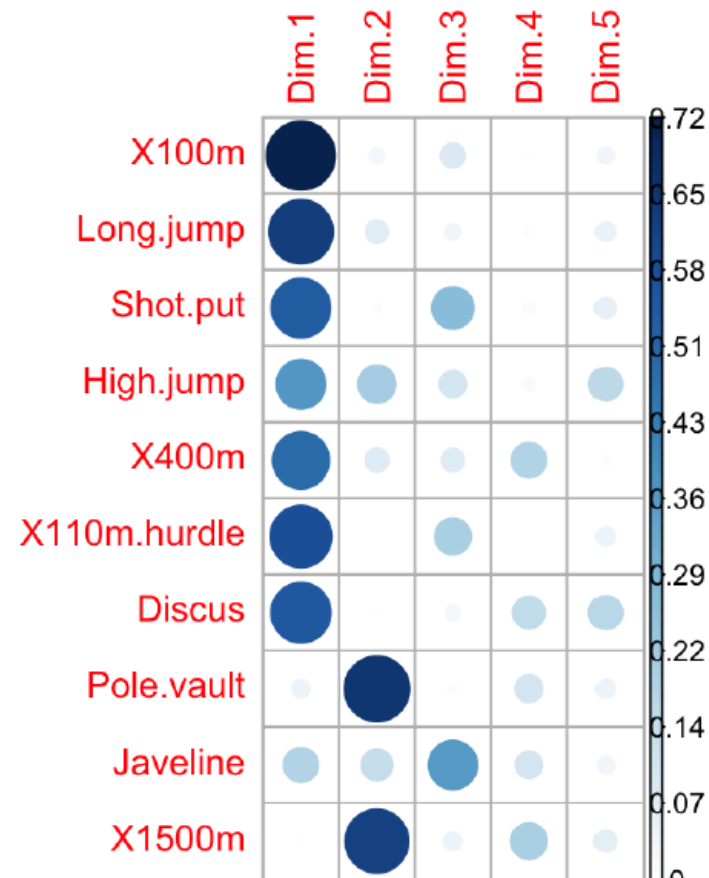
##	eigenvalue	variance.percent	cumulative.variance.percent
## Dim.1	4.124	41.24	41.2
## Dim.2	1.839	18.39	59.6
## Dim.3	1.239	12.39	72.0
## Dim.4	0.819	8.19	80.2
## Dim.5	0.702	7.02	87.2
## Dim.6	0.423	4.23	91.5
## Dim.7	0.303	3.03	94.5
## Dim.8	0.274	2.74	97.2
## Dim.9	0.155	1.55	98.8
## Dim.10	0.122	1.22	100.0



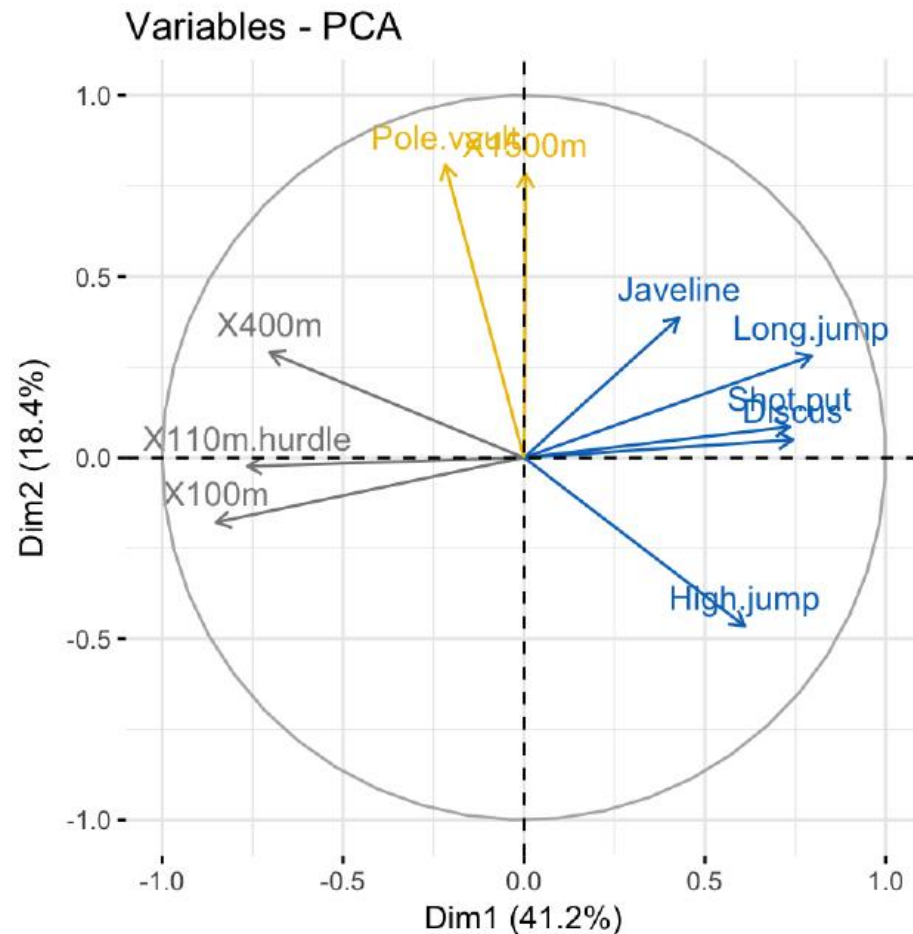
- Correlation circle



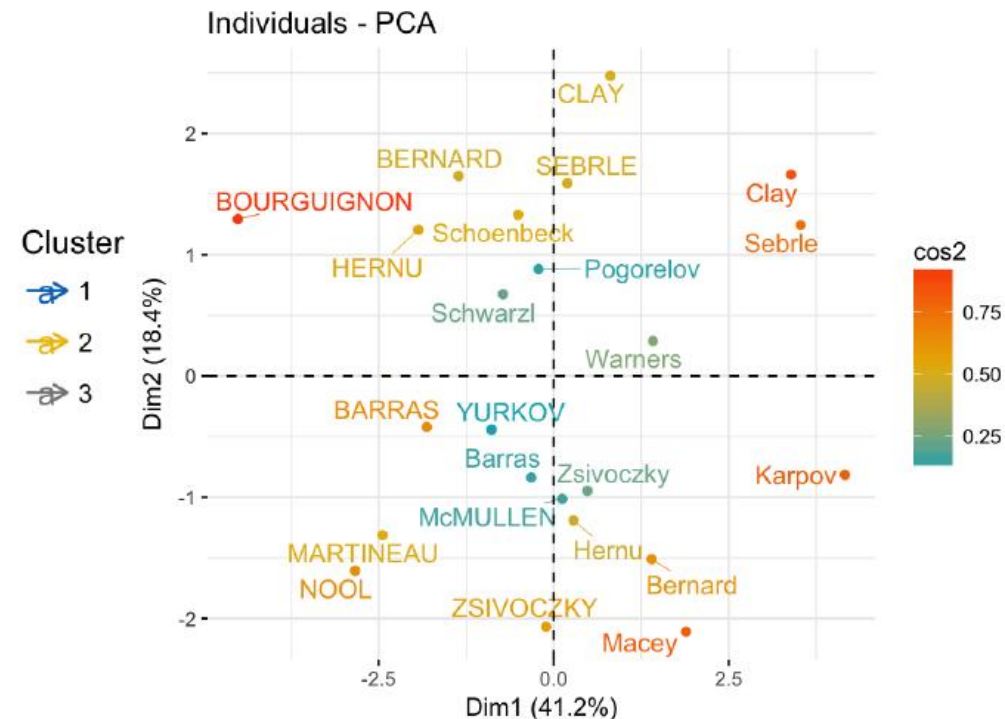
var.cos2 =
var.coord * var.coord



- Which events are similar?

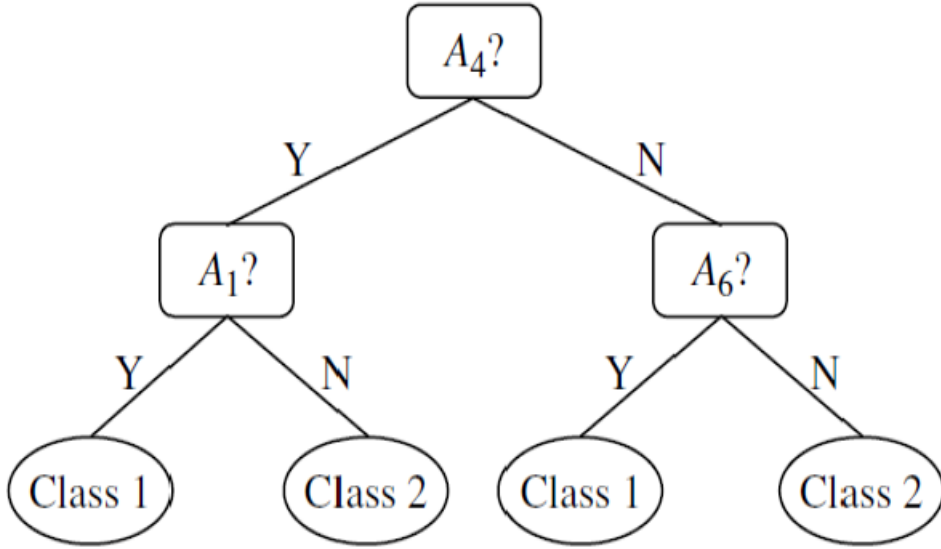


- Which athletes are similar?



- Attribute Subset Selection reduces the dataset size by removing irrelevant or redundant attributes.
- Redundant attributes : Information is contained or can be extracted from other attributes. Example : MRP of a product and the corresponding sales tax paid.
- Irrelevant attributes : They contain no information which is useful for the data analysis task at hand. Example : SRN is irrelevant to predict students' GPA.
- The goal of attribute subset selection is to find the minimum set of attributes such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes.
- Analysis on a reduced set of attributes implies the reduction of the number of attributes appearing in the discovered patterns , thus helping to make the patterns easier to understand.

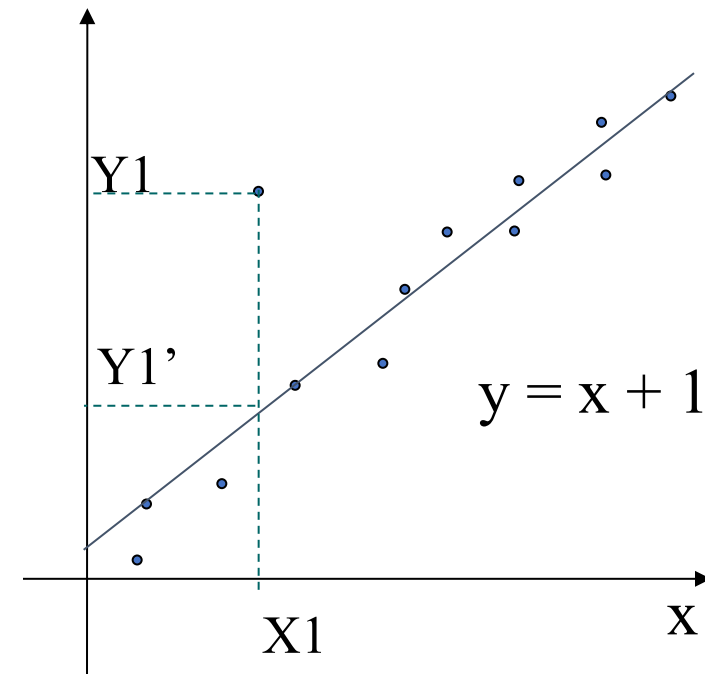
- For n attributes , there are 2^n possible subsets. An exhaustive search is infeasible.
- The *best* and the *worst* attributes are determined using tests of statistical significance , which assume that the attributes are independent of each other.
- **Stepwise forward selection** : The procedure starts with an empty set as the reduced set. In each iteration , the best of the original attributes is selected and added to the reduced set.
- **Stepwise backward elimination** : The procedure starts with the full set of attributes. In each iteration , the worst attribute remaining in the set is removed.
- **Combination of forward and backward selection** : In each iteration, the best attribute is selected and the worst attribute is removed.
- **Decision tree induction** : A decision tree is constructed using the given data. All attributes which do not appear in the tree are judged to be irrelevant. Thus the set of attributes appearing in the tree form the reduced subset.

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ $\Rightarrow \{A_1\}$ $\Rightarrow \{A_1, A_4\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  <pre> graph TD A4["A4?"] -- Y --> A1["A1?"] A4 -- N --> A6["A6?"] A1 -- Y --> C1_1((Class 1)) A1 -- N --> C2_1((Class 2)) A6 -- Y --> C1_2((Class 1)) A6 -- N --> C2_2((Class 2)) </pre> $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$

- Creating new features that can capture important information in the dataset more effectively than the original ones.
- For example , an attribute *area* can be added based on the attribute's *height* and *width*. By combining attributes, accuracy can be improved and missing information about the relationships between the attributes can be discovered.
- Three general methodologies
 - Attribute extraction – Domain specific.
 - Mapping data to a new space – Fourier transforms, wavelet transforms and manifold approaches
 - Attribute construction
 - Combining features
 - Data Discretization

- Reduce data volumes by choosing alternative, smaller forms of data representation.
- **Parametric methods**
 - Assume the data fits some model. Estimate the model parameters and store only the parameters , discarding the data (except the possible outliers).
 - Examples : linear regression , multiple regression , log-linear model
- **Non-Parametric methods**
 - Do not assume models.
 - Examples : histograms, clustering ,sampling

- A collective name for techniques for the modeling and analysis of numerical data consisting values of a *dependent variable* and of one or more *independent variables*.
- The parameters are estimated so as to give a *best fit* of the data. The best fit is often evaluated using the least squares method.
- Regression analysis is used for prediction (including forecasting of time series data) , inference, hypothesis testing and modeling of causal relationships.

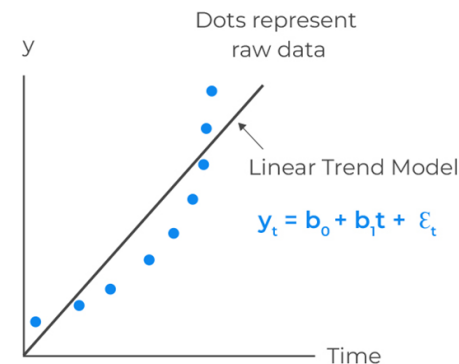


- A log-linear model is a mathematical model that takes the form of a function whose logarithm equals a linear combination of the parameters of the model.
- It approximates discrete multidimensional probability distributions for a set of discretized attributes based on smaller subset of dimensional combinations.
- It is useful for dimensionality reduction and data smoothing.

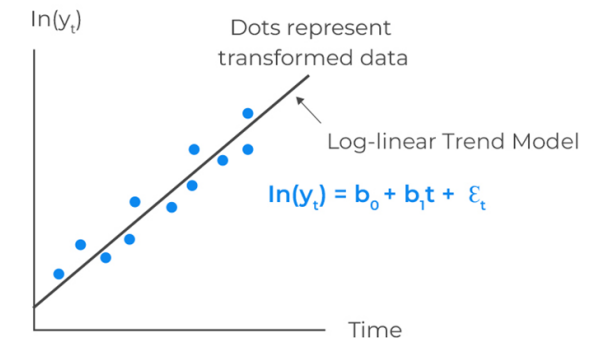


Linear vs. Log-linear Trend Models

Linear Trend Model

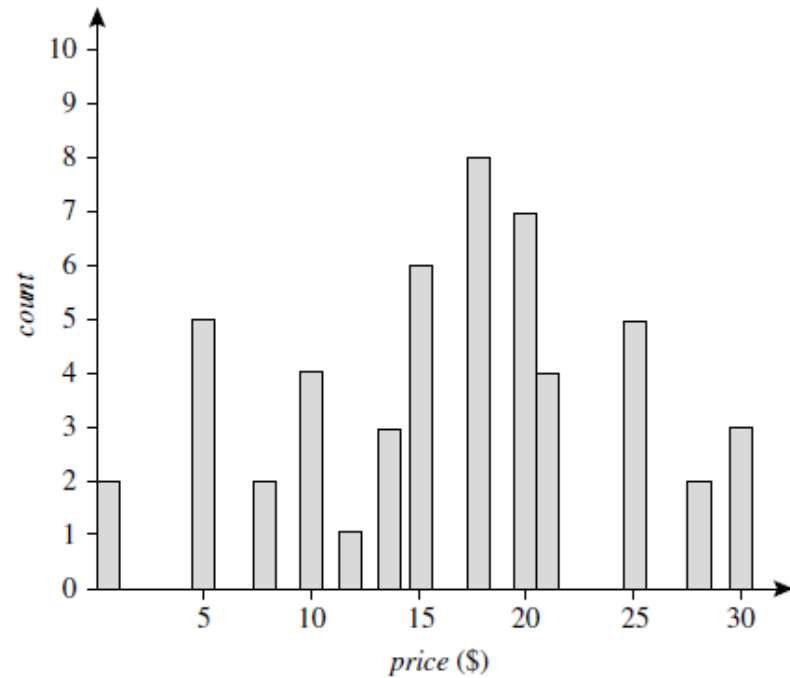


Log-linear Trend Model

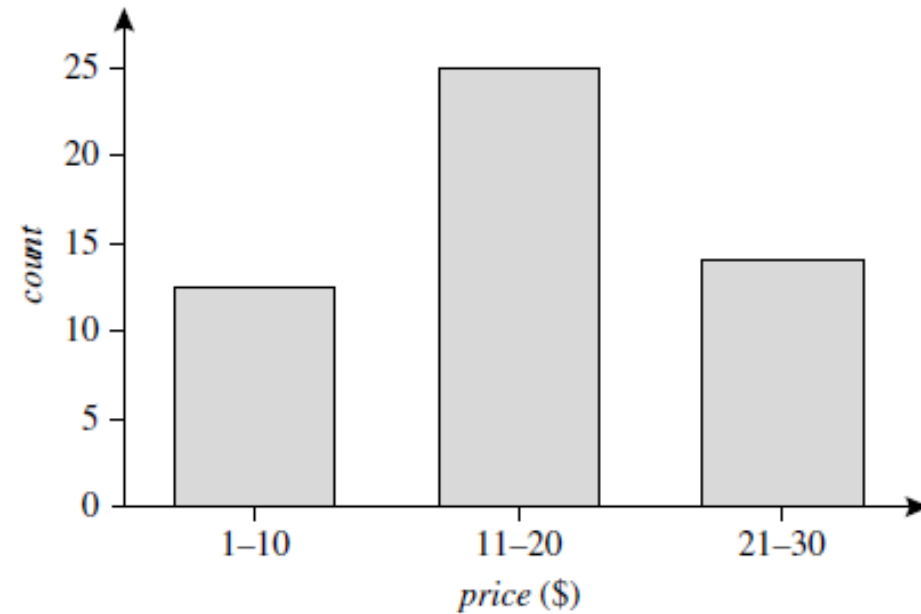


- A histogram for an attribute partitions data distribution into disjoint subsets referred to as bins or buckets.
- If each bucket represents only a single attribute value , it is called singleton bucket.
- Partitioning rules :
 - Equal-width : equal bucket range
 - Equal-frequency : equal depth
- The following data are a list of AllElectronics prices for commonly sold items in \$. The numbers have been sorted: 1, 1, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 28, 28, 30, 30, 30.

Draw a histogram using singleton buckets and equal-width bin of 10\$.

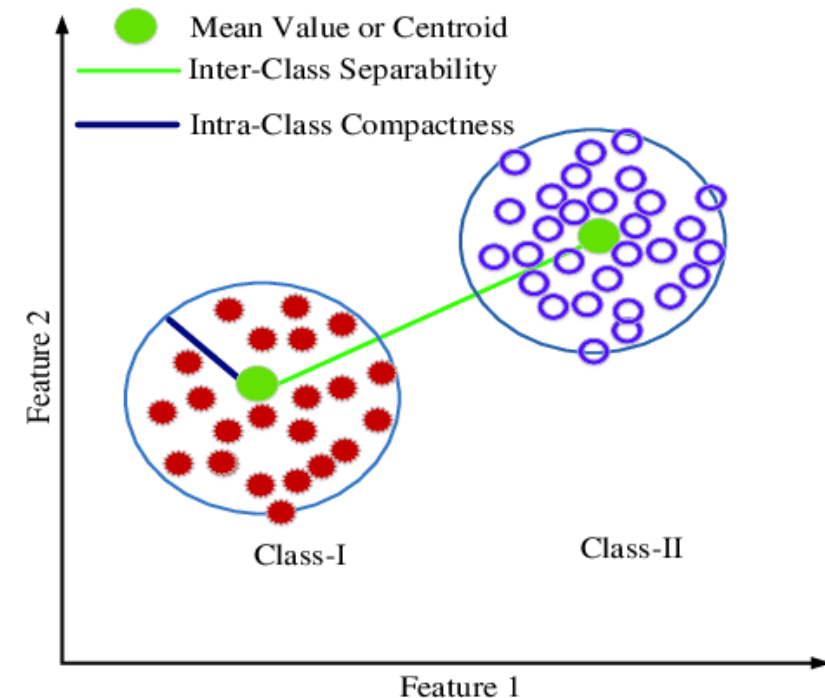


Using singleton buckets

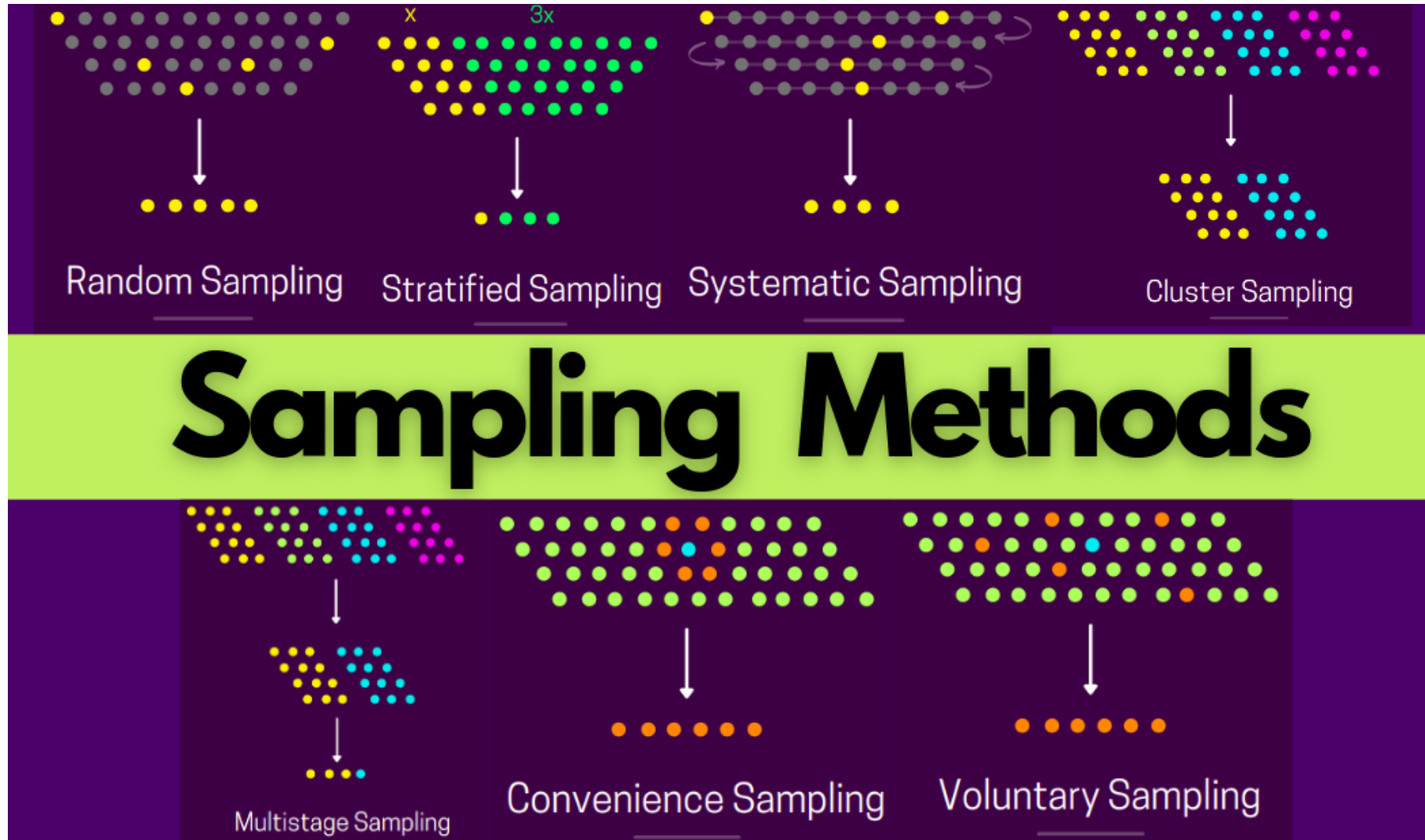


Using a width of 10\$

- Records are partitioned into clusters where records within the cluster are similar to one another and dissimilar to the ones in another cluster.
- Cluster representation of data is used to replace the actual data.
- The cluster representation can be the centroid and the diameter (intra-cluster distance) of the cluster.
- It is very effective if the data is *clustered* but not if it is *smeared*.
- Can use hierarchical clustering and be stored in multi-dimensional index trees.



- Sampling can be used as a data reduction technique as it allows a large dataset to be represented by a much smaller random subset.
- An advantage of sampling for data reduction is that the cost of obtaining a sample is proportional to the size of the sample.
- In the context of data reduction , sampling is most commonly used to estimate an answer to an aggregate query.
- Using central limit theorem (recall from Statistics for Data Science!!) , it is possible to determine a sufficient sample size for estimating a given function within a specified degree of error.
- Important to remember :
 - Choose a representative subset of the data.
 - Simple random sampling might have a poor performance in presence of a skew.
 - Develop adaptive sampling methods like stratified sampling



DATA ANALYTICS

Data Cube Aggregation



IMAGINE YOU HAVE TO PERFORM AN ANALYSIS ON YEARLY SALES AT DUNDER MIFFLIN PAPER COMPANY.



THE DATA YOU RECEIVE HAS SALES PER QUARTER FROM THE YEAR 2008 TO 2010.



SINCE YOU CARE ABOUT ANNUAL METRICS, THE DATA CAN BE *AGGREGATED* SO THAT THE RESULTING DATA SUMMARIZES THE ANNUAL SALES RATHER THAN QUARTERLY SALES.



THE RESULTING DATASET IS SMALLER IN VOLUME WITHOUT A LOSS OF INFORMATION NECESSARY TO THE TASK AT HAND!



USUALLY DATA CUBES ARE USED TO STORE MULTIDIMENSIONAL AGGREGATED INFORMATION.

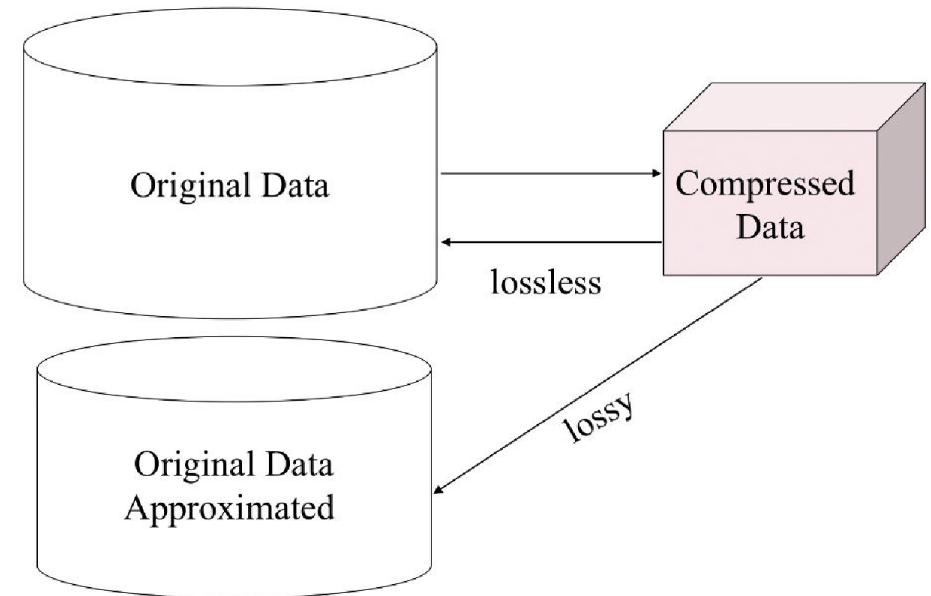
Year 2010	
Quarter	Sales
Q1	\$224,000
Q2	\$408,000
Q3	\$350,000
Q4	\$586,000

Year 2009	
Quarter	Sales
Q1	\$224,000
Q2	\$408,000
Q3	\$350,000
Q4	\$586,000

Year 2008	
Quarter	Sales
Q1	\$224,000
Q2	\$408,000
Q3	\$350,000
Q4	\$586,000

Year	Sales
2008	\$1,568,000
2009	\$2,356,000
2010	\$3,594,000

- Transformations are applied so as to obtain a reduced or *compressed* representation of the original data.
- If the original data can be reconstructed from the compressed data without any information loss, the data reduction is called lossless. But if we can reconstruct only an approximation of the original data, then the data reduction is called lossy.
- There are several lossless algorithms for string compression, however they allow only limited data manipulation.
- Dimensionality reduction and numerosity reduction can also be considered as data compression.



- Is PCA lossy or lossless?

Lossy

- Amongst **PCA** and **Attribute subset selection**, which data reduction method has more interpretability?

Attribute Subset Selection

- Which matrix is computed before finding eigenvalues and eigenvectors in PCA?

Covariance matrix

- For an Electrocardiography (ECG) wave which is a better transform: Fourier or wavelet?

Wavelet

Because ECG's have signals have short intervals of characteristic oscillation and Fourier transforms can only capture frequencies that persist over an entire signal which is not suitable here.

- _____ is a nonzero vector that stays parallel after matrix multiplication

Eigen Vectors

- [Data Mining: Concepts and Techniques](#) by Jiawei Han, Micheline Kamber and Jian Pei, The Morgan Kaufmann Series in Data Management Systems, 3rd Edition Chapter : 3.3 – 3.4
- <https://www.st-andrews.ac.uk/~wjh/dataview/tutorials/dwt.html>
- <https://towardsdatascience.com/the-wavelet-transform-e9cfa85d7b34>
- https://www.cs.unm.edu/~mueen/Teaching/CS_521/Lectures/Lecture2.pdf
- <https://medium.com/analytics-vidhya/understanding-principle-component-analysis-pca-step-by-step-e7a4bb4031d9>



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DATA ANALYTICS

UE23CS342AA2

UNIT-1

Lecture 9: Data Preprocessing – Transformations and Discretization - 2

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 9: Data Preprocessing – Transformations and Discretization - 2

Slides excerpted from: Data Mining : Concepts and Techniques by Han, Kamber and Pei, 3rd Edition

Gowri Srinivasa

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With grateful thanks for contribution of slides to:

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A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values.

Methods

- Smoothing
- Attribute construction
- Aggregation
- Normalization
- Discretization
- Concept hierarchy generation

- Divides the range of a continuous attribute into intervals.
- Interval labels are then used to replace the actual data values.
- Data size can be reduced by discretization.
- If the discretized process uses class information , it is called *supervised discretization* else it is called *unsupervised discretization*.
- **Top-down discretization** : Process starts by finding one or few points (called split or cut points) to split the entire attribute range and then repeat this process recursively on the resulting intervals.
- **Bottom-up discretization** : Also called as *merging* , starts by considering all the continuous values as potential splits. It removes few split points by merging neighborhood values to form intervals. This process is recursively applied to the resulting intervals.

- Binning :
 - Top-down split , unsupervised
- Histogram analysis :
 - Top-down split , unsupervised
- Clustering analysis :
 - Unsupervised , top-down split or bottom-up merge
- Decision-tree analysis :
 - Supervised , top-down split
- Correlation analysis :
 - Unsupervised , bottom-up merge

Note : All these methods can be applied recursively.

- **Equal-width (distance) partitioning**
 - Divides the range into N intervals of equal size.
 - The width of the interval is $w = (\text{Maximum} - \text{Minimum})/N$.
 - Is susceptible to outliers and skewed data.
- **Equal-depth (frequency) partitioning**
 - Divides the range into N intervals, each containing approximately same number of samples.
 - Ensures good data scaling but managing categorical attributes can get tricky.

- **Data :** 0, 4, 12, 16, 16, 18, 24, 26, 28

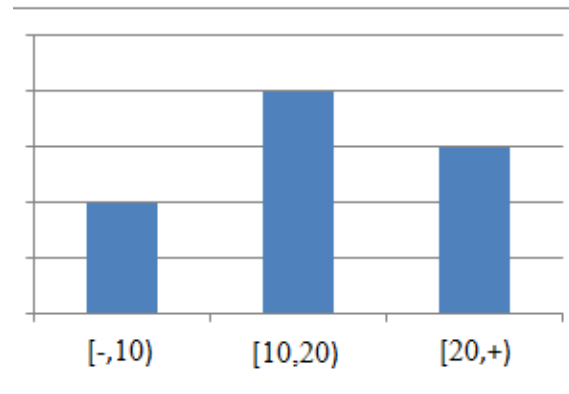
- **Equal width**

- Bin 1: 0, 4 [-,10)
- Bin 2: 12, 16, 16, 18 [10,20)
- Bin 3: 24, 26, 28 [20,+)

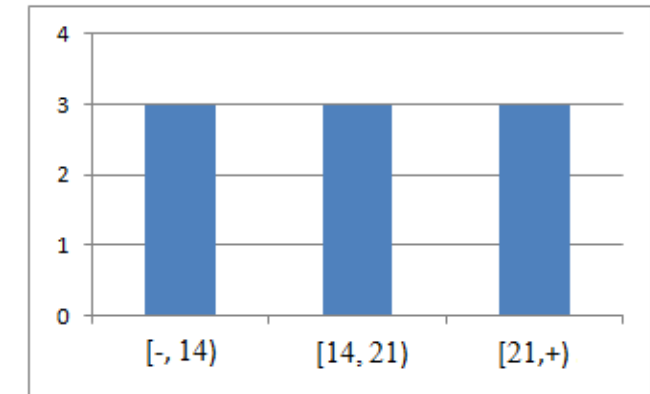
- **Equal frequency**

- Bin 1: 0, 4, 12 [-, 14)
- Bin 2: 16, 16, 18 [14, 21)
- Bin 3: 24, 26, 28 [21,+)

Equal width



Equal frequency



Sorted data for price (in \$) : 4,8,9,15,21,21,24,25,26,28,29,34

Partition into equal-depth (frequency) bins

Bin-1 : 4,8,9,15

Bin-2 : 21,21,24,25

Bin-3 : 26,28,29,34

- Smoothing by **bin means** :

Bin-1 : 9,9,9,9

Bin-2 : 23,23,23,23

Bin-3 : 29,29,29,29

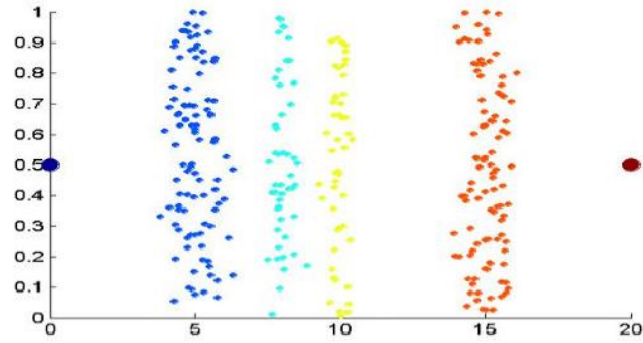
- Smoothing by **bin boundaries** :

Bin-1 : 4,4,4,15

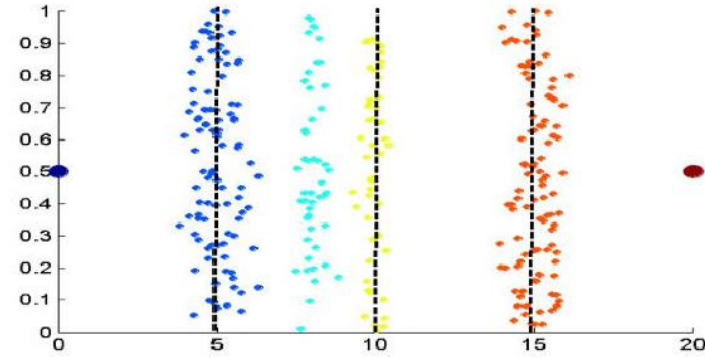
Bin-2 : 21,21,25,25

Bin-3 : 26,26,26,34

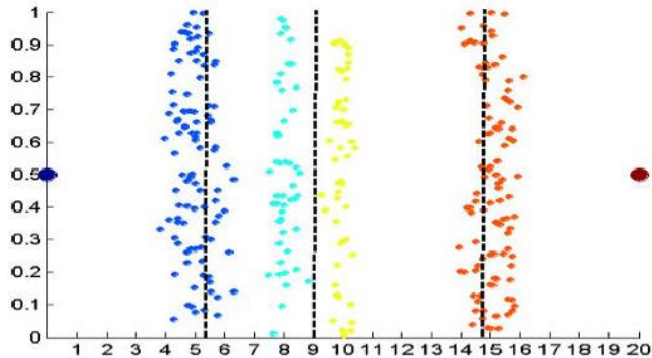
Binning vs Clustering (Unsupervised Discretization)



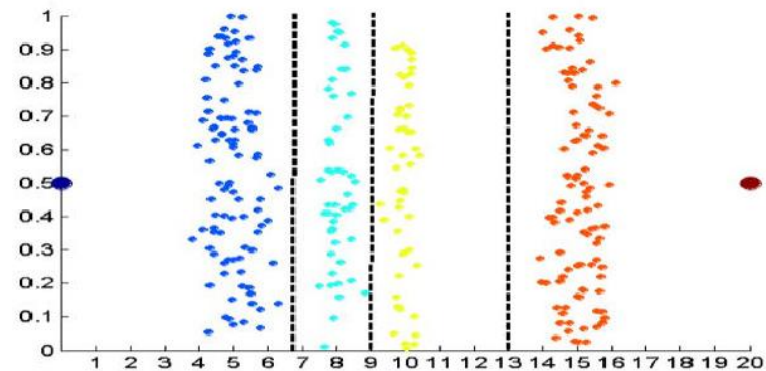
Original Data



Equal Width(binning)

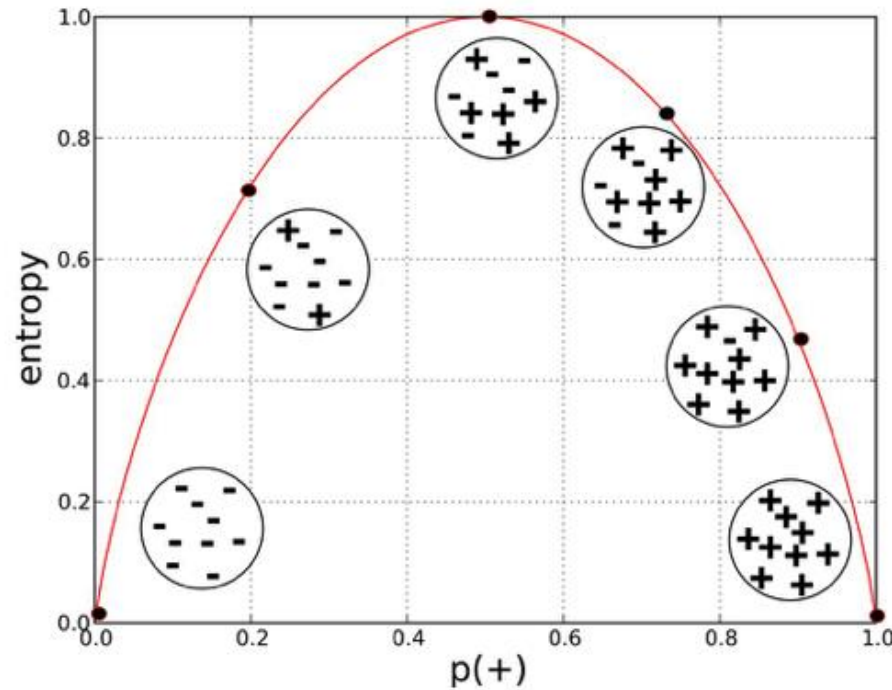


Equal frequency (binning)



K-means clustering leads to
better results

- Supervised : Class labels are used in determining the split point.
- It is a top-down discretization where recursive split is applied.
- Example : Decision tree analysis.
- Entropy is used to determine the split point. Lower the entropy , better the split. It is given by
$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$



- Supervised : Class labels are used in determining the split point.
- Example , Chi-merge: χ^2 -based discretization. It is a bottom-up merge.
- Initially each distinct value of the attribute is considered to be one interval.
- χ^2 tests are performed for every pair of adjacent intervals.
- Adjacent intervals with the least χ^2 values are merged as it indicates similar class distributions.
- This merging process proceeds recursively until a pre-defined threshold for χ^2 is met.

Discretization by Correlation Analysis - Example

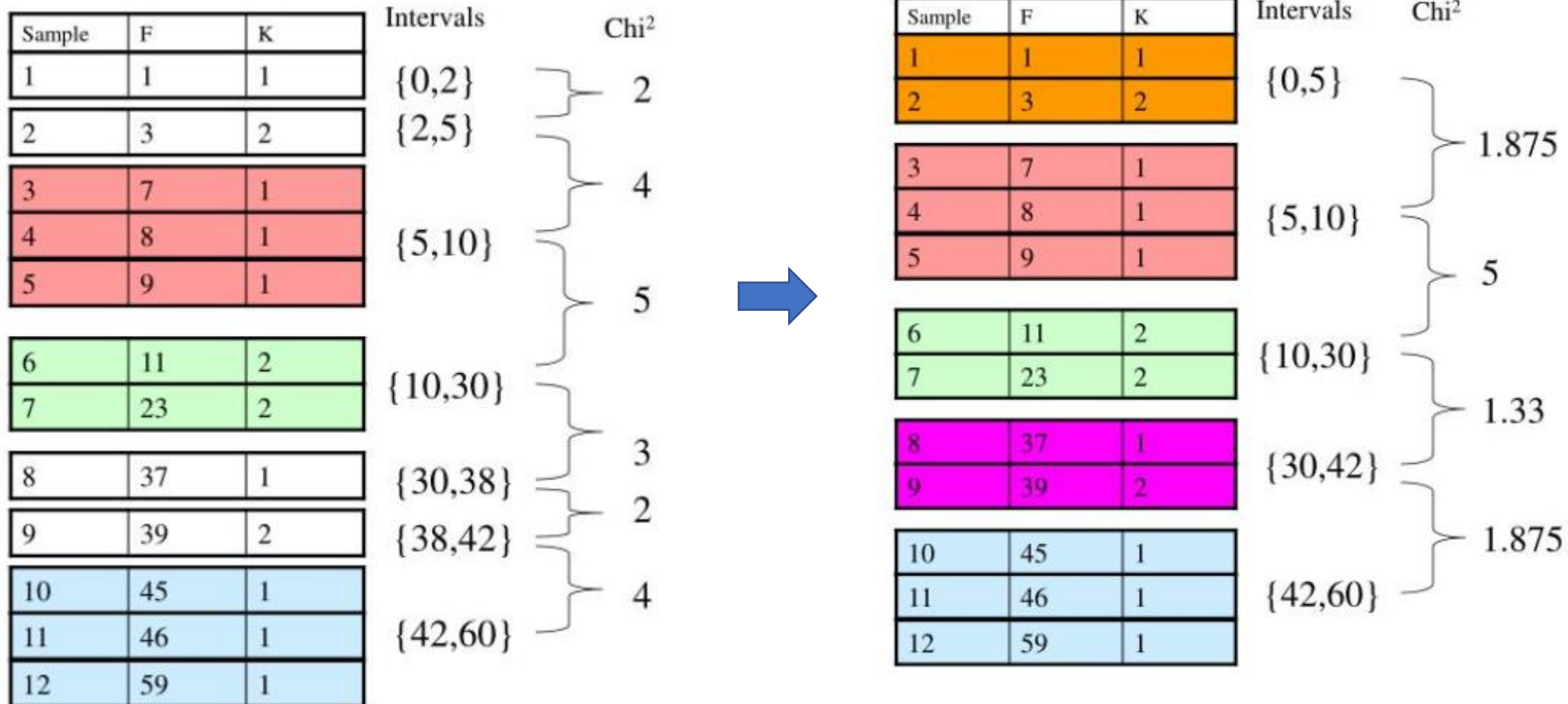
Sample	F	K
1	1	1
2	3	2
3	7	1
4	8	1
5	9	1
6	11	2
7	23	2
8	37	1
9	39	2
10	45	1
11	46	1
12	59	1



Sample	F	K	Intervals	Chi ²
1	1	1	{0,2}	2
2	3	2	{2,5}	2
3	7	1	{5,7.5}	0
4	8	1	{7.5,8.5}	0
5	9	1	{8.5,10}	2
6	11	2	{10,17}	0
7	23	2	{17,30}	2
8	37	1	{30,38}	2
9	39	2	{38,42}	2
10	45	1	{42,45.5}	0
11	46	1	{45.5,52}	0
12	59	1	{52,60}	0

•Calculate all the Chi Square value for all intervals

•Merge the intervals with the smallest Chi values



Sample	F	K	Intervals	Chi ²
1	1	1	{0,5}	1.875
2	3	2		
3	7	1	{5,10}	3.93
4	8	1		
5	9	1		
6	11	2	{10,30}	3.93
7	23	2		
8	37	1		
9	39	2		
10	45	1	{42,60}	3.93
11	46	1		
12	59	1		



Sample	F	K	Intervals	Chi ²
1	1	1	{0,10}	2.72
2	3	2		
3	7	1		
4	8	1		
5	9	1		
6	11	2	{10,30}	3.93
7	23	2		
8	37	1		
9	39	2		
10	45	1	{42,60}	3.93
11	46	1		
12	59	1		

•There are no more intervals that can satisfy the Chi Square test.

- Which method of normalization must one choose if they are dealing with a lot of outliers and don't know the range of their data?

Solution

Z-Score Normalization

- Which split-point is preferred for discretization using decision trees?

Solution

A Split point which results in least entropy.

- Which normalization method strictly works in the range of input data?

Solution

Min-Max Normalization

Test your understanding!

- Consider a set of Unsorted data for price in dollars
8 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

- 1) Smooth the data by equal frequency bins
- 2) On the results of part (1) apply smoothing by bin means

Solution:

1)

Bin 1: 8, 9, 15, 16

Bin 2: 21, 21, 24, 26,

Bin 3: 27, 30, 30, 34

For Bin 1:

$$(8 + 9 + 15 + 16 / 4) = 12$$

(4 indicating the total values like 8, 9, 15, 16)

Bin 1 = 12, 12, 12, 12

For Bin 2:

$$(21 + 21 + 24 + 26 / 4) = 23$$

Bin 2 = 23, 23, 23, 23

For Bin 3:

$$(27 + 30 + 30 + 34 / 4) = 30$$

Bin 3 = 30, 30, 30, 30

2)

- [Data Mining: Concepts and Techniques](#) by Jiawei Han, Micheline Kamber and Jian Pei, The Morgan Kaufmann Series in Data Management Systems, 3rd Edition Chapter 3.5
- <https://t4tutorials.com/binning-methods-for-data-smoothing-in-data-mining/>



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DATA ANALYTICS

UE24CS342AA2

UNIT-1

Lecture 5: Data preprocessing - Transformations and Discretization - 1

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 5: Data preprocessing - Transformations and Discretization - 1

Slides excerpted from: Data Mining : Concepts and Techniques by Han, Kamber and Pei, 3rd Edition

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A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values.

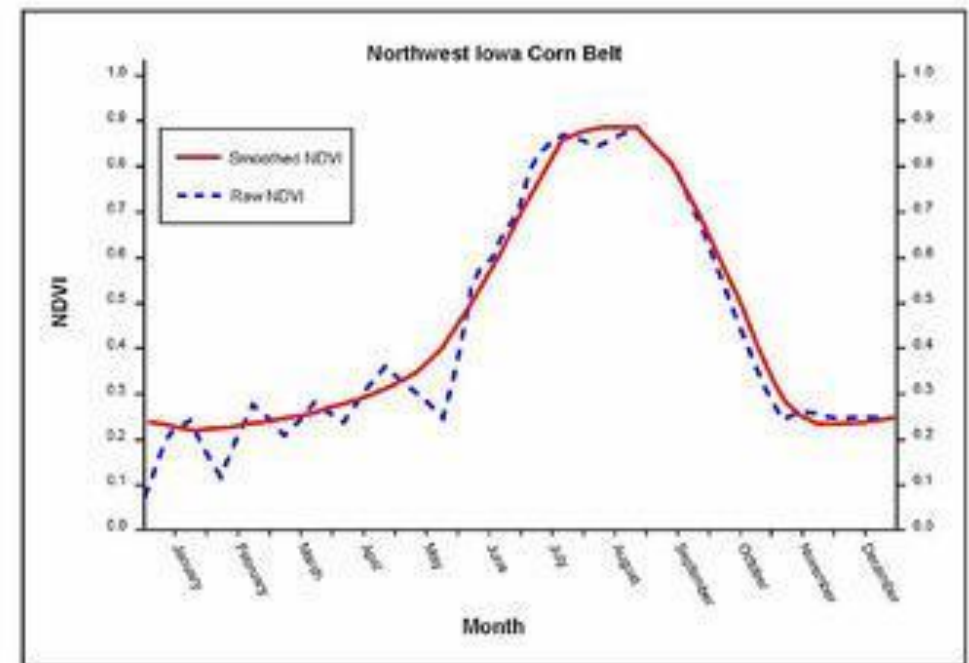
Methods

- Smoothing
- Attribute construction
- Aggregation
- Normalization
- Discretization
- Concept hierarchy generation

Removal of noise in data.

Techniques:

- Binning
- Regression
- Clustering
- Simple average – Time series data
- Weighted average – Time series data
- Exponential smoothing – Time series data
- Gaussian filter - Image



Need for normalization :

- The unit of measurement can affect data analysis.
- For example , changing the unit of measurement of height from inches to cm can lead to different results.
- An attribute in smaller units results in a larger range. This attribute will be given a higher weight(preference) by few distance-based algorithms like KNN(K-nearest neighbors).
- For example , income of a person is generally a few orders higher than their height.
- To overcome this , data is generally transformed to a range such as $[-1,1]$ or $[0,1]$.
- **Normalization by Decimal Scaling** : A number v is normalized by $v' = \frac{v}{10^j}$

Where j is the smallest integer such that $\text{Max}(|v'|) < 1$

For example , $[-616,973]$ is normalized to $[-0.616,0.973]$ where $j=3$.

- **Min-Max Normalization** : Performs a linear transformation. It transforms the values from $[\min_A, \max_A]$ to $[\text{new_min}_A, \text{new_max}_A]$. A value v is transformed by

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

- Example : Let the income range \$12,000 to \$98,000 be normalized to $[0.0, 1.0]$. Find out the mapping for \$73,600

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$$

- This preserves the relationship among the original data values. It will encounter an “out-of-bounds” error if an input which is outside the range of original data is provided.

- **Z-score normalization** : The values for an attribute A are normalized based on the mean and standard deviation of A. A value v can be normalized by

$$v' = \frac{v - \mu_A}{\sigma_A}$$

Where μ_A is the mean and σ_A the standard deviation.

- Example : Let $\mu = 54,000$ and $\sigma = 16,000$. Then z-score of 73,600 is

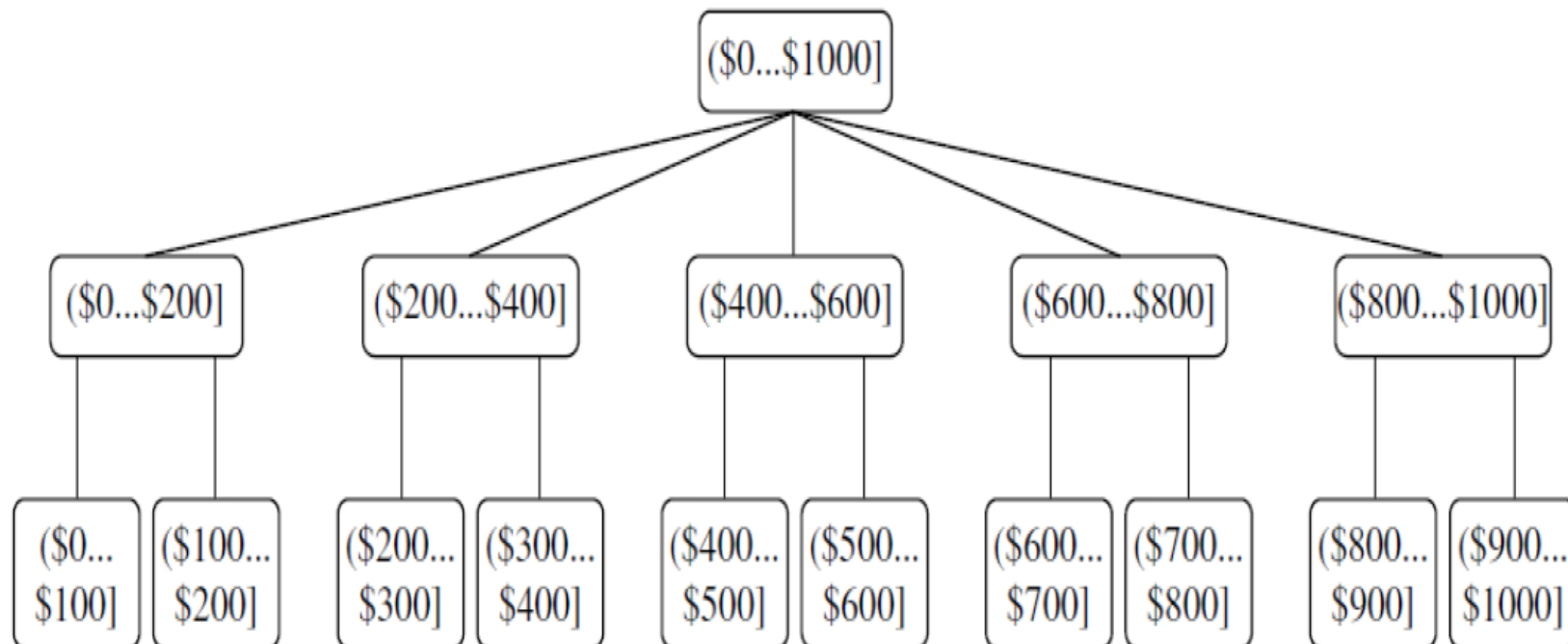
$$\frac{73,600 - 54,000}{16,000} = 1.225$$

- This method of normalization is useful when the actual minimum or maximum value of A is unknown or there are outliers in A.
- A variation of this is obtained by using Mean Absolute Deviation (MAD) instead of standard deviation. It is less susceptible to outliers.

- **Concept hierarchy** organizes concepts (Attribute values) hierarchically by representing a series of mappings from a set of low-level concepts to a high-level , generalized concepts.
- It facilitates drilling and rolling in data warehouses to view data in multiple granularity.
- Method : Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for age) by higher level concepts (such as kids, teenagers, adults, senior citizens).
- They can be explicitly specified by domain experts and/or data warehouse designers.

Discretization methods discussed till now can be used for numeric data.

Example:



1) Specification of a partial ordering of attributes explicitly at the schema level

- A user or an expert defines a concept hierarchy by specifying a partial or a total ordering of attributes at the schema level.
- For example , suppose a relational database contains the attributes *street*, *city*, *state* and *country* . *Location* dimension of the data warehouse may contain the same attributes.
- A hierarchy can be defined by specifying the total ordering among these attributes at the schema level

street < city < state < country

2) Specification of a portion of hierarchy by explicit data grouping

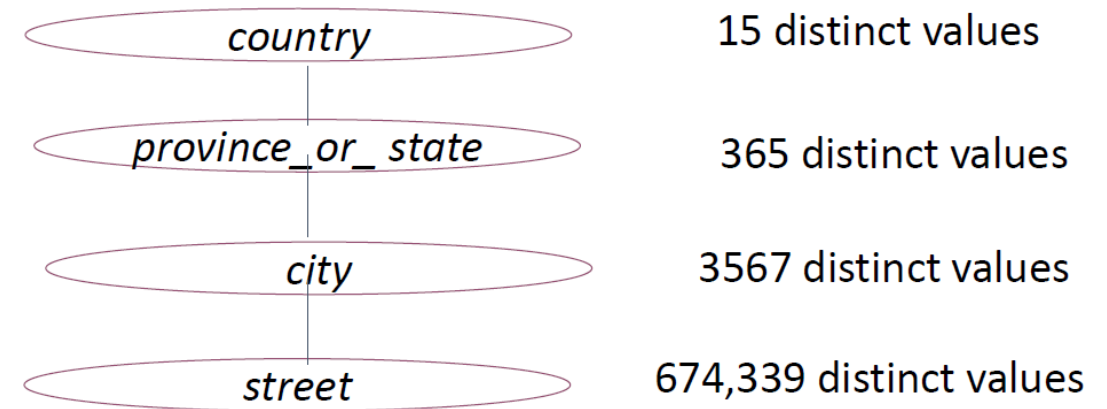
- A portion of the concept hierarchy is manually defined.
- In a large database , it is unrealistic to define the entire concept hierarchy by explicit value enumeration.
- However , we can easily specify explicit groupings for a small portion of intermediate-level data.
- For example , after specifying state and country form a hierarchy at the schema level , a user can define few intermediate levels manually
{Karnataka , Tamil Nadu, Kerala, Andhra Pradesh, Telangana} < South India

3) Specification of only a partial set of attributes.

- At times , a user can have a vague idea about what should be included in the hierarchy.
- The user may have included only a small subset of the relevant attributes in the hierarchy specification.
- For example, instead of including all the hierarchically relevant attributes for *location* , the user might have specified only *street* and *city*.
- To handle this , embed data semantics into the database schema. Hence one attribute will trigger a whole group of linked attributes to be added to the hierarchy. For example , when *city* is added , it would automatically include *state* and *country* as they are semantically related.

4) Automatic generation of hierarchies by analysis of distinct values per attribute

- Few hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the dataset.
- The attribute with the most distinct values is placed at the lowest level of the hierarchy.



Note : This method is not foolproof. For example, a time dimension in a database might contain 20 distinct *years* , 12 distinct *months* and 7 distinct *days of the week*. However , this doesn't suggest that the time hierarchy should be *year<month<days of the week* .

- Which method of normalization must one choose if they are dealing with a lot of outliers and don't know the range of their data?

Solution

Z-Score Normalization

- Which normalization method strictly works in the range of input data?

Solution

Min-Max Normalization

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DATA ANALYTICS

UE24CS342AA2

UNIT-1

Lecture 6: Data Preprocessing – Transformations and Discretization - 2

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 6: Data Preprocessing – Transformations and Discretization - 2

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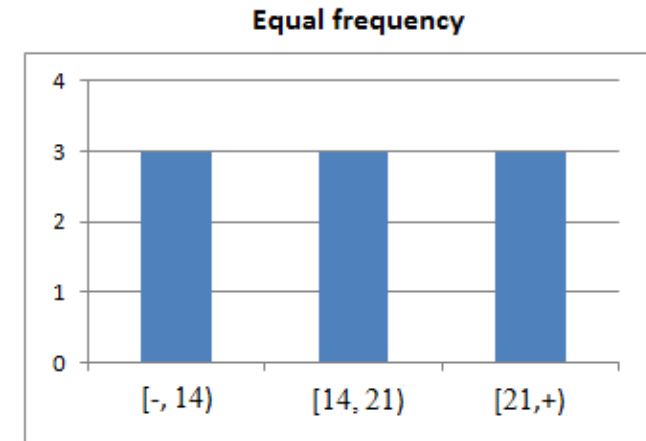
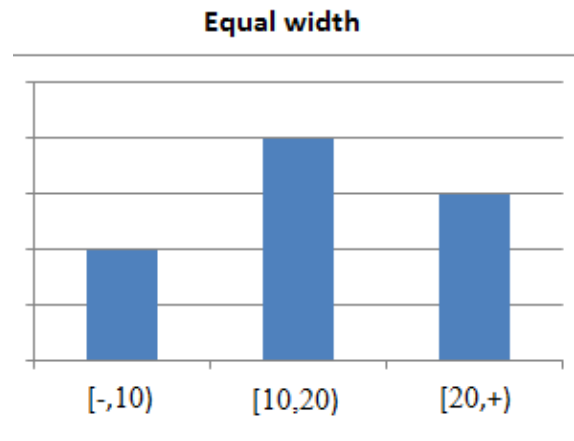
- **Data :** 0, 4, 12, 16, 16, 18, 24, 26, 28

- **Equal width**

- Bin 1: 0, 4 [-,10)
- Bin 2: 12, 16, 16, 18 [10,20)
- Bin 3: 24, 26, 28 [20,+)

- **Equal frequency**

- Bin 1: 0, 4, 12 [-, 14)
- Bin 2: 16, 16, 18 [14, 21)
- Bin 3: 24, 26, 28 [21,+)



Sorted data for price (in \$) : 4,8,9,15,21,21,24,25,26,28,29,34

Partition into equal-depth (frequency) bins

Bin-1 : 4,8,9,15

Bin-2 : 21,21,24,25

Bin-3 : 26,28,29,34

- Smoothing by **bin means** :

Bin-1 : 9,9,9,9

Bin-2 : 23,23,23,23

Bin-3 : 29,29,29,29

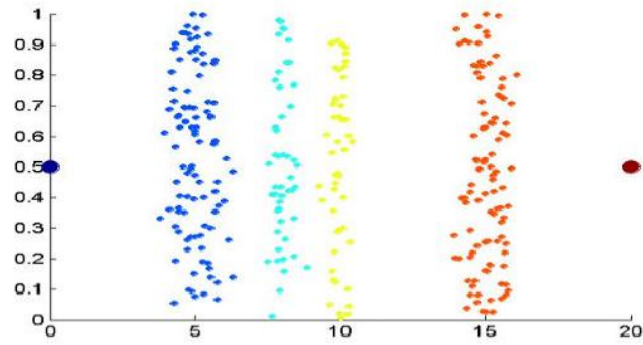
- Smoothing by **bin boundaries** :

Bin-1 : 4,4,4,15

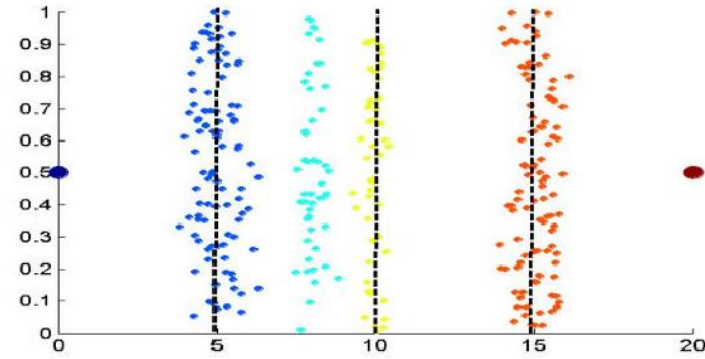
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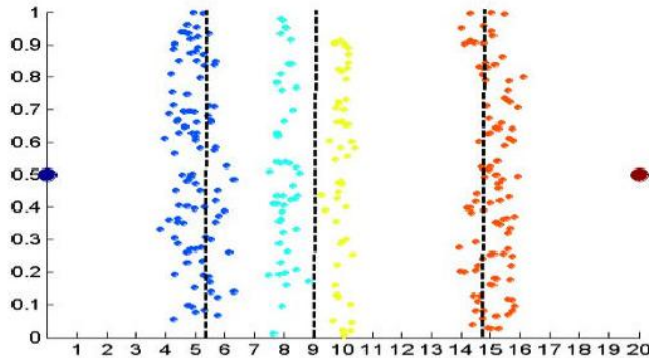
Binning vs Clustering (Unsupervised Discretization)



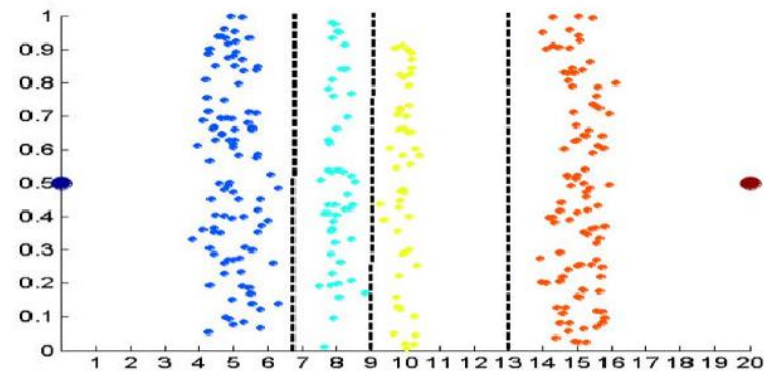
Original Data



Equal Width(binning)

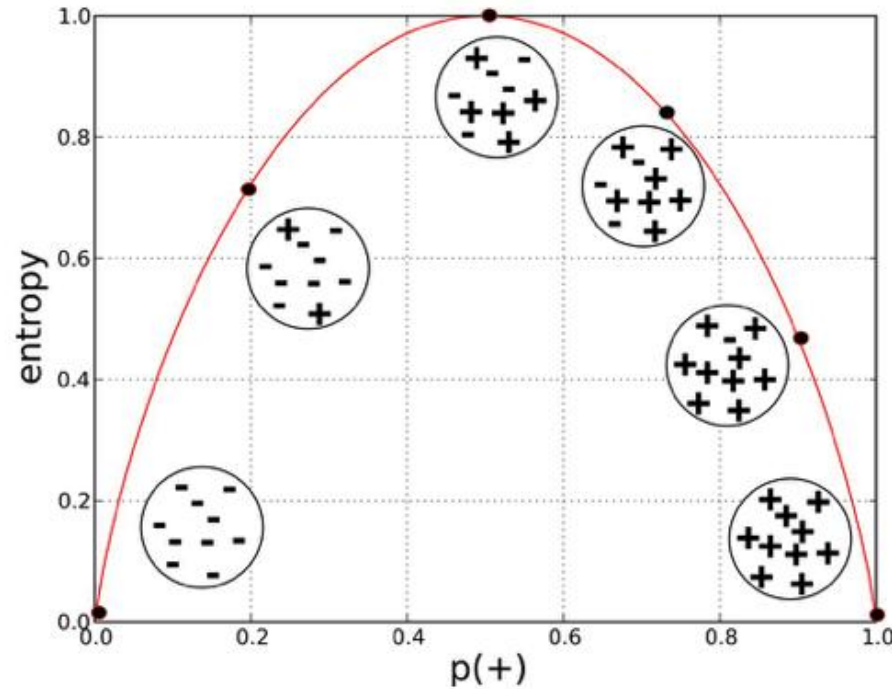


Equal frequency (binning)



K-means clustering leads to
better results

- Supervised : Class labels are used in determining the split point.
- It is a top-down discretization where recursive split is applied.
- Example : Decision tree analysis.
- Entropy is used to determine the split point. Lower the entropy , better the split. It is given by
$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$



- Supervised : Class labels are used in determining the split point.
- Example , Chi-merge: χ^2 -based discretization. It is a bottom-up merge.
- Initially each distinct value of the attribute is considered to be one interval.
- χ^2 tests are performed for every pair of adjacent intervals.
- Adjacent intervals with the least χ^2 values are merged as it indicates similar class distributions.
- This merging process proceeds recursively until a pre-defined threshold for χ^2 is met.

Discretization by Correlation Analysis - Example

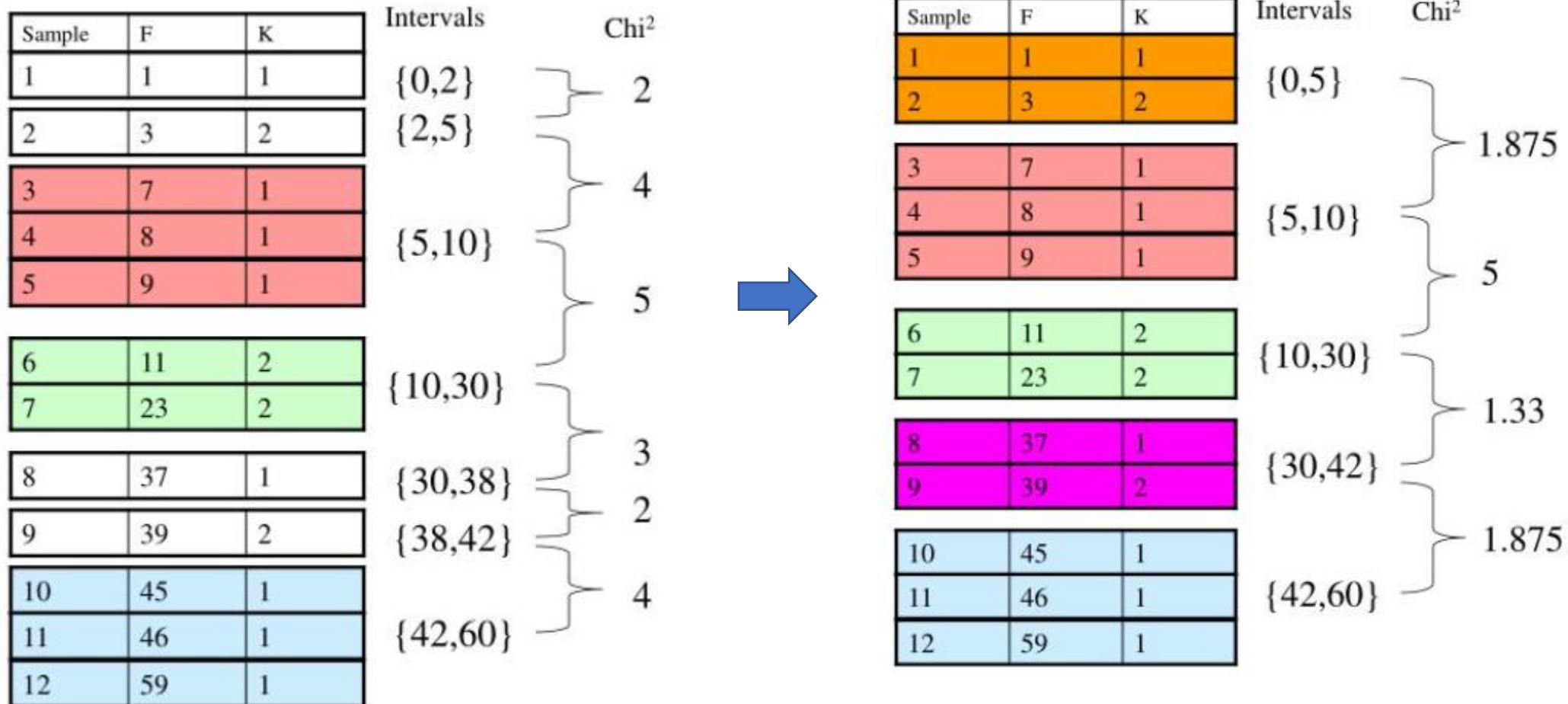
Sample	F	K
1	1	1
2	3	2
3	7	1
4	8	1
5	9	1
6	11	2
7	23	2
8	37	1
9	39	2
10	45	1
11	46	1
12	59	1



Sample	F	K	Intervals	Chi ²
1	1	1	{0,2}	2
2	3	2	{2,5}	2
3	7	1	{5,7.5}	0
4	8	1	{7.5,8.5}	0
5	9	1	{8.5,10}	2
6	11	2	{10,17}	0
7	23	2	{17,30}	2
8	37	1	{30,38}	2
9	39	2	{38,42}	2
10	45	1	{42,45.5}	0
11	46	1	{45.5,52}	0
12	59	1	{52,60}	0

•Calculate all the Chi Square value for all intervals

•Merge the intervals with the smallest Chi values



Sample	F	K	Intervals	Chi ²
1	1	1	{0,5}	1.875
2	3	2		
3	7	1	{5,10}	3.93
4	8	1		
5	9	1		
6	11	2	{10,30}	3.93
7	23	2		
8	37	1		
9	39	2		
10	45	1	{42,60}	3.93
11	46	1		
12	59	1		



Sample	F	K	Intervals	Chi ²
1	1	1	{0,10}	2.72
2	3	2		
3	7	1		
4	8	1		
5	9	1		
6	11	2	{10,30}	3.93
7	23	2		
8	37	1		
9	39	2		
10	45	1	{42,60}	3.93
11	46	1		
12	59	1		

•There are no more intervals that can satisfy the Chi Square test.

- Which split-point is preferred for discretization using decision trees?

A Split point which results in least entropy.

- Equal-width partitioning is more susceptible to outliers than equal-depth partitioning. True or False

True

- Consider a set of Unsorted data for price in dollars
8, 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

- 1) Discretize the data into three equal-frequency (equal-depth) bins.
- 2) On the results of part (1) apply smoothing by bin means

Solution:

1)

Bin 1: 8, 9, 15, 16

Bin 2: 21, 21, 24, 26,

Bin 3: 27, 30, 30, 34

For Bin 1:

$$(8 + 9 + 15 + 16 / 4) = 12$$

(4 indicating the total values like 8, 9, 15, 16)

Bin 1 = 12, 12, 12, 12

For Bin 2:

$$(21 + 21 + 24 + 26 / 4) = 23$$

Bin 2 = 23, 23, 23, 23

For Bin 3:

$$(27 + 30 + 30 + 34 / 4) = 30$$

Bin 3 = 30, 30, 30, 30

2)

- [Data Mining: Concepts and Techniques](#) by Jiawei Han, Micheline Kamber and Jian Pei, The Morgan Kaufmann Series in Data Management Systems, 3rd Edition Chapter 3.5
- <https://t4tutorials.com/binning-methods-for-data-smoothing-in-data-mining/>



THANK YOU

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DATA ANALYTICS

UE24CS342AA2

UNIT-1

Lecture 07: Analysis of Variance - 1

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 07: Analysis of Variance - 1

Slides excerpted from: U. Dinesh Kumar,
“Business Analytics”, Wiley, 2nd Edition 2022

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Nishanth M S, Department of CSE

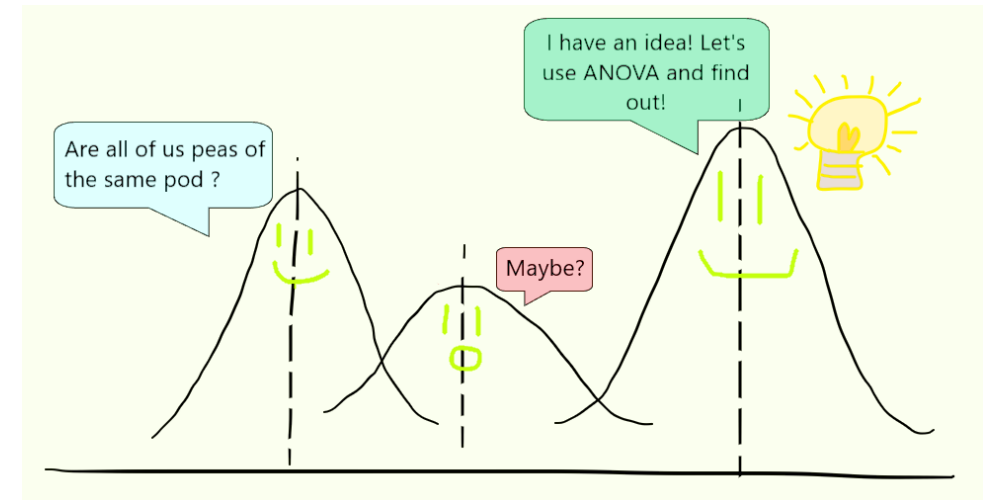
Harshitha Srikanth, PESU, Department of CSE

Karthik Namboori, PESU, Department of CSE

With grateful thanks for contribution of slides to:

Dr. Mamatha H R, Professor at the Department of CSE, PESU

- **Analysis of Variance (ANOVA)** is a statistical technique that is used to check if the means of two or more groups are significantly different from each other.
- ANOVA checks the impact of one or more factors by comparing the means of different samples.
- For example , assume there are 3 classes. For a given exam , you want to find out if the marks of the student (dependent variable) depends on the class they study in. So , we are trying to find out if *class they study in* is a **factor**.



<https://www.geeksforgeeks.org/one-way-anova/>

To understand whether the factor (different levels of factor) has any statistical significance on the population parameter, we compare to models

1) Means Model :

- It is given by $Y_{ij} = \mu + \varepsilon_{ij}$
- Y_{ij} is the value of the outcome variable of j^{th} observation for i^{th} factor level, μ is the overall mean value of all observations, ε_{ij} is the error assumed to be a normal distribution with mean 0 and standard deviation σ .
- Model defined in above equation is often called the **reduced model**, in which the mean μ is common for all levels of the factor.

2) Factor Effect Model :

- It is given by $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$
- μ is the overall mean and τ_i is the effect of factor i (or factor effect). τ_i is the difference between overall mean and the factor level mean.
- A non-zero τ_i implies that a factor has an influence on the value of the outcome variable Y_{ij} .
- Our objective is to verify if the variation is due to factors is different from the variations due to randomness.
- Model defined in above equation is called **full model**.

Multiple t-tests for comparing several means

- Consider a retail store who would like to study the impact of different levels of price discounts (factors) on the sales (outcome variable). Let's say they are analyzing the levels of discounts of 0%, 10% and 20%.
- If we had only 2 levels of discount, we could have used a t-test directly to check whether a statistically significant relationship exists between price discount and average sales quantity.
- One option is to use 3 different 2 sample t-test:
 - Test between 0% and 10%
 - Test between 0% and 20%
 - Test between 10% and 20%

Test	Null Hypothesis	Alternative Hypothesis	Significance (α)
A	$H_0: \mu_0 = \mu_{10}$	$H_A: \mu_0 \neq \mu_{10}$	$\alpha = 0.05$
B	$H_0: \mu_0 = \mu_{20}$	$H_A: \mu_0 \neq \mu_{20}$	$\alpha = 0.05$
C	$H_0: \mu_{10} = \mu_{20}$	$H_A: \mu_{10} \neq \mu_{20}$	$\alpha = 0.05$

- Let,

$$P(A) = P(\text{Retain } H_0 \text{ in test A} | H_0 \text{ in test A is true})$$

$$P(B) = P(\text{Retain } H_0 \text{ in test B} | H_0 \text{ in test B is true})$$

$$P(C) = P(\text{Retain } H_0 \text{ in test C} | H_0 \text{ in test C is true})$$

- Note : values of $P(A) = P(B) = P(C) = 1 - \alpha = 1 - 0.05 = 0.95$
- The conditional probability of simultaneously retaining all 3 null hypotheses when they are true is $P(A \cap B \cap C) = 0.95^3 = 0.8573$.
- Now consider the following null hypothesis:

$$H_0: \mu_0 = \mu_{10} = \mu_{20}$$

If we retain the null hypothesis based on the three individual t -tests, then the significance or Type I error is not α -value but much higher than α (Lunney, 1969; Siegel, 1990).

- For the case discussed , if we retain the null hypothesis based on 3 individual tests, then the Type I error is $1 - 0.8573 = 0.1426$.
- When more than 2 groups are involved, checking the population parameter values simultaneously using *t*-tests is inappropriate since the Type I and Type II errors will be estimated incorrectly.
- For this reason, we use analysis of variance (ANOVA) whenever we need to compare 3 or more groups for population parameter values simultaneously.

One-way ANOVA is appropriate under the following conditions:

- 1) We would like to study the impact of a single treatment (also known as factor) at different levels (thus forming different groups) on a continuous response variable (or outcome variable). For the example discussed, the variable 'price discount' is the treatment (or factor) and 0%, 10%, and 20% price discounts are the different levels (3 levels in this case), different levels of discount is likely to have varying impact on the sales of the product, where sales is the outcome variable. We would like to understand the impact of different levels of price discount on the response variable, sales.

- 2) In each group, the population response variable follows a normal distribution and the sample subjects are chosen using random sampling.
- 3) The population variances for different groups are assumed to be same. That is, variability in the response variable values within different groups is same

Although conditions 2 and 3 are necessary for one-way ANOVA, the model is robust and minor violations of the assumptions may not result in incorrect decision about the null hypothesis.

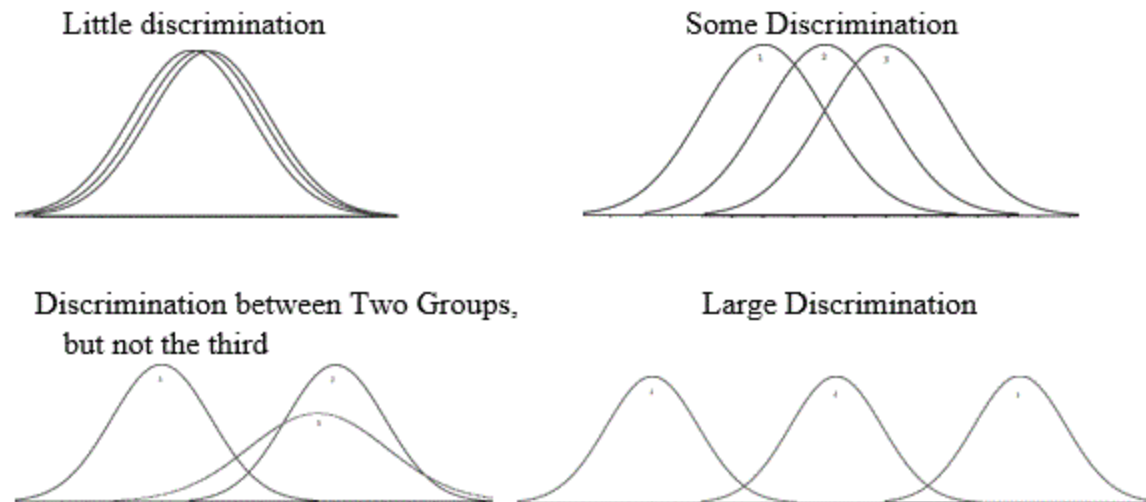
- Assume that we would like to study the impact of a factor (such as discount) with k levels on a continuous variable (such as sales quantity).
- Then the null and alternative hypotheses for one way ANOVA are given by

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

$$H_A: \text{Not all } \mu \text{ values are equal}$$

- Note that the alternative hypothesis, 'not all μ values are equal', implies that some of them could be equal.
- The null hypothesis is equivalent to stating that the factor effects $\tau_1, \tau_2, \dots, \tau_k$ defined in the equation $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ are zero.

The hypothesis test can be visualized as follows. Large discrimination in the means mean the factor levels have an impact on the outcome. If there is a little discrimination, it means that the factor levels don't have statistically significant impact.



- We are interested in analyzing single factor effect with k levels, thus we will have k groups.

Let

k = Number of groups (or samples)

n_i = Number of observations in group i ($i = 1, 2, \dots, k$)

n = Total number of observations ($= \sum_{i=1}^k n_i$)

Y_{ij} = Observation j in group i

- μ_i = Mean of group $i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}$
- μ = Overall mean = $\frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij}$

To arrive at the statistic, we calculate the following measures, which are variations within groups and between groups.

- **Sum of Squares of Total Variation (SST)**

Total variation is the sum of squared variation of all values of response variable (Y_{ij}) from the overall mean (μ) and is given by

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \mu)^2$$

The degrees of freedom for SST is $(n - 1)$ since only the value of μ is estimated from n observations and thus only one degree of freedom is lost. Mean Square

Total (MST) variation is given by $MST = \frac{SST}{n-1}$.

- Sum of Squares of Between (SSB) Group Variation

Sum of squares of between variation is the sum of squared variation between the group mean (μ_i) and the overall mean (μ) of the data and is given by

$$SSB = \sum_{i=1}^k n_i \times (\mu_i - \mu)^2$$

The degrees of freedom is $(k-1)$. Since the overall mean is estimated from the data, one degree of freedom is lost. Mean square between variation (MSB) is given by

$$MSB = \frac{SSB}{k - 1}$$

- Sum of Squares of Within (SSW) Group Variation

Sum of squares of within the group variation is the sum of squared variation of all observations (Y_{ij}) from that group mean (μ_i) and is given by

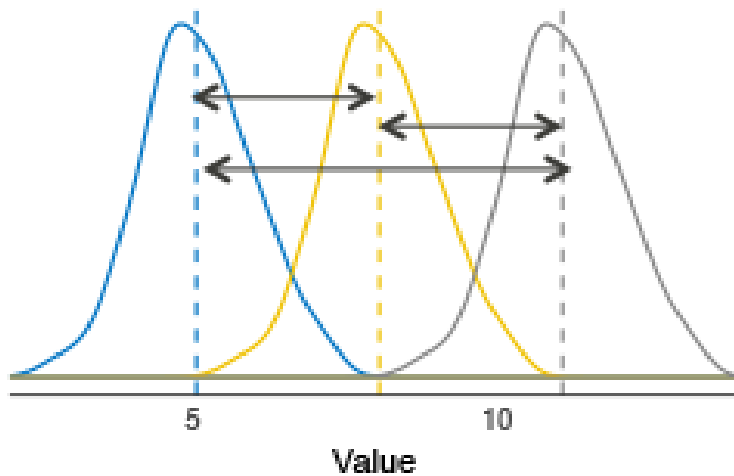
$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \mu_i)^2$$

The degrees of freedom for SSW is $(n - k)$. Here k degrees of freedom are lost since we estimate k group means (μ_i). The mean square of variation within the group is

$$MSW = \frac{SSW}{n - k}$$

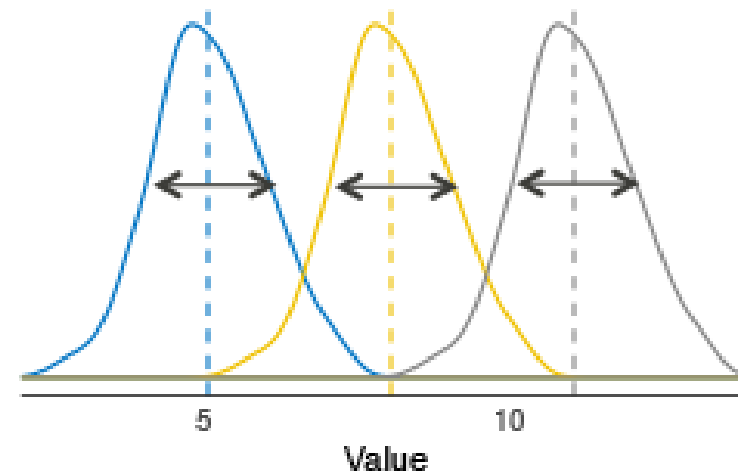
A

Between-group variation
(i.e. Differences among group means)



B

Within-group variation
(i.e. Variability within each group)



We can prove algebraically

$$\sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \mu)^2 = \sum_{i=1}^k n_i \times (\mu_i - \mu)^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \mu_i)^2$$

That is,

$$\mathbf{SST=SSB+SSW}$$

According to Cochran's theorem (Kutner *et al.*, 2013, page 70):

'If $Y_1, Y_2, \dots, Y_n$ are drawn from a normal distribution with mean μ and standard deviation σ and sum of squares of total variation is decomposed into k sum of squares (SS_r) with degrees of freedom df_r , then the ratio (SS_r/σ^2) are independent χ^2 variables with df_r degrees of freedom if

$$\sum_{r=1}^k df_r = n - 1$$

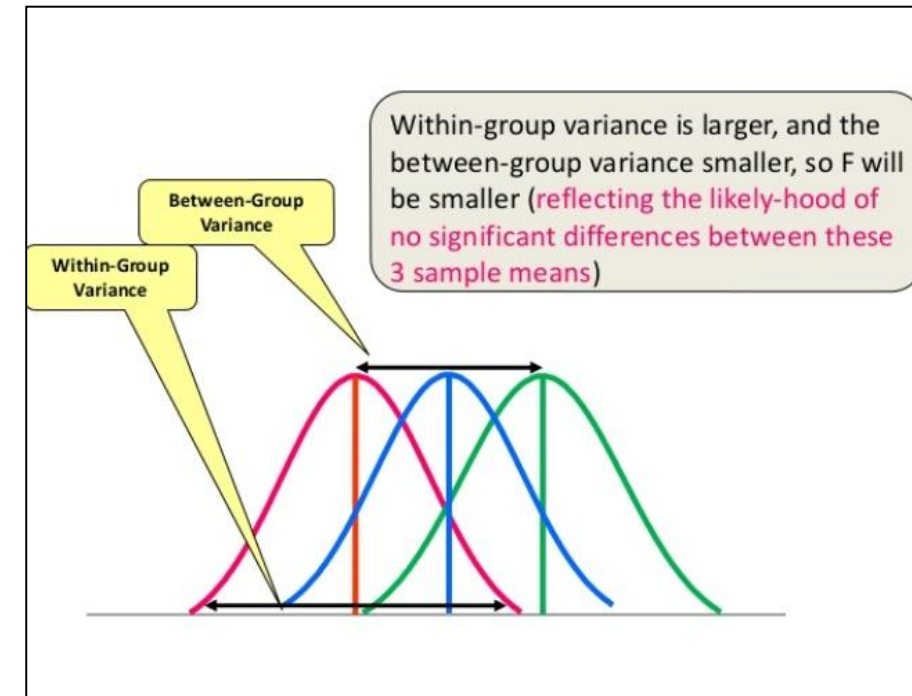
Note that, in the equation **$SST = SSB + SSW$** , the SST is decomposed into two sums of squares (SSB and SSW) and thus, **SSB/σ^2** and **SSW/σ^2** are chi-square variables.

- If the null hypothesis is true, then there will be no difference in the mean values which will result in no difference between MSB and MSW .
- Alternatively, if the means are different, then MSB will be larger than MSW .
- That is the ratio MSB/MSW will be close to 1 if there is no difference between the mean values and will be larger than 1 if the means are different.

- Following Cochran's theorem (Kirk, 1995) MSB/MSW is a ratio of two chi-square variate which is an F -distribution. Thus the statistic for testing the null hypothesis is

$$F = \frac{SSB/(k - 1)}{SSW/(n - k)} = \frac{MSB}{MSW}$$

- Note that the test statistic is a one-tailed test (right tailed) since we are interested in finding whether the variation between groups is greater than variation within the groups.
- It is important to note that rejecting the null hypothesis will not tell us exactly which means differ from each other, it will only indicate that there is a difference in at least one of the group means. We may have to conduct two-sample t-tests to find out which mean values are different.



Example (an Experimental Study)

Ms Rachael Khanna the brand manager of ENZO detergent powder at the 'one stop' retail was interested in understanding whether the price discounts has any impact on the sales quantity of ENZO. To test whether the price discounts had any impact, price discounts of 0% (no discount), 10% and 20% were given on randomly selected days. The quantity (in kilograms) of ENZO sold in a day under different discount levels is shown in below. Conduct a one-way ANOVA to check whether discount had any significant impact on the sales quantity at $\alpha = 0.05$.

No Discount (0% discount)									
39	32	25	25	37	28	26	26	40	29
37	34	28	36	38	38	34	31	39	36
34	25	33	26	33	26	26	27	32	40
10% Discount									
34	41	45	39	38	33	35	41	47	34
47	44	46	38	42	33	37	45	38	44
38	35	34	34	37	39	34	34	36	41
20% Discount									
42	43	44	46	41	52	43	42	50	41
41	47	55	55	47	48	41	42	45	48
40	50	52	43	47	55	49	46	55	42

- In this case, the number of groups $k = 3$; $n_1 = n_2 = n_3 = 30$; $\mu_1 = 32$, $\mu_2 = 38.77$, $\mu_3 = 46.4$; and $\mu = 39.05$.

- The sum of squares of between groups variation (SSB) is given by

$$SSB = \sum_{i=1}^k n_i \times (\mu_i - \mu)^2 = 30 \times [(32 - 39.05)^2 + (38.77 - 39.05)^2 + (46.4 - 39.05)^2]$$
$$= 3114.156$$

- Therefore

$$MSB = \frac{SSB}{k - 1} = \frac{3114.156}{2} = 1557.078$$

- The sum of squares of within the group variation is given by

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \mu_i)^2 = \sum_{j=1}^{30} (Y_{1j} - 32)^2 + \sum_{j=1}^{30} (Y_{2j} - 38.77)^2 + \sum_{j=1}^{30} (Y_{3j} - 46.4)^2$$
$$= 2056.567$$

- Therefore

$$MSW = \frac{SSW}{n - k} = \frac{2056.567}{90 - 3} = 23.63$$

- The F -statistic value is

$$F_{2,87} = \frac{MSB}{MSW} = \frac{1557.078}{23.6387} = 65.86$$

- The critical F -value with degrees of freedom (2, 87) for $\alpha = 0.05$ is **3.101**
- The p -value for $F_{2,87} = 65.86$ is 3.82×10^{-18}
- Since the calculated F -statistic is much higher than the critical F -value, we reject the null hypothesis and conclude that the mean sales quantity values under different discounts are different.

Anova: Single Factor

SUMMARY				
Groups	Count	Sum	Average	Variance
No Discount	30	960	32	27.17241
10% Discount	30	1163	38.76667	20.46092
20% Discount	30	1392	46.4	23.28276

ANOVA

Source of Variation	SS	df	MS	F	p-value	F crit
Between Groups	3114.15556	2	1557.078	65.86986	3.82E-18	3.101296
Within Groups	2056.56667	87	23.6387			
Total	5170.72222	89				

Excel output of ANOVA for this data

- What would happen if instead of using ANOVA to compare 7 groups , you performed multiple t-tests?
 - a) Making multiple comparisons with a t-test increases the probability of making a Type-1 error.
 - b) Sir Ronald Fischer would be turning over in his grave; He put all that work into developing ANOVA and you used multiple t-tests 🤖
 - c) Nothing apart from making multiple comparisons with a t-test requires more computation than ANOVA
 - d) Nothing , both are the same.

Solution

- a) **Making multiple comparisons with a t-test increases the probability of making a Type-1 error.**
- What kind of a hypothesis test is used for one-way ANOVA?

Solution

Right-tailed test

- For an experiment with a single factor of k levels with n observations, the degrees of freedom for sum of squares of variation within the group is ?
- a) $n - 1$
 - b) $k - 1$
 - c) $n - k + 1$
 - d) $n - k$

Solution

- d) $n - k$

- An investigator used ANOVA to compare 4 groups of students on numerical ability based on a class test. After analysis of raw scores, the following results were obtained. Calculate the value of the F-statistic

Source of Variation	DoF	Sum of squares
Between groups	3	625
Within Groups	36	2128

- a) 3.4
- b) 3.52
- c) 3.88
- d) 3.97

Solution:

3.52

- Why ANOVA and what issue of the multiple T-test does it address?
- Mean model
- Factors effect model
- Setting up 1 way ANOVA:
 - Appropriate conditions where 1-way ANOVA is applicable
 - Understanding all the variables and subscripts used
 - Deriving SST,SSB and SSW(corresponding MST,MSB and MSW based on DoF)
 - Cochran's theorem
 - F-statistic for ANOVA
 - Finally, when to accept and reject the null hypothesis(based on calculated F value and critical F-value)

- Business Analytics by U. Dinesh Kumar – Wiley 2nd Edition, 2022
Chapter : 7.1 - 7.3.3
- <https://www.analyticsvidhya.com/blog/2018/01/anova-analysis-of-variance/>
- <https://www.analyticsvidhya.com/blog/2020/06/introduction-anova-statistics-data-science-covid-python/>
- <https://www.geeksforgeeks.org/one-way-anova/>



THANK YOU

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DATA ANALYTICS

UE24CS342AA2

UNIT-1

Lecture 08: Analysis of Variance - 2

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 08: Analysis of Variance - 2

Slides excerpted from: U. Dinesh Kumar,
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With grateful thanks for contribution of slides to:

Dr. Mamatha H R, Professor at the Department of CSE, PESU

One-Way ANOVA : Example (Observational Study)

Share Raja Khan (SRK) is a top stockbroker and believes that the average annual stock return depends on the industrial sector. To validate his belief, SRK collected annual return of shares from three different industrial sectors – consumer goods, services, and industrial goods. The annual return of shares in 2015–2016 for different sectors is shown below. Conduct an ANOVA test at 5% significance level.

Annual return on 30 consumer goods stocks

6.32%	14.73%	11.95%	12.36%	10.28%	3.81%	10.15%	11.06%	6.29%	5.15%
8.44%	14.28%	8.89%	5.98%	6.96%	11.62%	5.22%	5.34%	5.93%	7.10%
10.91%	8.20%	10.19%	9.04%	8.61%	9.39%	2.63%	2.77%	4.76%	9.60%

Annual return on 30 services stocks

13.70%	3.58%	1.36%	17.41%	10.01%	10.88%	15.63%	−0.04%	10.32%	7.40%
11.48%	9.71%	11.19%	8.21%	1.64%	1.45%	10.12%	13.85%	−10.27%	5.26%
12.05%	4.47%	8.71%	5.59%	10.02%	7.65%	10.03%	7.87%	6.59%	13.60%

Annual return on 30 industrial goods stocks

6.74%	7.11%	5.69%	2.48%	5.42%	8.00%	2.55%	8.34%	4.99%	3.39%
8.73%	13.85%	5.29%	9.06%	2.84%	5.82%	7.66%	4.12%	9.10%	8.76%
10.77%	1.48%	4.71%	10.66%	0.44%	2.94%	6.55%	2.84%	3.90%	7.28%

- In this case, the number of cases $k = 3$; $n_1 = n_2 = n_3 = 30$; $\mu_1 = 0.082$, $\mu_2 = 0.079$, $\mu_3 = 0.0605$; and $\mu = 0.0743$
- The sum of squares of between (*SSB*) groups variation is given by

$$SSB = \sum_{i=1}^k n_i \times (\mu_i - \mu)^2 = 30 \times [(0.082 - 0.0743)^2 + (0.079 - 0.0743)^2 + (0.0605 - 0.0743)^2]$$
$$= 0.0087$$

- Therefore

$$MSB = \frac{SSB}{k - 1} = \frac{0.0087}{2} = 0.0043$$

- The sum of squares of within the group variation is given by

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \mu_i)^2 = \sum_{j=1}^{30} (Y_{1j} - 0.082)^2 + \sum_{j=1}^{30} (Y_{2j} - 0.079)^2 + \sum_{j=1}^{30} (Y_{3j} - 0.0605)^2$$
$$= 0.1463$$

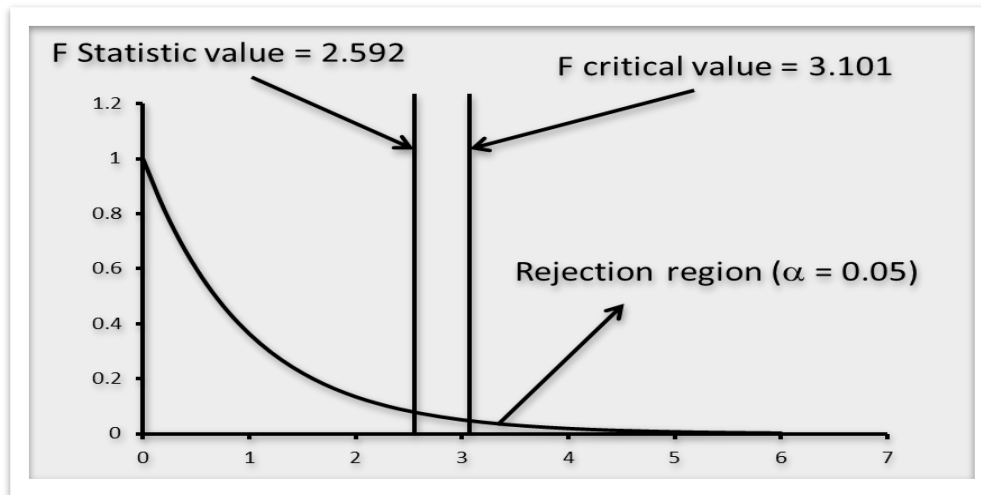
- Therefore

$$MSW = \frac{SSW}{n - k} = \frac{0.1463}{90 - 3} = 0.0016$$

- The F-statistic value is

$$F_{2,87} = \frac{MSB}{MSW} = \frac{0.0043}{0.0016} = 2.592$$

- The critical F -value with degrees of freedom (2, 87) for $\alpha = 0.05$ is 3.101
- The p -value for $F_{2,87} = 2.592$ is 0.0805
- Since the calculated F -statistic is less than the critical F -value, we retain the null hypothesis and conclude that the average annual returns under industrial sectors consumer goods, services, and industrial goods are not different.



F-distribution with critical value

ANOVA: Single Factor

SUMMARY				
Groups	Count	Sum	Average	Variance
Consumer Goods	30	2.4796	0.082653	0.00101
Services	30	2.3947	0.079823	0.003073
Industrial Goods	30	1.8151	0.060503	0.000963

ANOVA

Source of Variation	SS	df	MS	F	p-value	F critical
Between Groups	0.008722	2	0.004361	2.59294	0.080572	3.101296
Within Groups	0.146317	87	0.001682			
Total	0.155039	89				

Excel output of ANOVA for this data

- The values of response variable may be influenced by several factors. For example, in addition to price discounts, location of the stores may also play an important role in the sales quantity.
- The discounts may not have much impact if the store is located near affluent community compared to stores located near non-affluent community.
- We would like to understand the impact of both factors (price discount and location) simultaneously on sales by trying to answer to the following questions:
 - Are there differences in the average sales quantity with different levels of price discounts?
 - Are there differences in the average sales quantity with respect to different locations?
 - Are there interactions between price discounts and location with respect to average sales quantity?

The two-way ANOVA model can be expressed as

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \alpha_i\beta_j + \varepsilon_{ijk}$$

Where,

Y_{ijk} = Value of the k^{th} observation ($k = 1, 2, \dots, K$) of the response variable at level i ($i = 1, 2, \dots, a$) of factor A and level j ($j = 1, 2, \dots, b$) of factor B.

μ = Overall mean value of the response variable Y_{ijk}

α_i = Level (effect) of factor A ($i = 1, 2, \dots, a$)

β_j = Level (effect) of factor B ($j = 1, 2, \dots, b$)

$\alpha_i\beta_j$ = Interaction of i^{th} level of factor A and j^{th} level of factor B

ε_{ijk} = Error associated with k^{th} of observation at level i of factor A and level j of factor B.

The hypothesis tests associated with two-way ANOVA are as follows:

Test of Factor A Main Effects:

$H_0: \alpha_i = 0$ for all i ($i = 1, 2, \dots, a$)

H_A : Not all α_i are zero

Test of Factor B Main Effects:

$H_0: \beta_j = 0$ for all j ($j = 1, 2, \dots, b$)

H_A : Not all β_j are zero

Test of Interaction Effects:

$H_0: \alpha_i\beta_j = 0$ for all i ($i = 1, 2, \dots, a$) and j ($j = 1, 2, \dots, b$)

H_A : Not all $\alpha_i\beta_j$ are zero

The sum of squares in the case of two-way ANOVA with equal sample sizes is given by

$$SST = SSA + SSB + SSAB + SSW$$

Various components in the above equation are provided as follows :

- ***Sum of squared of total deviation (SST):***

$$SST = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c (Y_{ijk} - \mu)^2$$

where c is the number of observations in each group and μ is the overall mean.

- ***Sum of squares of deviation due to factor A (SSA):***

$$SSA = b \times c \times \sum_{i=1}^a (\mu_i - \mu)^2$$

where μ_i is the mean of all observations in level i of factor A and c is the number of observations in each group (assumed to be same for all groups).

- ***Sum of squares of deviation due to factor B (SSB):***

$$SSB = a \times c \times \sum_{j=1}^b (\mu_j - \mu)^2$$

Here μ_j is the mean of all observations in level j of factor B.

- ***Sum of squares of deviation due to interaction of factors A and B (SSAB)***

$$SSAB = c \times \sum_{i=1}^a \sum_{j=1}^b (\mu_{ij} - \mu_i - \mu_j + \mu)^2$$

where μ_{ij} is the average of i^{th} level of factor A and j^{th} level of factor B.

- ***Sum of squares of deviation within a group (SSW):***

$$SSW = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c (Y_{ijk} - \mu_{ij})^2$$

Setting up Two-Way ANOVA

Sum of squares of deviation for various effects and the corresponding F -statistic in a two-way ANOVA with equal sample size

Sum of Squared Variation	Degrees of Freedom	Mean Squared Variation	F-Statistics
SSA	$a - 1$	$MSA = SSA / (a - 1)$	$F = MSA / MSW$
SSB	$b - 1$	$MSB = SSB / (b - 1)$	$F = MSB / MSW$
SSAB	$(a - 1)(b - 1)$	$MSAB = SSAB / (a - 1)(b - 1)$	$F = MSAB / MSW$
SSW	$ab(c - 1)$	$MSW = SSW / ab(c - 1)$	

The table next slide shows the sales quantity of detergents at different discount values and different locations collected over 20 days. Conduct a two-way ANOVA at $\alpha = 0.05$ to test the effects of discounts and location on the sales.

Location 1			Location 2		
Discount			Discount		
0%	10%	20%	0%	10%	20%
20	28	32	20	19	20
16	23	29	21	27	31
24	25	28	23	23	35
20	31	27	19	30	25
19	25	30	25	25	31
10	24	26	22	21	31
24	28	37	25	33	31
16	23	33	21	26	23
25	26	27	26	22	22
16	25	31	22	28	32
18	22	37	25	24	22
20	24	28	23	23	29
17	26	25	23	26	25
26	28	23	24	16	34
16	21	26	20	30	30
21	27	33	23	22	25
24	25	28	18	16	39
19	20	30	19	25	32
19	26	30	19	34	29
21	26	26	30	23	22



The two-way ANOVA with replication (since the data is repeated for locations) output from Microsoft Excel is shown

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Sample (Location)	7.008333	1	7.008333	0.443898	0.506593	3.92433
Columns (Discount)	1240.317	2	620.1583	39.27997	1.06E-13	3.075853
Interaction	84.81667	2	42.40833	2.686085	0.07246	3.075853
Within	1799.85	114	15.78816			
Total	3131.992	119				

- In the table , the sample stands for the row factor (which in this case is location), column stands for the column factor (discount in this case), and interaction stands for interaction effect (location $\times$ discount).
- The p -value for locations (data in rows) is 0.5065, thus it is not statistically significant (we retain the null hypothesis that the locations have no statistical influence on sales), whereas for discount rates (data in column) the p -value is 1.06×10^{-13} , so we reject the null hypothesis (that is discount rate has influence on sales).
- The p -value for the interaction effect is 0.0724 and is not significant. That is only the factor discount is statistically significant at $\alpha = 0.05$.

- Analysis of Variance (ANOVA) is a hypothesis testing procedure used for comparing means from several groups simultaneously.
- In one-way ANOVA, we test whether the mean values of an outcome variable for different levels of factor are different. Using multiple two sample t-test to simultaneously test group means will result in incorrect estimation of Type- I error and ANOVA overcomes this problem.
- ANOVA plays an important role in multiple linear regression model diagnostics. The overall significance of the model is tested using ANOVA.
- In a two-way ANOVA we check the impact of more than one factor simultaneously on several groups.

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Test your understanding!

- A two-way ANOVA means that the experimental design includes
 - a) Two dependent variables
 - b) Two independent variables
 - c) Two types of variance
 - d) All of these

Solution

- b) **Two independent variables.**



- In a two-way ANOVA, the interaction effect is the
 - a) Effect of changing the levels of one factor on the dependent scores
 - b) Effect of changing the levels of one factor on the dependent scores, ignoring all other factors in the study
 - c) Extent to which the influence one factor has on scores depends on the level of the other factor
 - d) Effect on the independent variables of changing the levels of a factor

Solution

- c) **Extent to which the influence one factor has on scores depends on the level of the other factor**

- Why 2-way ANOVA(2 factors considered)
- Setting up 2-way ANOVA:
 - Understanding all the variables and subscripts used
 - Test of Factor A(Main effect)
 - Test of Factor B(Main effect)
 - Interaction effect
 - SST,SSA,SSB,SSAB,SSW and corresponding DoF's
 - F-statistics
 - Finally, when to accept and reject the null hypothesis(based on calculated F value and critical F-value and p value)

- Business Analytics by U. Dinesh Kumar – Wiley 2nd Edition, 2022
Chapter : 7.4
- <https://www.analyticsvidhya.com/blog/2018/01/anova-analysis-of-variance/>
- <https://www.analyticsvidhya.com/blog/2020/06/introduction-anova-statistics-data-science-covid-python/>



THANK YOU

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DATA ANALYTICS

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UNIT-1

Lecture 09: Correlation Analysis – 1

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 09: Correlation Analysis - 1

Slides excerpted from: U. Dinesh Kumar,
“Business Analytics”, Wiley, 2nd Edition 2022

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With grateful thanks for contribution of slides to:

Dr. Mamatha H R, Professor at the Department of CSE, PESU

- Covariance is similar to correlation

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

Correlation coefficient: $r_{A,B} = \frac{Cov(A, B)}{\sigma_A \sigma_B}$

Where n is the number of tuples, $\bar{A}$ and $\bar{B}$ are the respective mean or *expected values* of A and B , σ_A and σ_B are the respective standard deviations of A and B .

- **Positive Covariance** : If $\text{Cov}_{A,B} > 0$, then A and B both tend to be larger than their expected values.
- **Negative Covariance** : If $\text{Cov}_{A,B} < 0$, then if A is larger than its expected value, B is likely to be smaller than its expected value.
- **Independence** : $\text{Cov}_{A,B} = 0$, but the converse is not true :

Some pairs of random variables may have a covariance of 0 but are not independent. Only under few additional assumptions (example , the data follows multivariate normal distributions) does $\text{Cov}_{A,B} = 0$ imply independence.

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

- It can be simplified in computation as

$$Cov(A, B) = E(A \cdot B) - \bar{A}\bar{B}$$

Suppose two stocks A and B have the following values in one week : (2, 5), (3, 8), (5, 10), (4, 11), (6, 14).

Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?

$$E(A) = (2 + 3 + 5 + 4 + 6) / 5 = 20/5 = 4$$

$$E(B) = (5 + 8 + 10 + 11 + 14) / 5 = 48/5 = 9.6$$

$$\text{Cov}(A,B) = (2 \times 5 + 3 \times 8 + 5 \times 10 + 4 \times 11 + 6 \times 14) / 5 - 4 \times 9.6 = 4$$

Thus, A and B rise together since $\text{Cov}(A, B) > 0$.

- Organizations collect data on several variables and generate several more through feature engineering which may bring its count to thousands.
- One of the most challenging tasks in predictive analytics is identifying features that may be associated with the response or the outcome variable of interest.
- It is important to reduce the number of features to improve the time taken for analysis as well as ensure a model isn't destabilised due to excess features.

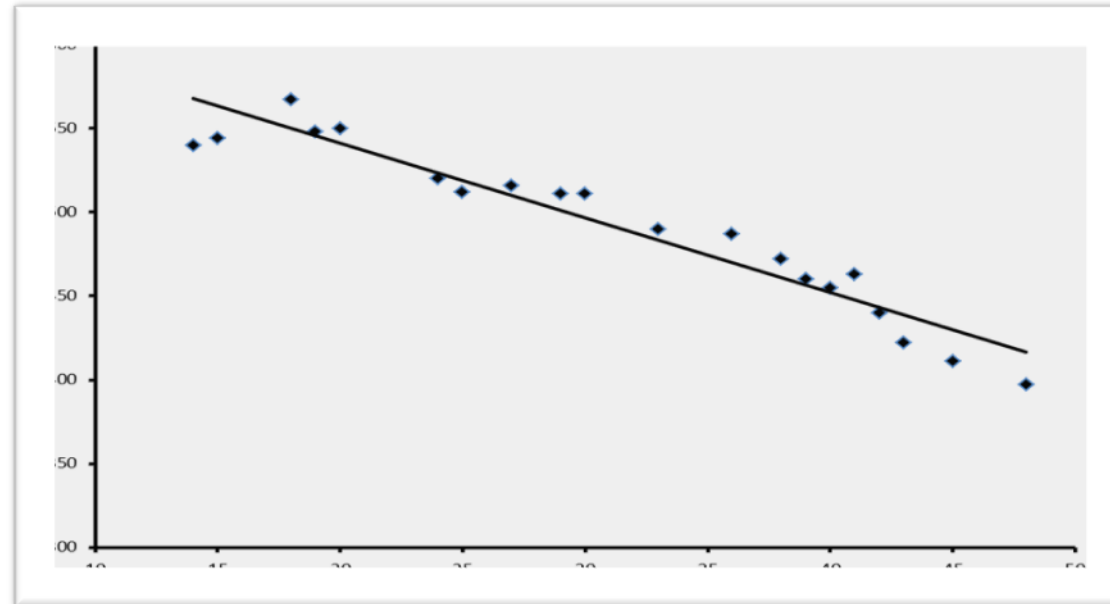
- **Correlation** is a statistical measure of an associative relationship that exists between two random variables.
- It is a measure of *strength and direction* of a relationship that exists between two random variables and is measured using a correlation coefficient.
- It is not necessarily a causal relationship.
- Correlation is important in analytics as it helps to identify variables that may be used in the model building and also useful for identifying issues such as multi-collinearity that can destabilise regression-based models.
- It is also useful for finding proxy variables in analytics model building.

- **Pearson product moment correlation**(in short Pearson correlation) is used for measuring the **strength** and **direction** of the **linear** relationship between two **continuous random variables**.

Consider the following data

Age	14	15	18	19	20	24	25	27	29	30
Call Duration	540	544	567	548	550	520	512	516	511	511
Age	33	36	38	39	40	41	42	43	45	48
Call Duration	490	487	472	460	455	463	440	422	411	397

In the graph , we can see that the average call duration (Y) decreases as the age of the customer (X) increases. We can measure the strength of the linear association relationship using a numerical measure called correlation coefficient.



Association relationship between age and average call duration

Calculating Pearson Correlation Coefficient

- To calculate Pearson correlation coefficient , the variables must belong to *ratio* or *interval* scale. The range of the variables can be different, thus we need to standardize the variables which then can be used to find the correlation.
- Let X_i be different values of the variable X and Y_i be different values of Y . Then the standardized values of X and Y are given by

$$Z_X = \left(\frac{X_i - \bar{X}}{\sigma_X} \right) \quad Z_Y = \left(\frac{Y_i - \bar{Y}}{\sigma_Y} \right)$$

Where $\bar{X}$ and $\bar{Y}$ are mean values of the random variables X and Y . σ_X and σ_Y are the corresponding standard deviations.

Calculating Pearson Correlation Coefficient

The Pearson's correlation coefficient is given by

$$r = \frac{\sum_{i=1}^n Z_X Z_Y}{n} = \frac{\sum_{i=1}^n (X_i - \bar{X}) \times (Y_i - \bar{Y})}{n\sigma_X\sigma_Y}$$

Where n is the number of records in the population. In case of working with the sample , to account for the degrees of freedom , the following formula is recommended

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X}) \times (Y_i - \bar{Y})}{(n - 1)S_X S_Y}$$

For large number of samples , the correlation coefficients calculated by both these equations converge.

- Whenever the value of X_i is greater than mean and if the corresponding value of Y_i is also greater than mean, then the numerator in equation will be positive.
- Whenever the value of X_i is lesser than mean and if the corresponding value of Y_i is also lesser than mean, then the numerator in equation will be positive.
- Whenever the value of X_i is lesser than mean (or greater than mean) and the corresponding value of Y_i is greater than mean (or lesser than mean), then the numerator in equation will be negative.
- It is possible that we may have combinations of three cases listed above in a data set. Thus the numerator in the equation is likely to be positive, negative, or zero.

- $$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \times \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$
- $$r = \frac{n \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{\sqrt{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} \times \sqrt{n \sum_{i=1}^n Y_i^2 - (\sum_{i=1}^n Y_i)^2}}$$
- $$r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Where $Cov(X,Y)$ is the covariance between the random variables X and Y and is given by

$$Cov(X,Y) = E\left(\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)\right)$$

- The value of correlation coefficient lies between -1 and $+1$. High absolute value of r , $|r|$, indicates strong relationship between the two variables.
- Positive value of r indicates positive correlation (as value of X increases, the value of Y also increases) and negative value of r indicates negative correlation (as the value of X increases, the value of Y decreases).
- The sign of correlation coefficient is same as the sign of covariance between the two random variables.

- Assume that the value of Pearson correlation coefficient between X and Y is r . Let Z_1 and Z_2 be the linear combinations of X and Y ($Z_1 = A + BX$ and $Z_2 = C + DY$). Then the correlation coefficient between Z_1 and Z_2 will be r when the signs of B and D are same (both are positive or negative) and $-r$ when the signs of B and D are opposite.
- Mathematically, square of correlation coefficient is equal to the co-efficient of determination (R^2) of the linear regression model, that is $r^2 = R^2$.
- Pearson correlation coefficient value may be zero even when there is a strong non-linear relationship between variables X and Y . Thus low correlation coefficient value cannot be taken as an evidence of no relationship.

Example

The average share prices of two companies over the past 12 months are shown below. Calculate the Pearson Correlation coefficient.

X	Y
274.58	219.50
287.96	242.92
290.35	245.90
320.07	256.80
317.40	240.60
319.53	245.23
301.52	232.09
271.75	222.65
323.65	231.74
259.80	214.43
263.02	201.86
286.03	204.23

The average values are :

$$\bar{X} = 292.9717 \qquad \bar{Y} = 229.8292$$

Correlation coefficient is calculated using the following equation:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \times \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

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Example



X_i	Y_i			XY	X^2	Y^2
274.58	219.50	-18.39	-10.33	189.97	338.25	106.6917
287.96	242.92	-5.01	13.09	-65.61	25.12	171.3699
290.35	245.90	-2.62	16.07	-42.13	6.87	258.2717
320.07	256.80	27.10	26.97	730.86	734.32	727.4259
317.40	240.60	24.43	10.77	263.11	596.74	116.0109
319.53	245.23	26.56	15.40	409.02	705.35	237.1857
301.52	232.09	8.55	2.26	19.33	73.07	5.111367
271.75	222.65	-21.22	-7.18	152.35	450.36	51.54043
323.65	231.74	30.68	1.91	58.62	941.16	3.651284
259.80	214.43	-33.17	-15.40	510.82	1100.36	237.1343
263.02	201.86	-29.95	-27.97	837.72	897.10	782.2743
286.03	204.23	-6.94	-25.60	177.70	48.19	655.3173
Sum				3241.77	5916.89	3351.98

Example



From the table , we have

$$\bullet \sum_{i=1}^{12} (X_i - \bar{X})(Y_i - \bar{Y}) = 3241.77$$

$$\bullet \sum_{i=1}^{12} (X_i - \bar{X})^2 = 5916.89$$

$$\bullet \sum_{i=1}^{12} (Y_i - \bar{Y})^2 = 3351.98$$

$$\text{Correlation coefficient } r = \frac{3241.77}{\sqrt{5916.89} \times \sqrt{3351.98}} = 0.7279$$

Hypothesis test for Correlation Coefficient

For any two sets of data , the Pearson correlation coefficient is most likely to give a value other than zero. Many thumb rules exist to group the correlation value as no correlation, low correlation, medium correlation, and high correlation (Monroe and Stuit, 1933).

Let ρ be the population correlation coefficient. The null and alternative hypotheses are given by

$H_0:$	$\rho = 0$ (there is no correlation between two random variables)
$H_A:$	$\rho \neq 0$ (there is a correlation between two random variables)

Hypothesis test for Correlation Coefficient



- The sampling distribution of correlation coefficient r follows an approximate t -distribution with $(n - 2)$ degrees of freedom (dof) where n is the number of cases in the sample for calculating the correlation coefficient.

- Two degrees of freedom are lost since we estimate two mean values from the data. The mean of the sampling distribution is ρ and the corresponding standard deviation is (Ezekiel,

$$1941) \sqrt{\frac{1-r^2}{n-2}}$$

- The t -statistic for null hypothesis is given by $t_{\alpha, n-2} = \frac{r-\rho}{\sqrt{\frac{1-r^2}{n-2}}}$

- When the null hypothesis is $\rho = 0$, the test statistic in above equation becomes $t_{\alpha, n-2} =$

$$r \sqrt{\frac{n-2}{1-r^2}}$$

Hypothesis test for Correlation Coefficient - Example

The average share prices of two companies over the past 12 months are shown in Table. conduct the following two hypothesis tests at $\alpha = 0.05$:

- (a) The correlation between share prices of two companies is zero.
- (b) The correlation between share prices of two companies is at least 0.5.

X	Y
274.58	219.50
287.96	242.92
290.35	245.90
320.07	256.80
317.40	240.60
319.53	245.23
301.52	232.09
271.75	222.65
323.65	231.74
259.80	214.43
263.02	201.86
286.03	204.23

(a) The null and alternative hypotheses are:

$$H_0: \rho = 0$$

$$H_A: \rho \neq 0$$

The corresponding t -statistic is

$$t = r \sqrt{\frac{n-2}{1-r^2}} = 0.7279 \sqrt{\frac{12-2}{1-0.7279^2}} = 3.3569$$

Note that this is a two-tailed test and the critical t -value at $\alpha = 0.05$ and $df = 10$ is 2.2281.

Since the calculated t -statistic is higher than the critical t -value, we reject the null hypothesis and conclude that there is a significant correlation between share prices of two companies.

The corresponding p -value is 0.0072.

(b) The null and alternative hypotheses are given by

$$H_0: \rho \leq 0.5$$

$$H_A: \rho > 0.5$$

The corresponding t -statistic is

$$t = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}} = \frac{0.7279 - 0.5}{0.2168} = 1.05$$

This is a right-tailed test and the corresponding t -critical value is 1.8124.

The calculated t -value is less than the critical value of t , and thus we retain the null hypothesis and conclude that the correlation between share prices of two companies is less than 0.5.

The corresponding p -value is 0.1592.

- Pearson correlation is applicable to what type of attribute(s)?

Solution

Attributes from ratio scale or interval scale

- What is the range of Pearson correlation coefficient?

Solution

[-1,1]

- Pearson correlation between 2 variables X and Y is 0.85. What is the Pearson correlation between $3X-100$ and $400-10Y$?

Solution

-0.85

- Business Analytics by U. Dinesh Kumar – Wiley 2nd Edition, 2022
Chapters : 8.1 – 8.2.4



THANK YOU

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DATA ANALYTICS

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UNIT-1

Lecture 10: Correlation Analysis – 2

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Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 10: Correlation Analysis – 2

Slides excerpted from: U. Dinesh Kumar,
“Business Analytics”, Wiley, 2nd Edition 2022

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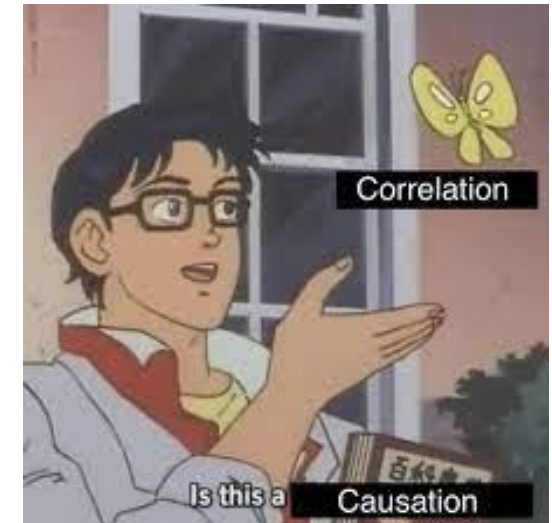
With grateful thanks for contribution of slides to:

Dr. Mamatha H R, Professor at the Department of CSE, PESU

- One of the major problem with correlation is the possibility of spurious correlation between two random variables which in many cases is caused due to some other latent variable (hidden variable) that influences both variables for which the correlation is calculated.

Following are few examples of spurious correlation between two random variables:

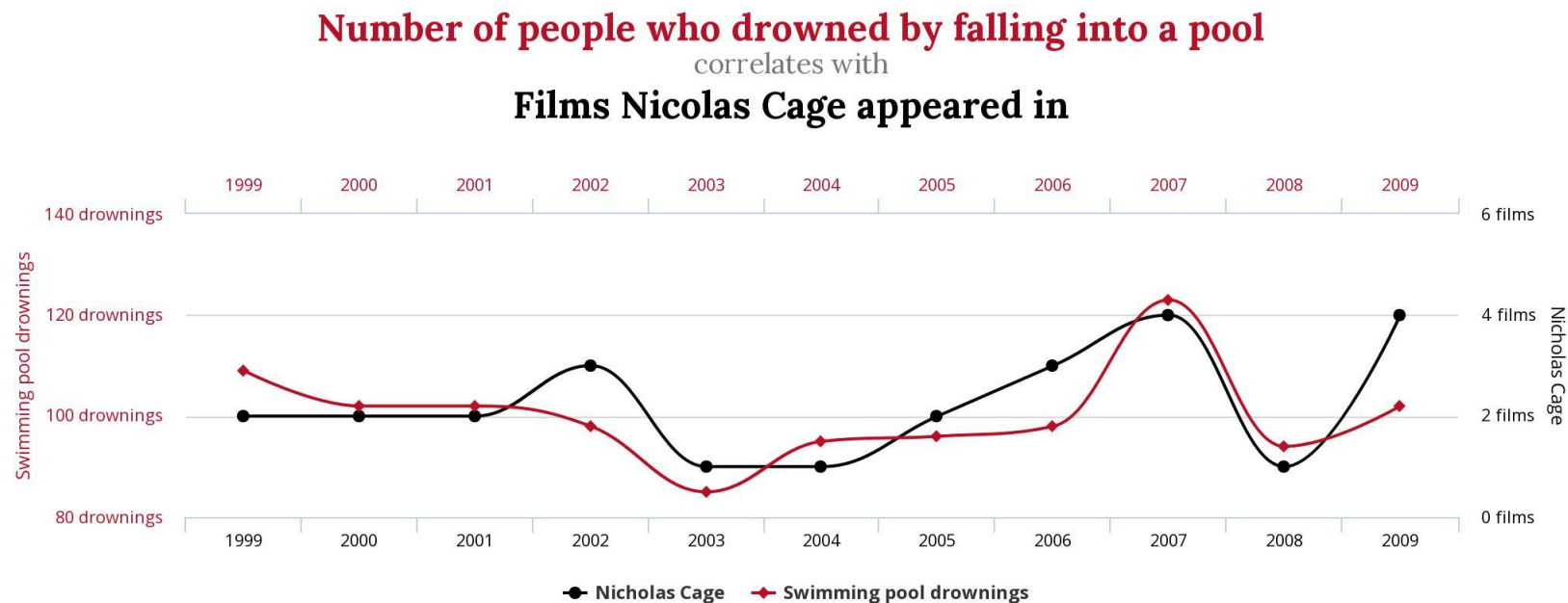
- **Crime rate versus ice cream sale:** It has been reported that the sale of ice cream and crime rates are positively correlated (Levitt and Dubner, 2009). Obviously, ice cream is not driving the crime rate. In this case the hidden variable is the temperature – Ice cream sales increase during the summer and since summer is the time of vacation , many homes are left empty , susceptible for theft thus driving the crime rates up.



DATA ANALYTICS

Spurious Correlation

- **Doctors and deaths:** Number of doctors is positively correlated with number of deaths in villages, that is, as the number of doctors increases, the deaths also increase. We can be sure that doctors are not causing the deaths to increase (Young, 2001). In this case, the need for doctors is more in villages with poor public healthcare.

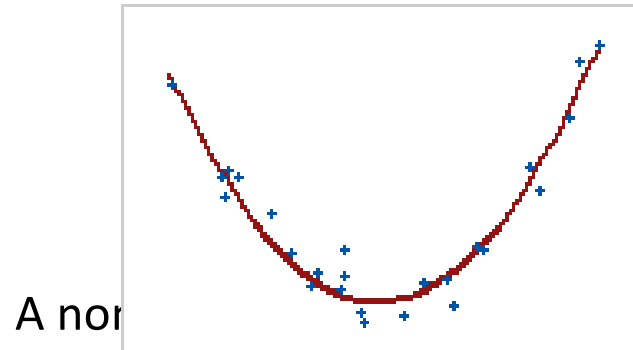


Should Nicolas Cage end his career to save lives?



Pearson Correlation with a non-linear relationship

- Pearson correlation finds a linear relationship between 2 random variables. If the resultant correlation is less between two variables, one cannot conclude the variables aren't correlated.
- For example, the correlation for a set of data turned out to be only 0.224. But on plotting a scatter plot, it was observed that the curve could be better modelled using a non-linear function like a quadratic or a cubic function. Thus it is always advisable to validate your results by visualizing the data.



<https://support.minitab.com/en-us/minitab/19/help-and-how-to/statistics/basic-statistics/supporting-topics/basics/linear-nonlinear-and-monotonic-relationships/>

- Pearson correlation is appropriate when the random variables involved are both from either ratio scale or interval scale.
- When both random variables are of ordinal scale, we use Spearman rank correlation (also known as Spearman's rho denoted by ρ_s).
- The Spearman rank correlation, r_s , estimated from a sample is given by (Yule and Kendall 1937, Woodbury, 1940)

$$r_s = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)}$$

where D_i = difference in the rank of case i under variables X and Y (that is $X_i - Y_i$).

- The sampling distribution of Spearman correlation r_s also follows an approximate t -distribution with mean ρ_s and standard deviation $\sqrt{\frac{1-r_s^2}{n-2}}$ with $n - 2$ degrees of freedom

Ranking of 12 countries under corruption and Gini Index (wealth discrimination) are shown in the table. Calculate the Spearman correlation and test the hypothesis that the correlation is at least 0.2 at $\alpha = 0.02$.

Countries	1	2	3	4	5	6	7	8	9	10	11	12
Corruption	1	4	12	2	5	8	11	7	10	3	6	9
Gini Index	2	3	9	5	4	6	10	7	8	1	11	12

The calculations are as follows

Country	Corruption Rank (X_i)	Gini Index (Y_i)	$D = X_i - Y_i$	D^2
1	1	2	-1	1
2	4	3	1	1
3	12	9	3	9
4	2	5	-3	9
5	5	4	1	1
6	8	6	2	4
7	11	10	1	1
8	7	7	0	0
9	10	8	2	4
10	3	1	2	4
11	6	11	-5	25
12	9	12	-3	9
$\sum_{i=1}^{12} D_i^2$				68

- The Spearman rank correlation is given by

$$r_s = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 68}{12(12^2 - 1)} = 0.7622$$

- The null and alternative hypotheses are

$$H_0: \rho_s \leq 0.2$$

$$H_A: \rho_s > 0.2$$

The corresponding t -statistic is

$$t = \frac{r_s - \rho_s}{\sqrt{\frac{1 - r_s^2}{n - 2}}} = \frac{0.7622 - 0.2}{0.2046} = 2.74$$

- The one-tailed t -critical value for $\alpha = 0.02$ and $dof = 10$ is 2.35. Since the calculated t -statistic value is more than the t -critical value, we reject the null hypothesis and conclude that Spearman rank correlation between two countries is at least 0.2.

- **Point Bi-Serial correlation** is used when we are interested in finding correlation between a **continuous random variable** and a **dichotomous (binary) random variable**.
- Assume that the random variable X is a continuous random variable and Y is a dichotomous random variable. Then the following steps are used for calculating the correlation between these two variables:
 1. Group the data into two sets based on the value of the dichotomous variable Y . That is, assume that the value of Y is either 0 or 1. Then we group the data into two subsets such that in one group the value of Y is 0 and in another group the value of Y is 1.
 2. Calculate the mean values of two groups: Let $\bar{X}_0$ and $\bar{X}_1$ be the mean values of groups with $Y = 0$ and $Y = 1$, respectively.
 3. Let n_0 and n_1 be the number of cases in a group with $Y = 0$ and $Y = 1$, respectively, and S_x be the standard deviation of the random variable X .

The point bi-serial correlation is given by (Pearson, 1909 and Soper, 1914)

$$r_b = \frac{\bar{X}_1 - \bar{X}_0}{S_X} \sqrt{\frac{n_0 n_1}{n(n-1)}}$$

where n is the total number of cases in the sample and S_X is the standard deviation of X estimated from sample and is given by

$$S_X = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Example

Ms. Sandra Ruth, data scientist at Airmobile, is interested in finding the correlation between the average call duration and sex of a person. The table provides the average call duration (measured in seconds) and sex of 30 customers of Airmobile. In the table, male is coded as 0 and Female is coded as 1. Calculate the point bi-serial correlation.

Sex	1	1	0	0	0	1	0	1	1	0
Call Duration	448	335	210	382	407	231	359	287	288	347
Sex	1	1	1	1	1	0	0	1	0	0
Call Duration	408	382	303	201	447	439	383	277	279	213
Sex	1	1	0	1	1	0	1	0	1	0
Call Duration	383	355	362	401	331	421	367	437	326	351

From the data we can calculate the following

$$\bar{X} = 345.33$$

$$\bar{X}_0 = 353.07$$

$$\bar{X}_1 = 339.4118$$

$$S_X = 71.7189$$

$$n_0 = 13$$

$$n_1 = 17$$

Point Bi-serial correlation is given by

$$r_b = \frac{\bar{X}_1 - \bar{X}_0}{S_X} \sqrt{\frac{n_0 n_1}{n(n-1)}} = \frac{339.4118 - 353.07}{71.7189} \sqrt{\frac{13 \times 17}{30(29)}} = -0.0960$$

There is a very low negative correlation between sex of a person and call duration.

Karl Pearson recommended the use the **Phi-coefficient** when **both variables are binary** for calculating the association relationship (Cramer, 1946). Let X and Y be two random variables both taking binary values (that is, X takes values 0 or 1 and similarly Y also takes values either 0 or 1). One can create a contingency table as shown in table below.

	Y = 0	Y = 1	Total
X = 0	N_{00}	N_{01}	$N_{X0} = N_{00} + N_{01}$
X = 1	N_{10}	N_{11}	$N_{X1} = N_{10} + N_{11}$
Total	$N_{Y0} = N_{00} + N_{10}$	$N_{Y1} = N_{01} + N_{11}$	

Contingency table for presence or absence of two categorical variables

	Drink Coffee	Don't drink Coffee
Drink Tea	N_{11}	N_{10}
Don't drink Tea	N_{01}	N_{00}

In the contingency table (Table in previous slide)

- N_{00} = Number of cases in the sample such that $X = 0$ and $Y = 0$
- N_{01} = Number of cases in the sample such that $X = 0$ and $Y = 1$
- N_{10} = Number of cases in the sample such that $X = 1$ and $Y = 0$
- N_{11} = Number of cases in the sample such that $X = 1$ and $Y = 1$
- N_{X0} = Number of cases in the sample such that $X = 0$
- N_{X1} = Number of cases in the sample such that $X = 1$
- N_{Y0} = Number of cases in the sample such that $Y = 0$
- N_{Y1} = Number of cases in the sample such that $Y = 1$
- The Phi-coefficient is given by

$$\varphi = \frac{N_{11}N_{00} - N_{10}N_{01}}{\sqrt{N_{X0}N_{X1}N_{Y0}N_{Y1}}}$$

Example

Joy Finance (JF) is a company that provides gold loans (in which gold is used as guarantee against the loan). Mr Georgekutty, Managing Director of JF, collected data to understand the relationship between loan default status (variable Y) and the marital status of the customer (variable X). Data is collected on past 40 loans and is shown in the table. Calculate the Phi-coefficient. In the table, $Y = 0$ implies non-defaulter, $Y = 1$ is defaulter, $X = 0$ is single, and $X = 1$ is married.

X	1	0	1	0	0	0	0	0	1	0
Y	0	1	0	1	0	0	0	1	1	1
X	0	1	1	0	0	1	0	0	0	1
Y	0	1	1	1	0	0	1	1	0	0
X	1	0	0	0	1	1	1	0	0	1
Y	0	0	0	1	0	0	0	0	0	0
X	1	0	0	0	1	0	1	0	1	1
Y	1	0	0	1	1	0	1	1	0	1

The resultant contingency table is as follows

From the table

$$N_{00} = 13, N_{01} = 10, N_{10} = 10, N_{11} = 7$$

$$N_{X0} = 23, N_{X1} = 17, N_{Y0} = 23, N_{Y1} = 17$$

	Y			Total
		0	1	
X	0	13	10	23
	1	10	7	17
Total		23	17	40

The Phi-coefficient is given by

$$\phi = \frac{N_{11}N_{00} - N_{10}N_{01}}{\sqrt{N_{X0}N_{X1}N_{Y0}N_{Y1}}} = \frac{7 \times 13 - 10 \times 10}{\sqrt{23 \times 17 \times 23 \times 17}} = -0.0230$$

Since the Phi-coefficient is very small, we conclude that there is not much association between the marital status and loan default.

- In a study conducted, it was found that the number of suicides by hanging, strangulation or suffocation showed a positive increase with increase in US spending on science, space and technology. Does this mean that the two are related?
 - a) Yes, positive correlation means that the two variables are related
 - b) No, the variables are only related if there is negative correlation
 - c) No, this is an instance of spurious correlation
 - d) The information in this statement is not sufficient to arrive at a conclusion

Solution

c) No, this is an instance of spurious correlation

- Which correlation measure is applicable to data which is in ordinal, ratio or interval scale?

Solution

Spearman Rank Correlation

- Business Analytics by U. Dinesh Kumar – Wiley 2nd Edition, 2022
Chapters : 8.3 – 8.5



THANK YOU

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DATA ANALYTICS

UE24CS342AA2

UNIT-1

Lecture 11: Hidden Variables and Confounding Variables

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 11 : Hidden and confounding variables

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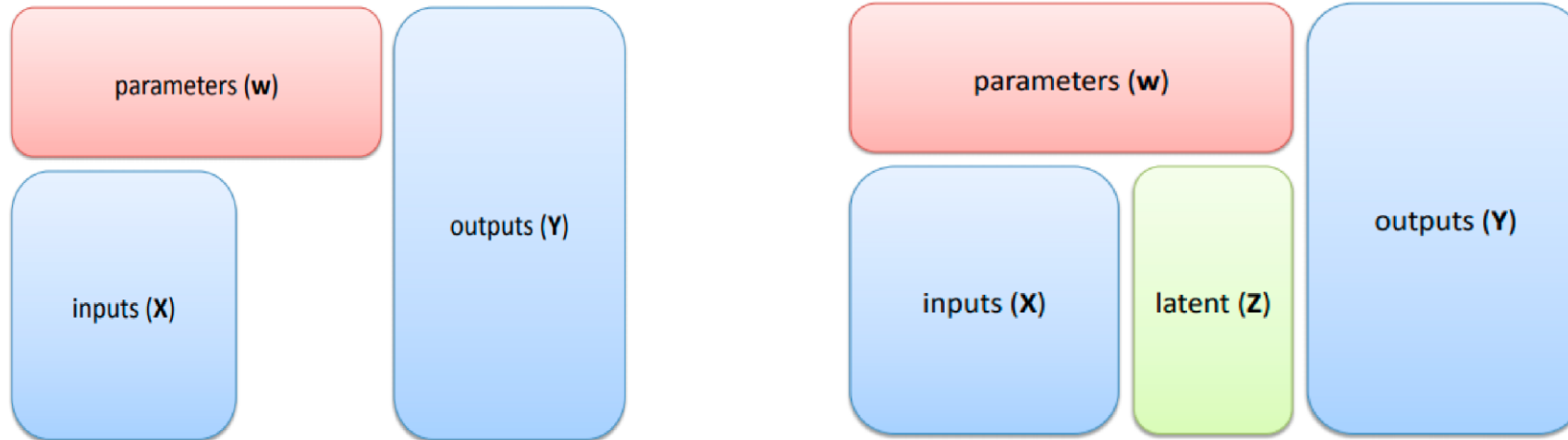
Department of Computer Science and Engineering

With grateful thanks for contribution of slides to:

Prof. Swati Pratap Jagdale,

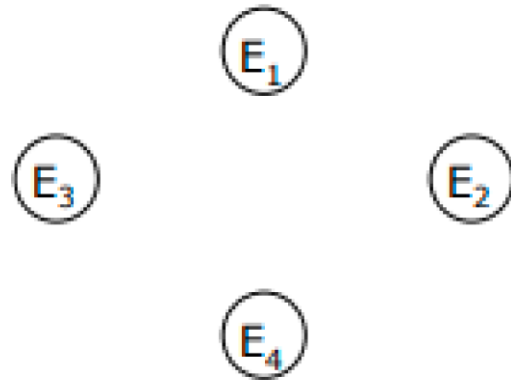
Professor at the Department of CSE, PESU

- Random variables in supervised learning



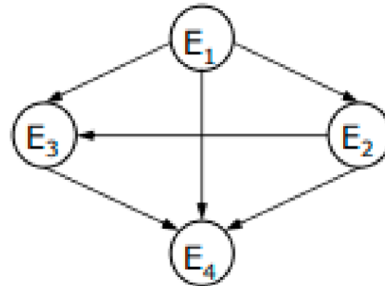
- 'Hidden' (or latent) variable is one that we never 'see'
- Not even in training
- Sometimes we believe they are real
and sometimes they approximate reality (as it happens in Physics)

- Consider a situation in which you can observe a whole bunch of different evidence variables, E_1 through E_n . Maybe they are all the different symptoms that a patient might have. Or maybe they represent different movies and whether someone likes them.



- If those variables are all conditionally dependent on one another, then we would need a highly connected graph that is capable of representing the entire joint distribution between the variables.
- Because the last node has $n-1$ parents, it will take on the order of 2^n parameters to specify the conditional probability tables in this network.

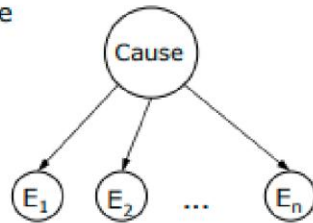
Without the cause,
all the evidence is
dependent on
each other



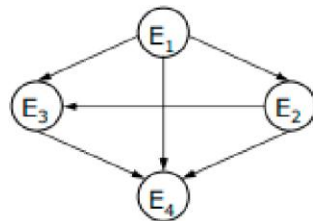
$O(2^n)$ parameters

- But, in some cases, we can get a considerably simpler model by introducing an additional “cause” node.
- It might represent the underlying disease state that was causing the patients’ symptoms or some division of people into those who like drama and those who like comedies.

Cause is unobservable



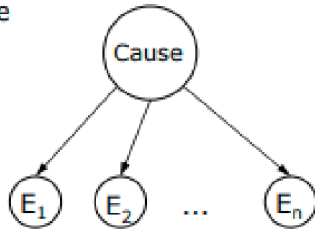
Without the cause,
all the evidence is
dependent on
each other



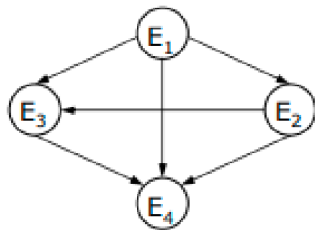
$O(2^n)$ parameters

- In the simpler model, the evidence variables are conditionally independent given the causes. That means that it would only require on the order of n parameters to describe all the CPTs in the network, because at each node, we just need a table of size 2 (if the cause is binary; or k if the cause can take on k values), and one (or $k-1$) parameter to specify the probability of the cause.

Cause is unobservable



$O(n)$ parameters



$O(2^n)$ parameters

Without the cause,
all the evidence is
dependent on
each other

- So, what if you think there's a hidden cause? How can you learn a network with unobservable variables?

- Edward Hugh Simpson, a statistician and former cryptanalyst at Bletchley Park, described the statistical phenomenon - Simpson's paradox
- The art of data science is seeing beyond the data — using and developing methods and tools to get an idea of what that hidden reality looks like.
- Simpson's paradox showcases the importance of skepticism and interpreting data with respect to the real world, and also the dangers of oversimplifying a more complex truth by trying to see the whole story from a single data-viewpoint.

- *Simpson's Paradox:*

A trend or result that is present when data is put into groups that reverses or disappears when the data is combined.

Example: UC Berkley's suspected gender-bias.

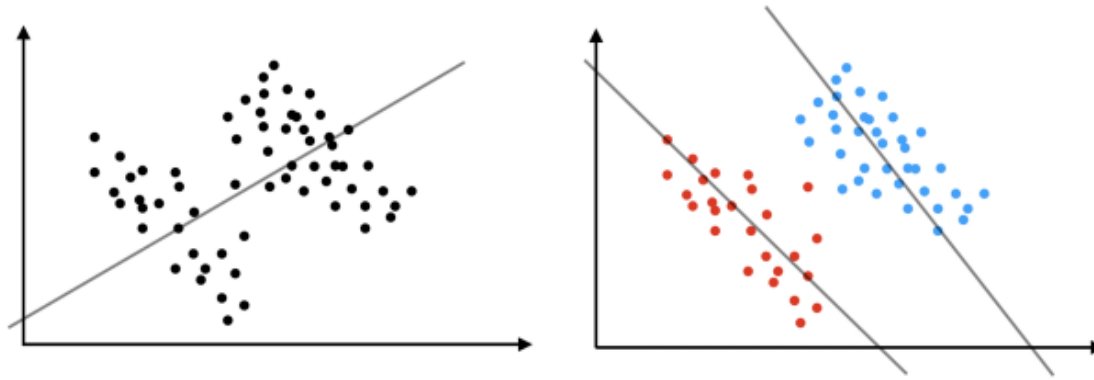
At the beginning of the academic year in 1973, UC Berkeley's graduate school had admitted roughly 44% of their male applicants and 35% of their female applicants.

Was there a discrimination against female applicants?

- When the data was studied department-wise, it was observed that:
- there was a statistically significant gender bias in favor of women for 4 out of the 6 departments, and no significant gender bias in the remaining 2
- It was discovered that women tended to apply to departments that admitted a smaller percentage of applicants overall, and that this hidden variable affected the marginal values for the percentage of accepted applicants in such a way as to reverse the trend that existed in the data as a whole

DATA ANALYTICS

Simpson's Paradox



A visual example: the overall trend reverses when data is grouped by some colour-represented category.

- A simple example in business:
- Suppose the soft drinks industry is trying to choose between two new flavors they have produced. We could sample public opinion on the two flavors

Flavour	Sample Size	# Liked Flavour
Sinful Strawberry	1000	800
Passionate Peach	1000	750

- 80% of people enjoyed 'Sinful Strawberry' whereas only 75% of people enjoyed 'Passionate Peach'. So 'Sinful Strawberry' is more likely to be the preferred

- Some other information while conducting the survey, such as the sex of the person sampling the drink. What happens if we split our data up by sex?
- 84.4% of men and 40% of women liked 'Sinful Strawberry' whereas 85.7% of men and 50% of women liked 'Passionate Peach'

Flavour	# Men	# Liked Flavour (Men)	# Women	# Liked Flavour (Women)
Sinful Strawberry	900	760	100	40
Passionate Peach	700	600	300	150

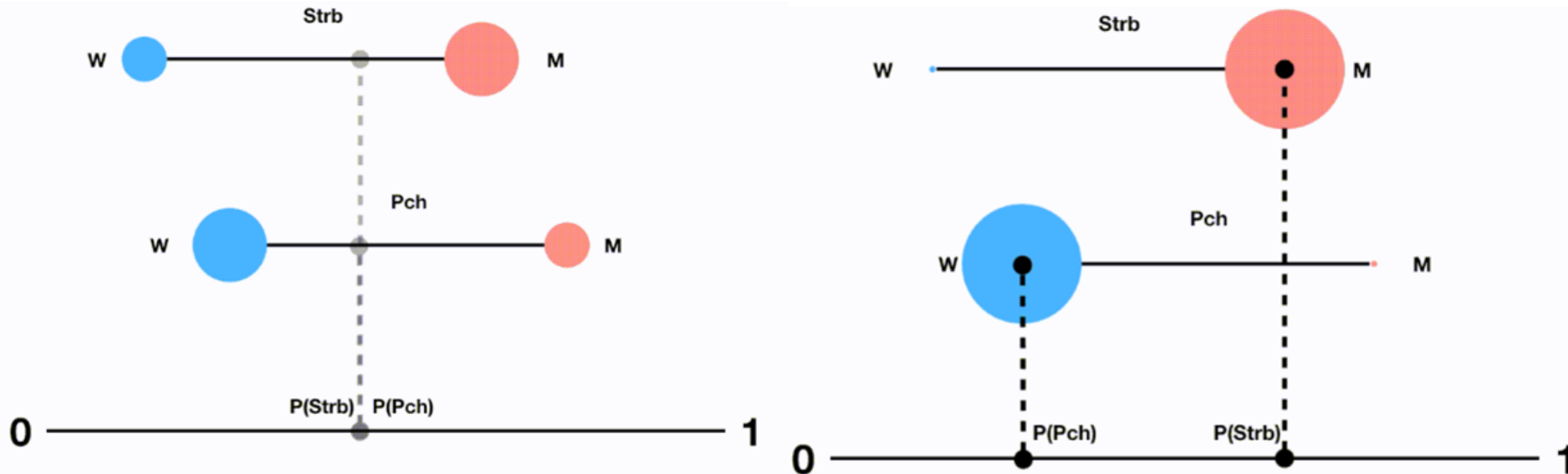
Simpson's Paradox – An example

- *According to our sample data, generally people prefer 'Sinful Strawberry', but both men and women separately prefer 'Passionate Peach'.*
- *This is an example of Simpson's Paradox!*

Flavour	# Men	# Liked Flavour (Men)	# Women	# Liked Flavour (Women)
Sinful Strawberry	900	760	100	40
Passionate Peach	700	600	300	150

- Lurking variables (Hidden Variables)
- Simpson's paradox arises when there are hidden variables that split data into multiple separate distributions.
- Such a hidden variable is aptly referred to as a lurking variable, and they can often be difficult to identify.
- Consider the lurking variable (sex) and a little bit of probability theory:-
- $$P(\text{Liked Strawberry}) = P(\text{Liked Strawberry} \mid \text{Man})P(\text{Man}) + P(\text{Liked Strawberry} \mid \text{Woman})P(\text{Woman})$$
$$= (760/900) \times (900/1000) + (40/100) \times (100/1000) = 800/1000$$
- $$P(\text{Liked Peach}) = P(\text{Liked Peach} \mid \text{Man})P(\text{Man}) + P(\text{Liked Peach} \mid \text{Woman})P(\text{Woman})$$
$$= (600/700) \times (700/1000) + (150/300) \times (300/1000) = 750/1000$$

- Lurking variables (Hidden Variables)
- We can think of the marginal probabilities of sex ($P(\text{Man})$ and $P(\text{Woman})$) as weights that, in the case of 'Sinful Strawberry', cause the total probability to be significantly shifted towards the male opinion.



- Each coloured circle represents either the men or women that sampled each flavour, the position of the centre of each circle corresponds to that group's probability of liking the flavour.

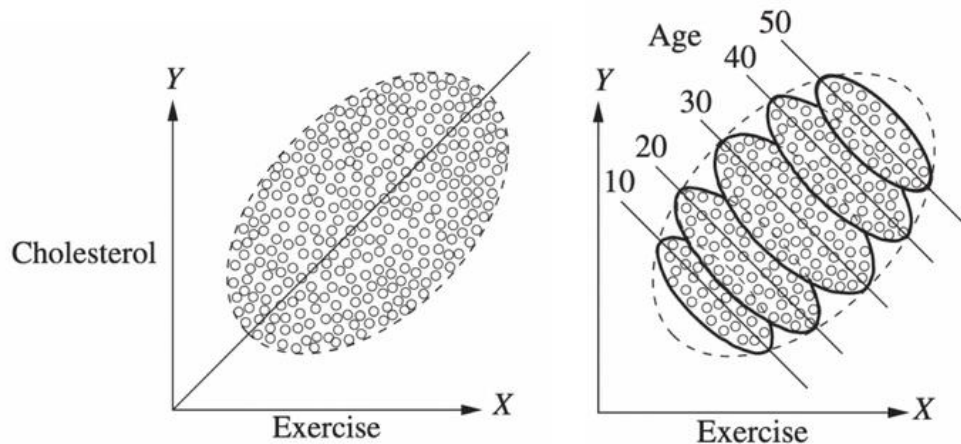
As the circles grow (i.e. sample proportions change) we can see how the marginal probability of liking the flavour changes.

The marginal distributions shift and switch as samples become weighted with respect to the lurking variable (sex).

Causal Inference is the process where causes are inferred from data, from other variables in the model i.e., from observations or evidence.

How do hidden variables link to causal inference...?

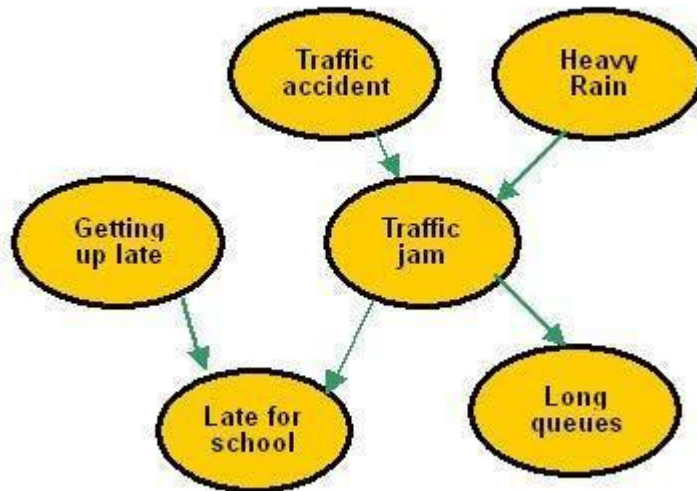
- Standard statistics is all about correlations, which are all good and fun, but correlations can lead to wrong assumptions



This is a graph showing the correlative relationship between Exercise and Cholesterol (which looks like a causal relationship but is not). If we just look at the correlative relationship between cholesterol and exercise, it looks like there's a causal relationship between the two. But this correlation actually happens because both cholesterol and exercise share a common cause or the confounder: age.

Confounder: hidden variable that relates to both the independent and dependent variable

- Independent variables:
 - Traffic accident, heavy rains, getting up late
- Dependent variables
 - Late for school, long queues
- Confounding variable
 - Traffic jam



Why do we care about confounding variables?

Confounding Bias

- Bias is usually a result of errors in data collection or measurement.
- However, one definition of bias is “***...the tendency of a statistic to overestimate or underestimate a parameter***”, so in this sense, confounding is a type of bias.
- Confounding bias is the result of having confounding variables in your model. It has a direction, depending on if it over- or underestimates the effects of your model:
 - **Positive confounding** is when the observed association is biased away from the null. In other words, it overestimates the effect.
 - **Negative confounding** is when the observed association is biased toward the null. In other words, it underestimates the effect.

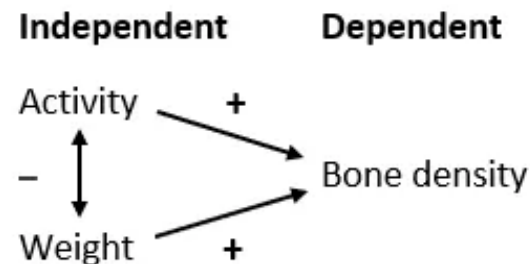
Confounding bias – another example

- Example of How Confounding Variables Can Produce Bias

Example:

- In a biomechanics lab, one study assessed the effects of physical activity on bone density.
- They measured various characteristics including the subjects' activity levels, their weights, and bone densities among many others.
- Theories about how our bodies build bone suggest that there should be a positive correlation between activity level and bone density. In other words, higher activity produces greater bone density.
- Simple regression analysis to determine whether there is a relationship between activity and bone density... **there was**

- They included activity level as the only independent variable, but it turns out there is another variable that correlates with both activity and bone density—the **subject's weight**.



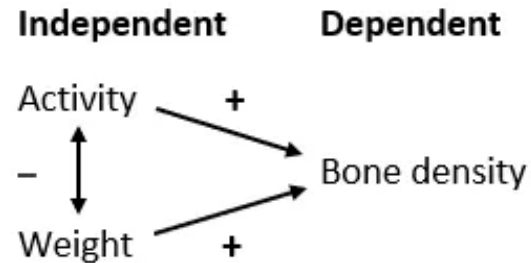
The diagram shows the signs of the correlations between the variables.

- Subjects who are more active tend to have higher bone density. Additionally, subjects who weigh more also tend to have higher bone density. However, there is a negative correlation between activity and weight. More active subjects tend to weigh less.

This correlation structure produces two opposing effects of activity. More active subjects get a bone density boost. However, they also tend to weigh less, which reduces bone density.

no relationship at all!

Confounding bias – another example



The diagram shows the signs of the correlations between the variables.

In this study 'weight' is the confounding variable.

If we try to build a model that predicts bone density based only on 'activity', then,

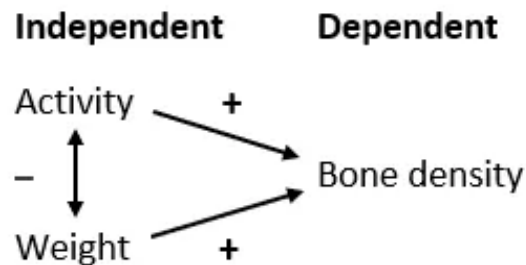
More activity → higher bone density
(for those who are more active)
More activity → lower bone density
(for those who weigh less)

Two opposing effects have to be attributed to one variable: 'activity'

A model that does not account for 'weight' but only uses 'activity' ends up with a negative bias: the model underestimates the importance of activity

In other words, without 'weight', 'activity' is deemed 'not important' for 'bone density' (as it has opposing effects)!

	Included and Omitted: Negative <u>Correlation</u>	Included and Omitted: Positive Correlation
Included and Dependent: Negative Correlation	Positive bias: coefficient is overestimated.	Negative bias: coefficient is underestimated.
Included and Dependent: Positive Correlation	Negative bias: coefficient is underestimated.	Positive bias: coefficient is overestimated.



Included (activity) and omitted (weight) are negatively correlated.
The included variable (activity) and the dependent variable (bone density) have a positive relationship, implies the result has a **negative bias**.

How do we identify a confounding variable?

- Confounding variables cause omitted variable bias
 - They are also called confounders
- A confounding variable is closely related to both the independent and dependent variables in a study. An independent variable represents the supposed *cause*, while the dependent variable is the supposed *effect*.
- A confounding variable is a third variable that influences both the independent and dependent variables. Failing to account for confounding variables can cause you to wrongly estimate the relationship between your independent and dependent variables.
- A lurking variable is one that was simply not included in the study
- How do we identify a confounding variable?
 1. There must be three or more variables in the study
two variables => one is the cause the other is the effect)
 2. The variable we suspect is a confounding variable, changes systematically with at least one of the variables we are measuring (either independent or dependent)
 3. Identify extraneous variables that relate to subjects (age, gender, etc.), the environment in which the study is conducted (weather, location, etc.) and to the two variables we are explicitly measuring to test for systematic changes to identify a confounding variable.

How do we reduce the effects of a confounding variable?

- Make sure you identify all of the possible confounding variables in your study.
- Make a list of everything you can think of and one by one, consider whether those listed items might influence the outcome of your study. Usually, someone has done a similar study before you. So check the academic databases for ideas about what to include on your list.
- Once you have figured out the variables, techniques to reduce the effect of those confounding variables:
 - Bias can be eliminated with random samples.
 - Introduce control variables to control for confounding variables. For example, you could control for age by only measuring 30 year olds.
 - Within subjects designs test the same subjects each time. Anything could happen to the test subject in the “between” period so this does not make for perfect immunity from confounding variables.
 - Counterbalancing can be used if you have paired designs.
In counterbalancing, half of the group is measured under condition 1 and half is measured under condition 2.

- <http://www.cs.cmu.edu/~nasmith/psnlp/lecture5.pdf>
- <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-825-techniques-in-artificial-intelligence-sma-5504-fall-2002/lecture-notes/Lecture18FinalPart1.pdf>
- <https://towardsdatascience.com/simpsons-paradox-and-interpreting-data-6a0443516765>
- <https://www.statisticshowto.com/experimental-design/confounding-variable/>
- <https://statisticsbyjim.com/regression/confounding-variables-bias/>

- It is observed that ice cream sales increase in the summer and number of robberies also increase in the summer. It is later discovered that ‘summer’ (high temperature) is when people have holidays and travel, hence their homes are vacant. Summer is also when ice cream sales increase. Is ‘summer’ (temperature) a hidden (lurking) variable in this study or a confounding one?

Hidden or lurking

(as it was not a part of the study)

- While studying the effect of sunlight on Vitamin D levels versus dietary supplements on the level of Vitamin D, group A subject to sunlight had a lot of orange juice (infused with Vitamin D). Is ‘orange juice (infused with Vitamin D)’ a hidden variable or a confounding variable in this study?

Confounding

(it was a part of the study, but does not allow us to isolate the effect of sunlight alone versus dietary supplement – is the increase in the level of vitamin D due to orange juice or sunlight? We need further studies to infer the effect of sunlight alone).

<http://www.cs.cmu.edu/~nasmith/psnlp/lecture5.pdf>

<https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-825-techniques-in-artificial-intelligence-sma-5504-fall-2002/lecture-notes/Lecture18FinalPart1.pdf>

<https://medium.com/data-science/simpsons-paradox-and-interpreting-data-6a0443516765>

<https://www.statisticshowto.com/experimental-design/confounding-variable/>

<https://statisticsbyjim.com/regression/confounding-variables-bias/>



THANK YOU

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DATA ANALYTICS

UE22CS342AA2

UNIT-1

**Lecture 2 : The R programming
environment and descriptive statistics**

Gowri Srinivasa

Department of Computer Science and Engineering

Data Analytics

Unit 1

Lecture 2 : The R programming environment and descriptive statistics.

Slides excerpted from: U. Dinesh Kumar,
“Business Analytics”, Wiley, 2nd Edition 2022

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With grateful thanks for contribution of slides to:

Dr. Mamatha H R, Professor at the Department of CSE, PESU

- **Free, open source interpreted language**
Ross Ihaka & Robert Gentleman in 1993.
- **Most widely used data analysis software**
Used by 2M+ data scientists, statisticians and analysts
- **Most powerful statistical programming language**
Flexible, extensible and comprehensive for productivity
- **Create beautiful and unique data visualizations**
As seen in New York Times, Twitter and Flowing Data
- **Thriving open-source community**
Leading edge of analytics research



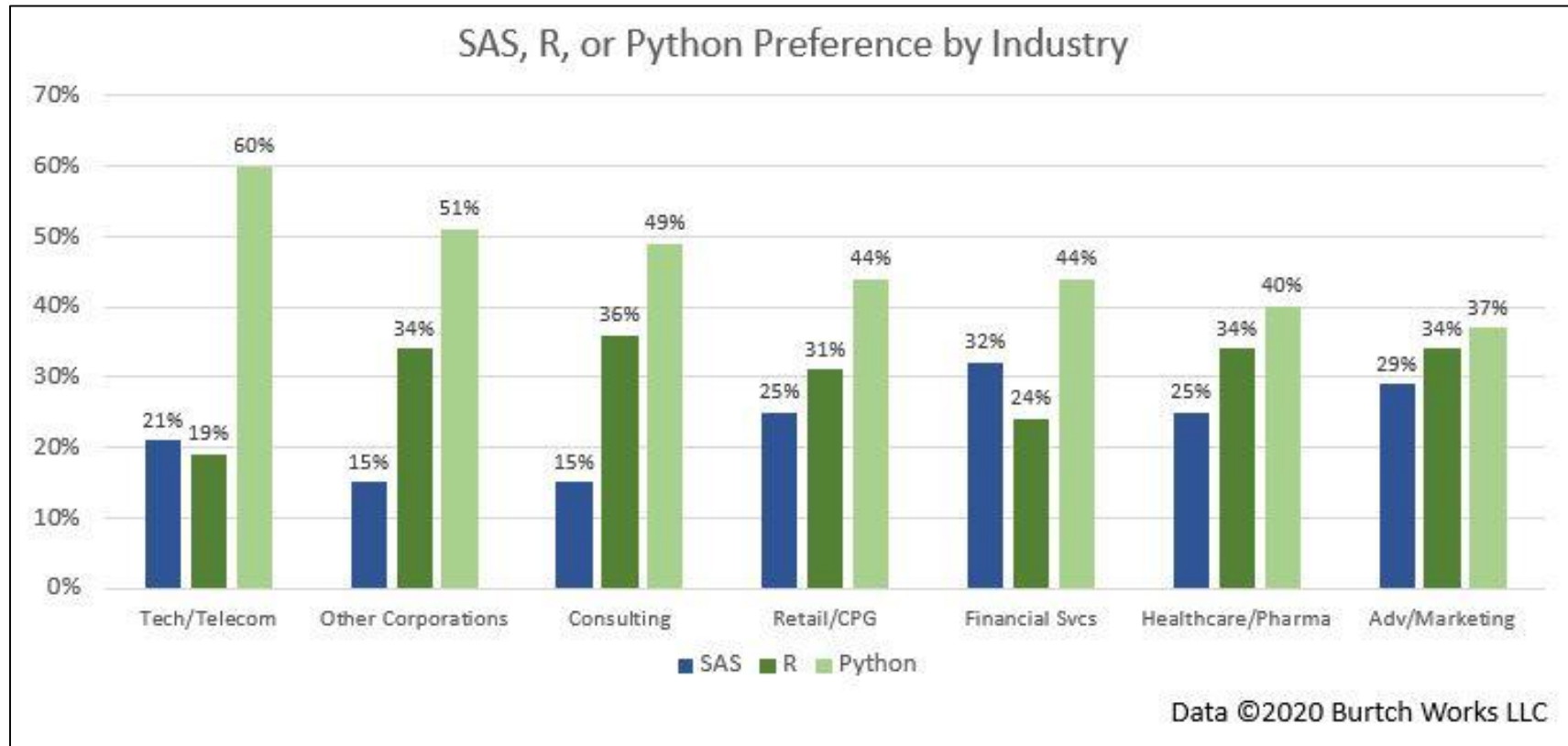
- R allows to collect data in real-time, perform statistical and predictive analysis, create visualizations and communicate actionable results to stakeholders
- It houses more than 9100 packages of statistical functions.
- R's expressive syntax allows to quickly import, clean and analyze data from various data sources.
- R also has charting capabilities, which means you can plot your data and create interesting visualizations from any dataset.

- R is used in predictive analytics and machine learning.
- Packages for ML tasks like linear and non-linear regression, decision trees, linear and non-linear classification and many more.
- R has been used primarily in academics and research and is great for exploratory data analysis.
- In recent years, enterprise usage has rapidly expanded. It is used by statisticians, engineers, and scientists without computer programming skills.
- It is popular in academia, finance, pharmaceuticals, media, and marketing.

- R is great for data analysis because of its huge number of packages, readily usable tests, and the advantage of using formulas.
- It can handle basic data analysis without needing to install packages.
- Big datasets require the use of packages such as `data.table` and `dplyr`.

DATA ANALYTICS

Preference by Industry



- The **S** language has been developed since the late 1970s by **John Chambers** and colleagues at Bell Labs as a **language for programming with data**.
- S language combines ideas from a variety sources (awk, lisp, APL,) and provides an environment for quantitative computations and visualization.
- Provides an explicit and consistent structure for manipulating, analyzing statistically and visualizing data.
- **S-Plus** is a commercialization of the Bell Labs framework. It is “S” plus “graphics”.
- R is a free implementation of a dialect of [the S language](#)

- R is an Open source statistical environment/platform developed by Robert Gentleman and Ross Ihaka (University of Auckland) during the 1990s.
- R is currently maintained by the R core-development team, a hard-working, international team of volunteer developers.
- [The primary R system is available from the Comprehensive R Archive Network, also known as CRAN.](#)
- CRAN also hosts many add-on packages that can be used to extend the functionality of R. Over 6,789 packages are available on CRAN that have been developed by users and programmers around the world.

Finding out the latest version of R:

- To find out what is the latest version of R, you can look at [the CRAN \(Comprehensive R Network\) website,](http://cran.r-project.org/)

Installing R on Windows:

- To install R on your Windows computer, follow the below steps :
- Go to <https://cran.rstudio.com/bin/windows/base/>
- Alternative : <http://ftp.heanet.ie/mirrors/cran.r-project.org>
- Under _“Download and Install R” , click on the “Windows” link.
- Download the required.exe file. The latest version of R is 4.3.1. Click on the link which says “Download R 4.3.1 for windows.”
- After downloading, double click on the executable to run it.
- Choose English as the language to install it.

DATA ANALYTICS

Download and install RStudio on Windows

Visit <https://www.rstudio.com/products/rstudio/download/>
and click on DOWNLOAD RSTUDIO DESKTOP

List of IDEs

- RStudio
- StatET for R (eclipse based)
- R-Brain IDE (RIDE)
- IntelliJ IDEA
- R Tool for Visual Studio



R getwd() Function

- Working directory is the directory where R finds all R files for reading and writing.
- getwd() function returns an absolute file path representing the current working directory of the R process.
- **getwd()** **Output:"C:/Users/*****/Documents"**

R setwd() Function

- setwd(dir) is used to set the working directory to dir.
- **setwd("e:/folder")** **Output:Sets pwd to "e:/folder"**

R dir() Function

- dir() function lists all the files in a directory.
- **dir()** **Output:Lists all files in the pwd**

R ls() Function

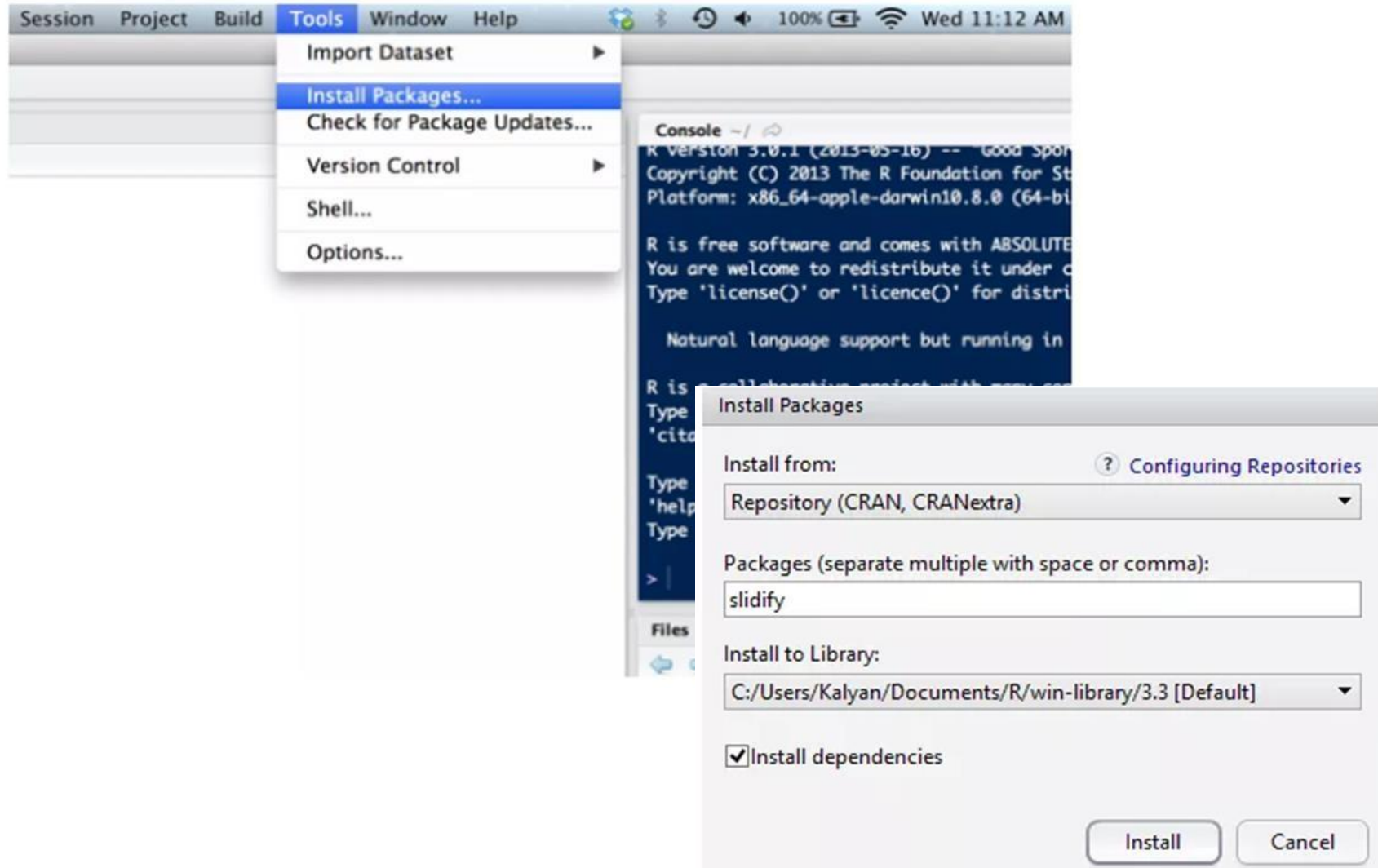
- ls() is a function in R that lists all the object in the working environment.
- **ls()** **Output:Lists all objects in the pwd**
- It can be used in the scenario where you want to clean the environment before running the code.The following command will remove all the object from R environment.
- **rm(list = ls())**

- To get general help just type the below command
 - **help.start()**
- To access documentation for the standard lm (linear model) function, for example, enter the command.
 - **help(lm) / help("lm") / ?lm /?"lm"**
- To see the list of pre-loaded data (datasets), type the function:
 - **data()**
- A guide to get started on working with built in datasets:

- The primary location for obtaining R packages is [CRAN](#).
- Information about the available packages on CRAN with the ***available.packages()*** function.
- **`a <- available.packages()`**
- Packages can be installed with the ***install.packages()*** function in R.
- **`install.packages("ggplot2")`**
- Install multiple R packages at once with a single call to `install.packages()`. Place the names of the R packages in a character vector.
- **`install.packages(c("caret","ggplot2","dplyr"))`**

DATA ANALYTICS

Installing an R Package in RStudio



- Installing a package does not make it immediately available to you in R; you must load the package. The `library()` function is used to load packages into R.
- **`library(ggplot2)`**
- After loading a package, the functions exported by that package will be attached to the top of the `search()` list (after the workspace).
- **`library(ggplot2)`**
- **`search()`**
- To save your workspace to a file, you may type **`save.image()`** or use **Save Workspace** in the **File** menu
- The default workspace file is called **`.RData`**



THANK YOU

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Established under the Karnataka Act No. 16 of 2013

UE20CS312 - Data Analytics

Worksheet 1a - Part 2: EDA with R | ANOVA

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Prerequisites

To download the data required for this worksheet, visit [this Github link](#). This worksheet has two parts, the first focuses on the basics of dealing with data and exploratory data analysis using R. The second deals with ANOVA. To help guide you through the worksheet, here are a few resources:

- Revise how to deal with DataFrames in R [here](#).
- [This online book](#) has everything you need to get started with visualizations in R.
- Check out [this](#) resource for an excellent deep-dive of visualizations using the `ggplot2` library (**optional**).
- The following are resources to learn about ANOVA:
 - [Anova in Python](#)
 - [Anova in R](#)

Part I. Exploratory Data Analysis with R

Book Club Marketing Dataset

Charles Book Club (CBC) is a book club that has an active database of 500,000 subscribers. The organization sends out monthly mailings to its database of members with the latest promotional offerings. Its marketing team would like to see if customer data can be used to reduce the cost of marketing activities to improve the profitability of their marketing operations. For an initial pilot of a predictive analytics solution, CBC decided to focus on its strongest customers and run a marketing test for a new book release of *'The Art History of Florence'*.

The dataset provided consists of information about customer purchases CBC has as its disposal after conducting the marketing test. Use the `CharlesBookClubDataset.csv` for Part I of the worksheet. This data was adapted from a famous business database called the 'Charles Book Club', dealt with in more detail in a case study from the 'Data Mining for Business Analytics' book.

Data Dictionary

ID#: Customer Identification number

Gender: Male, Female

M: Monetary - Total money spent on books

R: Recency - Months since last purchase

F: Frequency - Total number of purchases
FirstPurch: Months since first purchase
ChildBks: Number of purchases from category of child books |
YouthBks: Number of purchases from category of youth books
CookBks: Number of purchases from category of cook books
DoItYBks: Number of purchases from category of DIY books
RefBks: Number of purchases from category of reference books
ArtBks: Number of purchases from category of art books
GeoBks: Number of purchases from category of geography books
ItalCook: Number of purchases of book title 'Secrets of Italian Cooking'
ItalAtlas: Number of purchases of book title 'Historical Atlas of Italy'
ItalArt: Number of purchases of book title 'Italian Art'
Related Purchase: Number of related books purchased
Florence: = 1 if 'Art History of Florence' was purchased; = 0 if not

Loading the Dataset

Use the following commands to load the dataset from CSV format and get a high-level overview of its fields:

```
library(tidyverse)
cbc_df <- read_csv(path_to_csv)
head(cbc_df)
```

Points

The problems for this part of the worksheet are for a total of 8 points, with a non-uniform weightage.

- *Problem 1* : 1 point
- *Problem 2* : 2 points
- *Problem 3* : 2 points
- *Problem 4.1* : 1 point
- *Problem 4.2* : 1 point
- *Problem 4.3* : 1 point

Problems

Problem 1 (1 point)

Generate an understanding of the dataset via a summary of its features. Find the count, missing count, minimum, 1st quartile, median, mean, 3rd quartile, max and standard deviation of all relevant columns. Separately, print the total number of missing values in each column.

Problem 2 (2 points)

Replace missing values within the Recency, Frequency, and Monetary features with suitable values. Explain your reasoning behind the method of substitution used. *Hint*: Try plotting the distribution of the values in each feature using the `hist` function. Think about how to best deal with data imputation. Also, plot the distribution of feature values after imputation.

Problem 3 (2 points)

Discretize the continuous values of Monetary, Recency, and Frequency into appropriate bins, and create three new columns `Mcode`, `Rcode` and `Fcode` respectively, for the discretized values. Explicitly mention the number of bins used and explain the choice for the bin size. Print out the summary of the newly created columns. *Hint*: Use the `cut` function to break on preset breakpoints. What are the most optimum breakpoints you can choose? Try to think of a statistical function that provides these breakpoints for optimum binning.

Problem 4

The marketing team heavily relies on the RFM variables of the recency of last purchase, total number of purchases, and total money spent on purchases to gauge the health of the members of the book club. Increases in either the frequency of purchases or monetary spend and decreases in time since last purchase across the customer base, will intuitively lead to more sales for the business.

4.1 Bar Graphs (1 point) Create and visualize histograms for the discretized Recency, Frequency, Monetary features. Also create one for the `FirstPurch` feature.

4.2 Box Plot (1 point) Transform the `Florence` variable into a categorical feature that can take up the values `True` or `False`. Create and visualize horizontal box plots for the original Recency, Frequency, Monetary and `FirstPurch` features against the `Florence` variable. *Hint:* To transform `Florence`, use the concept of factors in R and set the labels `True` and `False`.

4.3 Density Plot (1 point) Create and visualize a density plot for Recency, Frequency, Monetary and `FirstPurch` features.

Part II. ANOVA

An Analysis of Variance Test, or ANOVA, can be thought of as a generalization of the t-tests for more than 2 groups. The independent t-test is used to compare the means of a condition between two groups. ANOVA is used when we want to compare the means of a condition between more than two groups. ANOVA tests if there is a difference in the mean somewhere in the model (testing if there was an overall effect), but it does not tell us where the difference is (if there is one). To find where the difference is between the groups, we have to conduct post-hoc tests.

To perform any tests, we first need to define the null and alternate hypothesis:

- **Null Hypothesis:** There is *no significant difference* among the groups.
- **Alternate Hypothesis:** There is a *significant difference* among the groups.

Points

The problems for this part of the worksheet are for a total of 6 points, with a non-uniform weightage.

- *Problem 1* : 2 points
- *Problem 2* : 3 points
- *Problem 3* : 1 point

Scenario 1

It's a brand new day in the 99th precinct of the New York Police Department. Lieutenant Terrance has had enough of Hitchcock and Scully being useless paper pushers and wanted to assign them work to help the investigations; they were assigned the duty of gaining insights from the different types of objects in the evidence log of an ongoing investigation focused on the New York Mafia.

Problems

Problem 1 (2 points)

Captain Holt provided a file containing the names of a few `People of Interest` and the number of items logged at various evidence lockers of various precincts pertaining to them. He also instructs Peralta and Diaz to look into the file as he was told it should contain more information.

Scully decided to use ANOVA.

For this problem, use the data file named **Scenario 1.csv** in the data repository. Load the following libraries before moving on and read the dataset,

```
library(ggpubr)
library(dplyr)
library(ggplot2)
library(ggpubr)
library(broom)
library(car)

data <- read.csv('Scenario 1.csv')
```

1. Consider the dataset. Which type of ANOVA can Scully use? (Justify why the particular test)
2. What function(s) could have been used by Scully for ANOVA if he uses the R programming language?
3. What does the output of this/these functions tell Scully? (Specify hypotheses and what each column in the summary of the output means considering 5% significance)

Problem 2 (3 points)

Peralta and Diaz find a member of the family, a certain Frank Pentangeli, through Doug Judy. They discovered that the *famiglia* had altered this file resulting in invalid results. The original file was then recovered by the squad and was sent to Scully and Hitchcock for analysis. To their surprise they discovered that the file also had additional column of which gives the priority.

The dataset has three columns:

- First column has the **Person of Interest(POI)** in the Mafia
- Second column has the number of evidence items collected in particular evidence locker (evidence lockers are present across the city and many precincts have multiple squads working on the mafia, so one POI has multiple entries).
- Third column gives the **Priority** given to collect the evidence by a particular squad with respect to a POI.

Read the dataset before moving on. For this problem, use the data file named **Scenario 2.csv** in the data repository.

```
data <- read.csv('Scenario 2.csv')
```

1. Consider the data. Which type of ANOVA can Scully use? (Justify why the particular test)
2. What function(s) could have been used by Scully for the ANOVA if he uses the R programming language?
3. What does the output of this/these functions tell Scully? (Specify hypotheses and what each column in the summary of the output means considering 5% significance)
4. Hitchcock thinks that Scully has missed a task which completes the ANOVA test. What should Scully have thought of? *Hint*: Philosophically, a hypothesis is a proposition made as a basis for reasoning, without any assumption of its truth.

Problem 3 (1 point)

Hitchcock also wanted to compare the number of items collected for each pair of Person of Interest and priority. He decided to follow the common practice of doing a **Tukey's HSD**. The [Tukey's Honestly-Significant-Difference](#)[TukeyHSD] test lets us see which groups are different from one another.

What insights did Hitchcock gain after doing the Tukey's HSD? (The **TukeyHSD** function can be used to do this test and the output of this function can be represented graphically using the **plot** function.)

PES University, Bangalore

Established under the Karnataka Act No. 16 of 2013

UE20CS312 - Data Analytics

Worksheet 1a - Part 1: Exploring Data with R

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Exploring Data with R

Prerequisites

This worksheet aims to develop your understanding of summary statistics and basic visualizations through a pragmatic approach. To download the data required for this worksheet, visit [this Github link](#).

Resources:

- Check out [this](#) beautifully comprehensive resource for everything you need to get started with R.
- [This online book](#) provides guided explanations about visualizations in R using the `ggplot2` library.

1. About the Data: Top 1000 Instagrammers

To make this worksheet a bit interesting for you all, we have picked a dataset from Kaggle which comprises of the details of the top 1000 influencers on Instagram. If you are on this list, send me an email ;P

This dataset has been taken from [this Kaggle dataset](#) by Syed Jafer.

Data Dictionary

Name: Name of the account

Rank: Overall rank in the world.

Category: Stream of the account (Music, Games, etc..)

Followers: Number of followers

Audience Country: country of the majority of audience.

Authentic Engagement: Engagement with the users.

Engagement Avg.: Average engagement of the users.

2. Assignment Submission Format

The following problems are to be completed using the R programming language and should be submitted as a R markdown file (`.rmd`). Since the dataset is public and many of you students will have the same numerical answers, the grades are allocated on the analysis of the problems and personalized answers within the conclusion section.

The markdown file should follow this format:

```
---
title: "UE20CS312 - Data Analytics - Worksheet 1a - Part 1 - Exploring data with R"
subtitle: "PES University"
author:
  - 'INSERT_NAME, Dept. of CSE - INSERT_SRN'
output: pdf_document
urlcolor: blue
editor_options:
  markdown:
    wrap: 72
---

## Solutions

### Problem 1
INSERT SOLUTION CODE IN MARKDOWN
INSERT SCREENSHOT OF R OUTPUT
INSERT ANALYSIS

### Problem 2
INSERT SOLUTION CODE IN MARKDOWN
INSERT SCREENSHOT OF R OUTPUT
INSERT ANALYSIS
(etc)

### Conclusion
INSERT SUMMARY
```

3. Loading the Dataset

Step 1: Ensure you are on the right working directory and the CSV exists in this directory.

```
# Get and print current working directory
print(getwd())

# Set current working directory
setwd("/UR/WORKING/DIRECTORY")

# Get and print current working directory
print(getwd())
```

Step 2: Read CSV File

```
# Load CSV
data <- read.csv("top_1000_instagrammers.csv")
```

4. Preliminary Guided Exercises

Make sure you have the R programming language installed on your system. It is also recommended to make sure RStudio, the popular IDE for R, is installed. RStudio provides a lot of useful functionality like R markdown, a script editor and GitHub integration. Use RStudio Projects as a great way of keeping each week's assignment work organized.

Data Import To import data from CSV files into a DataFrame:

```
data <- read.csv('top_1000_instagrammers.csv', header=TRUE)
```

The `header = TRUE` argument specifies that the first row of your data contains the variable names. If this is not the case you can specify `header = FALSE` (this is the default value so you can omit this argument entirely).

Compact Summary Use the `str()` function to return a compact and informative summary of the DataFrame.

```
str(data)
```

```
## 'data.frame': 1000 obs. of 8 variables:
## $ X : int 0 1 2 3 4 5 6 7 8 9 ...
## $ Name : chr "cristiano" "leomessi" "kendalljenner" "arianagrande" ...
## $ Rank : int 1 2 3 4 5 6 7 8 9 10 ...
## $ Category : chr "Sports with a ball" "Sports with a ballFamily" "ModelingFashion" "Mus
## $ Followers : num 4.63e+08 3.47e+08 2.48e+08 3.21e+08 1.47e+08 ...
## $ Audience.Country : chr "India" "Argentina" "United States" "United States" ...
## $ Authentic.Engagement: num 5500000 3600000 3000000 2400000 4300000 1700000 2400000 1200000 1400000
## $ Engagement.Avg. : num 6600000 4800000 4900000 3400000 5800000 2500000 3200000 1900000 1900000
```

Here we see that `flowers` is a 'data.frame' object which contains 1000 rows and 8 variables (columns). Each of the variables are listed along with their data class and the first 10 values.

Summary Statistics To access the data in any of the variables (columns) in our data frame we can use the `$` notation. Indexing in R starts at 1, which means the first element is at index 1. Access the first 10 values of the `Name` column:

```
data$Name[1:10]
```

```
## [1] "cristiano" "leomessi" "kendalljenner" "arianagrande"
## [5] "zendaya" "kimkardashian" "taylorswift" "kyliejenner"
## [9] "selenagomez" "thv"
```

We can assign a column to another variable and calculate a mean of a numeric variable or get a summary of a variable using the `summary()` function.

```
names <- data$Name
summary(names)
```

```
## Length Class Mode
## 1000 character character
```

```
auth_eng <- data$Authentic.Engagement
mean(auth_eng)
```

```
## [1] 566199.2
```

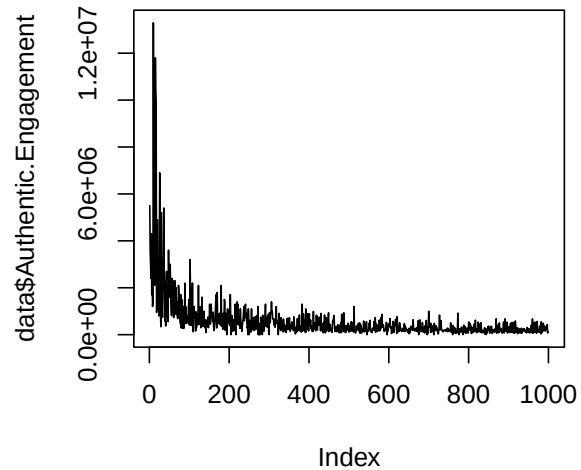
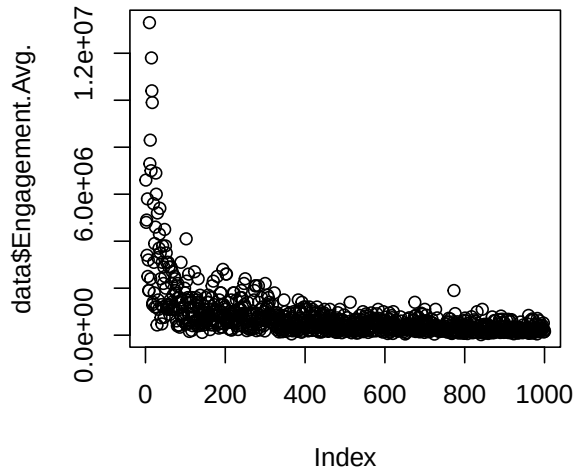
```
summary(auth_eng)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0 169000 316050 566199 604275 13300000
```

Notice how the behavior of the `summary` function changes with different types of variables. Let's now try to explore how we can visualize our data.

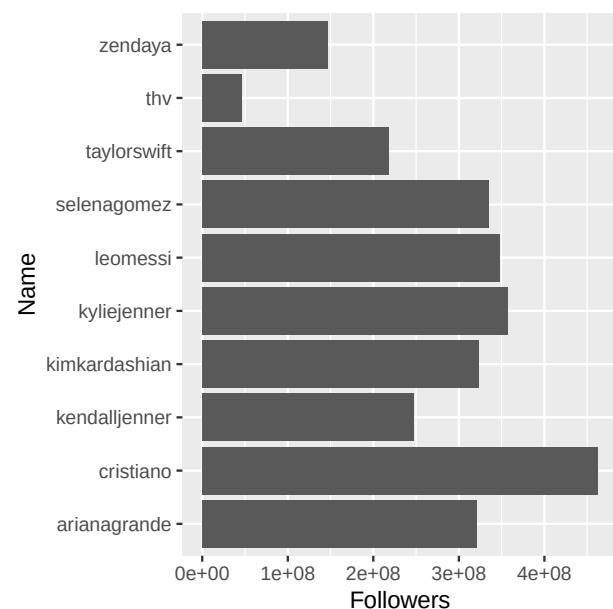
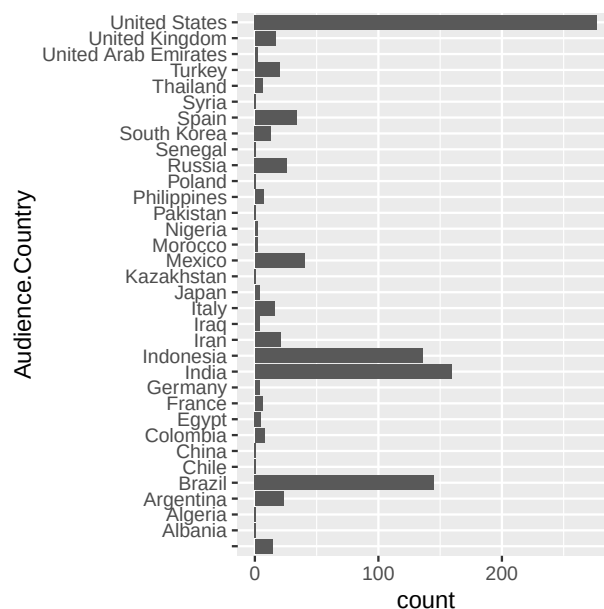
Scatterplots The most common high level function used to produce plots in R is the `plot` function.

```
plot(data$Engagement.Avg., type="p")
plot(data$Authentic.Engagement, type="l")
```



Using the ggplot2 Library Create a bar graph for the distribution of the categorical variable `Audience.Country`.

```
# Import the library for visualization
library(ggplot2)
# Create a bar graph
ggplot(data, aes(x=Audience.Country)) + geom_bar() + coord_flip()
# Number of followers of the top 10 most followed instagrammers
ggplot(data[1:10,], aes(x=Name, y=Followers)) + geom_col() + coord_flip()
```



5. Points

The problems for this part of the worksheet are for a total of 8 points, with a non-uniform weightage.

- *Problem 1* : 1 point
- *Problem 2* : 2 points
- *Problem 3* : 1 points
- *Problem 4* : 3 point
- *Conclusion* : 1 point

6. Problems

Problem 1 (1 point)

Get the summary statistics (mean, median, mode, min, max, 1st quartile, 3rd quartile and standard deviation) for the dataset. Calculate these only for the numerical columns [Audience Country, Authentic Engagement and Engagement Average]. What can you determine from the summary statistics? How does your Instagram stats hold up with the top 1000 :P ?

Problem 2 (2 points)

What are the top 3 audience countries that follow most of the top 1000 instagrammers? *Hint*: Go back to bar graph created earlier. Use R to calculate the percentage of the top 1000 instagrammers that have the top 1 audience country.

Problem 3 (1 point)

Create a horizontal box plot using the column `Authentic.Engagement`. What inferences can you make from this box and whisker plot?

Problem 4 (3 points)

Create a histogram where the x-axis contains the Audience Country and y-axis contains the total follower count for accounts with that Audience Country. Which country is associated with the most amount of followers? *Hint*: Recall the concept of `groupby()` in Pandas. Try using the `aggregate()` function in R to achieve the same goal. What is the total for India and what rank does it fall compared to other countries?

Conclusion (1 point)

In a few short sentences, describe your Instagram profile (category, followers, estimated engagement). Compare your profile to the analysis done of the top 1000 profiles. If you were tasked to becoming an influencer, what would be the best way for you to increase your followers and user engagement?